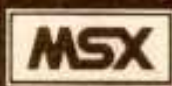


MSX BIOS

The Complete
MSX BASIC
I/O Listing



QEST PUBLISHING INC.

Edited: January 1985
by Steven M. Ting
Graphic design: Mervin Fong.

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MSX BIOS

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```
( MSX ROM BASIC BIOS ) Macro-80      3.44    01-Jan-85      PAGE    1
-BIOS header- BIOS calls (Basic Interpreter, Slot I/O)
```

1 .list
2 ;
3 ;
4 ;
5 (C) Copyright by ASCII Corp., 1983
6 Proprietary information. All rights reserved.
7 ;
8 File: BIOHDR.MAC
9 USE: Restart calls and ROM entries table
10 Written by Jey Suzuki, Rick Yamashita
11 ASCII Corporation, Japan
12 ;
13 Edit: January, 1985
14 Reason: Zilog Z80 Mnemonic version and cleanup
15 Edited by: Steven M. Ting
16 ;
17 ; Labels referenced in this listing, are the absolute locations
18 ; within the MSX ROM. However, "ONLY" this BIOS entry point table,
19 ; and RAM variables are guaranteed to be permanent.
20 ;
21 ; All other locations in the ROM, will be changed without notice.
22 ;
23 SUBTTL -BIOS header- BIOS calls (Basic Interpreter, Slot I/O)

```

24
25
26 ;
27 ; The following RST's (RST 0 thru RST 5) are reserved for BASIC
28 ; interpreter, RST 6 for inter-slot calls, and RST 7 for
29 ; hardware interrupt
30 ;
31 BEGIN: DI ;Fail safe
32 0000 F3 ;Finds all connected RAM
33 0001 C3 02D7 ;and cartridges
34
35 ;
36 ; ** Special information for the VDP. **
37 ; Any program that accesses the VDP hardware directly
38 ; should read the I/O port address found here, to be certain
39 ; the software is compatible with future versions of the VDP.
40 ;
41
42
43 0004 1BBF DW CGTABL ;Address of character generator table,
44 ;to allow use of other character ROM.
45 ;
46
47
48 0006 98 DB 98H ;Current port address for VDP Data read
49 0007 98 DB 98H ; " " " " " write
50 ;
51
52 0008 C3 2683 JP SYNCHR ;Check byte following the RST 8, see
53 ;if equal to the byte pointed by HL
54
55 000B 00 DB 0 ;Read a byte from another slot
56 000C C3 01B6 JP RDSLT
57 000F 00 DB 0 ;Fetch next char from BASIC text
58 0010 C3 2686 JP CHRCTR
59 0013 00 DB 0
60 0014 C3 01D1 JP WRSILT
61 0017 00 DB 0 ;Write a byte to another slot

```

55	0018	C3 1B45	JP	OUTDO		
56	001B	00	DB	0		;Output a char to the Console or printer
57	001C	C3 0217	JP	CALSLT		;Perform Inter-slot call
58	001F	00	DB	0		
59	0020	C3 146A	JP	DCOMPR		;Compares [HL] to [DE]
60	0023	00	DB	0		
61	0024	C3 025E	JP	ENASLT		;Permanently enables a slot
62	0027	00	DB	0		
63	0028	C3 2689	JP	GETYPR		;Returns the [FAC] type
64	002B	00	DB	0		;ID Byte (1)
65						;Format:
66						; B7 B6 B5 B4 B3 B2 B1 B0
67						; + + + + + + + +
68						; + + + + + - - - - - Type of character
69						; + + + + + generator.
70						; + + + + + 0:Japanese
71						; + + + + + 1:International
72						; + + + + + 2:Korea
73						; + + - - - - - Date format
74						; + + 0: Y-M-D 1: M-D-Y
75						; + + 2: D-M-Y
76						; - - - - - Interrupt frequency
77						; 0: 60 Hz 1: 50 Hz
78	002C	00	DB	0		; ID Byte (2)
79						;Format:
80						; B7 B6 B5 B4 B3 B2 B1 B0
81						; + + + + + + + +
82						; + + + + + - - - - - Type of Keyboard
83						; + + + + + 0:Japanese 2:French
84						; + + + + + 1:Int 3:UK
85						; + + + + + 4:DIN

-BIOS header- BIOS calls (Basic Interpreter, Slot I/O)

```

86
87
88
89      002D      00 00 00      DB      0,0,0
90      0030      C3 0205      JP      CALLF
91      0033      00 00 00 00      DB      0,0,0,0,0
92      0037      00
93
94      ;
95      ; Following are used for I/O initialization
96      ;
97      0038      C3 0C3C      JP      KEYINT      ;Handlers for hardware interrupt
98      003B      C3 049D      JP      INITIO      ;Do device initialization
99      003E      C3 139D      JP      INIFNK      ;Reset all function key's text
100
101      SUBTTTL -BIOS header- BIOS calls (Video display processor)

```

```

; - - - - - Version of BASIC
;
;      0: Japanese
;      1: International
;
; Performs Far-call (i.e., Inter-slot)

```

```
102 ;
103 ;
104 ; The following entry points provides control of the
105 ; VDP's registers, screen mode settings, and memory block
106 ; move between DRAM and VRAM.
107 ;
108 0041 C3 0577 JP DISSCR ;Disables screen display
109 0044 C3 0570 JP ENASCR ;Enables screen display
110 0047 C3 057F JP WRTVDP ;Write a byte to any VDP register
111 004A C3 07D7 JP RDVRM ;Read VRAM addressed using [HL]
112 004D C3 07CD JP WRTVRM ;Write VRAM addressed using [HL]
113 0050 C3 07EC JP SETRD ;Sets up VDP for read
114 0053 C3 07DF JP SETWRT ;Sets up VDP for write
115 0056 C3 0815 JP FILVRM ;Fills VRAM with specified data
116 0059 C3 070F JP LDIRMV ;Moves block of data from VRAM to memory
117 005C C3 0744 JP LDIRVM ; " " " " memory to VRAM
118 005F C3 084F JP CHGMOD ;Change screen mode of VDP to [SCRMOD]
119 0062 C3 07F7 JP CHGCLR ;change Foreground, background,
120 ;border, color
121 0065 00 DB 0
122 ;
123 ;
124 0066 C3 1398 JP NMI ;Handler for non-maskable interrupt
125 ;
126 0069 C3 06A8 JP CLRSPR ;Init sprite data
127 006C C3 050E JP INITXT ;Init VDP for 40 X 24 text mode (SCREEN 0)
128 006F C3 0538 JP INIT32 ; " " 32 X 24 text mode (SCREEN 1)
129 0072 C3 05D2 JP INIGRP ; " " High resolution mode (SCREEN 2)
130 0075 C3 061F JP INIMLT ; " " Multi color mode (SCREEN 3)
131 0078 C3 0594 JP SETTXT ;Sets VDP to display 40 X 24 text mode
132 007B C3 05B4 JP SETT32 ; " " 32 X 24 text mode
```

```

( MSX ROM BASIC BIOS ) Macro-80          3.44  01-Jan-85          PAGE  3-1
-BIOS header- BIOS calls (Video display processor)

133      007E      C3 0602      JP      SETGRP      ;      "      "      "      High-res mode
134      0081      C3 0659      JP      SETMLT      ;      "      "      "      Multi color mode
135      0084      C3 06E4      JP      CALPAT      ;Get address of sprite pattern table
136      0087      C3 06F9      JP      CALATR      ;      "      "      "      attribute table
137      008A      C3 0704      JP      GSPSIZ      ;Returns current sprite size
138      008D      C3 1510      JP      GRPPRT      ;Print a character on the graphic screen
139
140      SUBTTL -BIOS header- BIOS calls (Programmable Sound Generator control)

```

```

141
142
143
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150
151

;
; Following entry points are used for PSG initialization,
; read and write PSG registers, and PLAY statement execution.
;

0090 C3 04BD JP GICINI ;Init PSG, and static data for PLAY
0093 C3 1102 JP WRTPSG ;Write data to PSG
0096 C3 110E JP RDPSG ;Read data from PSG
0099 C3 11C4 JP STRTMS ;Checks and start background task for PLAY
;
SUBTTL -BIOS header- BIOS calls (Keyboard, CRT, and Printer)

```

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
 -BIOS header- BIOS calls (Keyboard, CRT, and Printer)

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;
; General INPUT and PRINT utilities.
;

009C C3 0D6A JP CHSNS ;Checks status of keyboard status
009F C3 10CB JP CHGET ;Return char typed, with wait
00A2 C3 08BC JP CHPUT ;Output character to console
00A5 C3 085D JP LPTOUT ; " " to printer, if possible
00A8 C3 0884 JP LPTSTT ;Checks status of line printer
00AB C3 089D JP CNVCHR ;Checks for graphic header byte
                                ;and convert code
00AE C3 23BF JP PINLIN ;Read line from keyboard to buffer
00B1 C3 23D5 JP INLIN ;Same as above, except in case of
                                ;AUTFLG is set
00B4 C3 23CC JP QINLIN ;Print a "?", then jump to INLIN
00B7 C3 046F JP BREAKX ;[Control-STOP] pressed??
00BA C3 03FB JP ISCNTC ;[Shift-STOP] pressed??
00BD C3 10F9 JP CKCNTC ;Same as ISCNTC, but used by BASIC
00C0 C3 1113 JP BEEP ;Buzz
00C3 C3 0848 JP CLS ;Clear screen
00C6 C3 088E JP POSIT ;Place cursor at Column [H], Row [L]
00C9 C3 0B26 JP FNKSB ;Display Function key, if necessary
00CC C3 0B15 JP ERAFNK ;Stop displaying the Function keys
00CF C3 0B2B JP DSPFNK ;Enable Function key display
00D2 C3 083B JP TOTEXT ;Force screen to text mode

SUBTTTL -BIOS header- BIOS calls (Game and Cassette I/O, Queue handler)
;

```

```
179 ;
180 ;
181 ; Following are used to read the value from Joysticks,
182 ; Graphic pad (tablet), and Paddles.
183 ;
184 ;
185 ;
186 ;
187 ;
188 ;
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206 ;
207 ;
208 ;

00D5 C3 11EE JP GTSTCK ;Return status of joystick
00D8 C3 1253 JP GTRIG ;Read joystick trigger button
00DB C3 12AC JP GTPAD ;Returns status of graphic pad
00DE C3 1273 JP GTPDL ;Read paddle
;

; Following are used to access the cassette tape,
; data read/write, and motor on/off
;

00E1 C3 1A63 JP TAPION ;Turn on motor and read tape header
00E4 C3 1ABC JP TAPIN ;Read tape data
00E7 C3 19E9 JP TAPIOF ;Stops reading from tape
00EA C3 19F1 JP TAPOON ;Turn on motor and write tape header
00ED C3 1A19 JP TAPOUT ;Write data to tape
00F0 C3 19DD JP TAPOFF ;Stops writing to tape
00F3 C3 1384 JP STMOTR ;Start, stop cassette motor, or
; ;flip motor(on to off, off to on)
;
; BASIC queues
;
00F6 C3 14EB JP LFTQ ;Bytes left in queue
00F9 C3 1492 JP PUTQ ;Send a byte to queue
;
SUBTTL -BIOS header- BIOS calls (Generalized graphics)
```

-BIOS header- BIOS calls (Generalized graphics)

```

209      ;
210      ;
211      ; For BASIC interpreter's GENGPR and ADVGRP modules use
212      00FC      C3 16C5      JP RIGHTC      ;Moves one pixel right
213      00FF      C3 16EE      JP LEFTC       ; " " " left
214      0102      C3 175D      JP UPC         ; " " " up
215      0105      C3 173C      JP TUPC        ; " " " "
216      0108      C3 172A      JP DOWNC       ; " " " down
217      010B      C3 170A      JP TDOWNC      ; " " " "
218      010E      C3 1599      JP SCALXY      ; Scales X Y coordinates
219      0111      C3 15DF      JP MAPXYC      ; Maps coordinates to physical address
220      0114      C3 1639      JP FETCHC      ; Get current physical address and
221                                     ; mask pattern
222      0117      C3 1640      JP STOREC       ; Put current physical address and
223                                     ; mask pattern
224      011A      C3 1676      JP SETATR       ; Sets the color attribute byte
225      011D      C3 1647      JP READC        ; Reads attribute of current pixel
226      0120      C3 167E      JP SETC         ; Sets current pixel to specified attribute
227      0123      C3 1809      JP NSETCX      ; Sets pixel horizontally
228      0126      C3 18C7      JP GTASPC      ; Returns aspect ratio
229      0129      C3 18CF      JP PNTINI      ; Do paint initialization
230      012C      C3 18E4      JP SCANR       ; Scan pixels to the right
231      012F      C3 197A      JP SCANL       ; " " " left
232      ;
233      SUBTTTL -BIOS header- BIOS calls (Misc. Entries)

```

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-BIOS header- BIOS calls (Misc. Entries)

```

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;
;
;
;
; Turn [CAPSLOCK] light, on/off
; Change status of 1 bit sound port
; Return output of primary slot register
; Write to primary slot register
; Read VDP status register
; Read a specified row in the
; keyboard matrix
; Performs operation for mass storage
; devices (such as disks)
; Initialize mass storage device
; Are we doing device I/O
; Output to line printer
; Used by Music background tasking
; " " " "
; Clear the keyboard buffer
; Performs far-call into BASIC
; RESERVED FOR EXPANSION

;
; SUBTTL - SLOT - Slot handler stuff

```

```
256
257      00A8      EQU      0A8h      ;A8H      read from PPI Port A
258      00A8      EQU      0A8h      ;A8H      Write to PPI Port A
259
260      ;
261      ; Every cartridge located at 0000-3FFFH must contain codes in
262      ; this module which are entered via following addresses.
263      ;
264      ;      000CH RDSLT
265      ;      0014H WRSLT
266      ;      001CH CALSLT
267      ;      0024H ENASLT
268      ;
269      ; ----- RDSLT -----
270      ;
271      ; Selects the appropriate slot according to the value given
272      ; through registers, and read the content of memory from the
273      ; slot.
274      ;
275      ; Input parameters:
276      ; A - FxxxSSPP
277      ;      | | | |
278      ;      | | | | primary slot # (0-3)
279      ;      | | +--- secondary slot # (0-3)
280      ;      | +----- 1 if secondary slot # specified
281      ;
282      ;      HL - address of target memory
283      ;      Returned value
284      ;      A - content of memory
285      ;
286      ; Note: Interrupts are disabled automatically but never enabled
```

```

287 ;
288 ;
289 ; by this routine.
290 ;
291 ; RDSLT:
292 ;
293 ; CD 027E
294 ; FA 01C6
295 ; DB A8
296 ; 57
297 ; A1
298 ; B0
299 ; CD F380
300 ; 7B
301 ; C9
302 ; E5
303 ; CD 02A3
304 ; E3
305 ; C5
306 ; CD 01B6
307 ; 18 1B
308 ;
309 ;
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;Calculate bit pattern and mask code
;Expanded slot specified
;Save current setting
;Cancel current setting for target address
;Add new setting
;Call read primitive routine (in system area)
;Return value via [Acc]

;Save target address
;Select secondary slot
;Restore target address and save [HL]

;Restore old slot select register
;Write slot)

```

```

307 ;
308 ;
309 ; ----- WRSLT -----
310 ;
311 ; Selects the appropriate slot according to the value given
312 ; through registers, and write to the memory in the specified
313 ; slot.
314 ;
315 ; Input parameters:
316 ; A - FxxxSSPP
317 ; | | | |
318 ; | | | | primary slot # (0-3)
319 ; | | | | secondary slot # (0-3)
320 ; +----- 1 if secondary slot # specified
321 ;
322 ; HL - address of target memory
323 ;
324 ; E - value to be written
325 ;
326 ; Note: Interrupts are disabled automatically but never enabled
327 ; by this routine.
328 ;
329 WRSLT:
330 01D1 D5 PUSH DE ;Save data to be written
331 01D2 CD 027E CALL SELPRM ;Calculate bit pattern and mask code
332 01D5 FA 01E1 JP M,WRSLT ;Expanded slot specified
333 01D8 D1 POP DE ;Restore data to be written
334 01D9 DB A8 IN A,(PPI.AR) ;Save current setting
335 01DB 57 LD D,A ;Cancel current setting for target address
336 01DC A1 AND C ;Add new setting
337 01DD B0 OR B

```

338	01DE	C3 F385	JP	WRPRIM	;Call write primitive routine (in system area)
339	01E1				
340	01E1	E3	EX	(SP),HL	;Save target address, get data to be written
341	01E2	E5	PUSH	HL	;Save data to be written
342	01E3	CD 02A3	CALL	SELEXP	;Select secondary slot
343	01E6	D1	POP	DE	;Restore data to be written
344	01E7	E3	EX	(SP),HL	;Restore target address and save [HL]
345	01E8	C5	PUSH	BC	
346	01E9	CD 01D1	CALL	WRSLT	
347	01EC				
348	01EC	C1	POP	BC	
349	01ED	E3	EX	(SP),HL	;Save target address and get old [HL]
350	01EE	F5	PUSH	AF	;Save value returned by RDSLT
351	01EF	78	LD	A,B	;Get current setting
352	01F0	E6 3F	AND	00111111B	;Cancel current setting for 0C000H..0FFFFH
353	01F2	B1	OR	C	
354	01F3	D3 A8	OUT	(PPI.AW),A	;Enable 0C000H..0FFFFH of target bank
355	01F5	7D	LD	A,L	;Restore old setting of slot register
356	01F6	32 FFFF	LD	(0FFFFH),A	
357	01F9	78	LD	A,B	;Finally restore old primary slot register
358	01FA	D3 A8	OUT	(PPI.AW),A	
359	01FC	F1	POP	AF	;Restore value returned by RDSLT
360	01FD	E1	POP	HL	;Restore target address
361	01FE	C9	RET		

WRESLT:

WRESED:

```

( MSX ROM BASIC BIOS ) Macro-80
-SLOT- Slot handler (Write slot)

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383

                                CALBAS:
                                LD 2A FCC0
                                18 12
                                JR
                                CALSLT
                                IY,(EXPTBL-1)
                                CALLF:
                                EX
                                PUSH
                                PUSH
                                LD
                                PUSH
                                POP
                                INC
                                LD
                                INC
                                LD
                                INC
                                PUSH
                                POP
                                POP
                                POP
                                EX
                                SUBTTL -SLOT-

                                (SP),HL
                                AF
                                DE
                                A,(HL)
                                AF
                                IY
                                HL
                                E,(HL)
                                HL
                                D,(HL)
                                HL
                                DE
                                IX
                                DE
                                AF
                                (SP),HL

                                ;Get return address, save [HL]
                                ;Save working registers
                                ;Get destination slot
                                ;Move it to IYH
                                ;Get destination address
                                ;Prepare true return address
                                ;Move it to IX
                                ;Restore working registers
                                ;Resture [HL], save true return address

```

```

384 ;
385 ;
386 ; ----- CALSLT -----
387 ;
388 ; Performs inter-slot call to specified address.
389 ;
390 ; Input parameters:
391 ; IY - FxxxSSPP
392 ;      |  ||||
393 ;      |  |||+--- primary slot # (0-3)
394 ;      |  ||+----- secondary slot # (0-3)
395 ;      |  +----- 1 if secondary slot # specified
396 ;
397 ; IX - address to call
398 ;
399 ; Note: Interrupts are disabled automatically but never enabled
400 ; by this routine.
401 ; You can never pass arguments via alternate registers
402 ; of Z80.
403 ;
404 CALSLT:
405     D9      EXX
406     08      EX      AF,AF'
407     FD E5   PUSH    IY
408     F1      POP     AF
409     DD E5   PUSH    IX
410     E1      POP     HL
411     CD 027E CALL    SELPRM
412     FA 022E JP      M,CALESL
413     DB A8   IN      A,(PPI.AR)
414     F5      PUSH    AF
                ;Save current value of primary slot register

                ;Save environments

                ;Get target slot information

                ;Get target address

                ;Call expanded slot

                ;Save current value of primary slot register

```

(MSX ROM BASIC BIOS) Macro-80		3.44	01-Jan-85	PAGE 12-1
415	0228	A1		
416	0229	B0	AND	;Cancel current setting for target address
417	022A	D9	OR	;Add new setting
418	022B	C3 F38C	EXX	;Restore environments except PSW
419	022E		JP	;Jump to primitive routine (in system area)
420	022E	CD 02A3		
421	0231	F5	CALL	;Select secondary slot register
422	0232	FD E1	PUSH	;Move primary slot # in [IYH]
423	0234	E5	POP	
424	0235	C5	PUSH	;Save [B,C,L] which contain information
425	0236	4F	PUSH	;for restoring slot environments
426	0237	06 00	LD	;Move primary slot # to [BC]
427	0239	7D	LD	
428	023A	A4	LD	;Re-calculate what is currently output
429	023B	B2	AND	;to expansion slot register
430	023C	21 FCC5	OR	
431	023F	09	LD	;Calculate address into SLTTBL
432	0240	77	ADD	
433			LD	;Set current value output to expansion
434	0241	E5		;slot register
435	0242	08	PUSH	;Remember this address
436	0243	D9	EX	;Restore possible arguments passed
437	0244	CD 0217	EXX	;via registers
438	0247	D9	CALL	;Call by primary slot #
439	0248	08	EXX	;Save possible values returned via
440	0249	E1	EX	;registers
441	024A	C1	POP	;Restore address into SLTTBL
442	024B	D1	POP	;Restore information about old slots
443	024C	78	POP	
444	024D	E6 3F	LD	;Get current setting
445	024F	B1	AND	;Cancel current setting for 0C000H..0FFFFH
			OR	

CALESL:

-SLOT-

446	0250	F3	DI		
447	0251	D3 A8	OUT	(PPI.AW),A	;Enable 0C000H..0FFFFH of target bank
448	0253	7B	LD	A,E	;Restore old setting of slot register
449	0254	32 FFFF	LD	(0FFFFH),A	
450	0257	78	LD	A,B	;Finally restore old primary slot register
451	0258	D3 A8	OUT	(PPI.AW),A	
452	025A	73	LD	(HL),E	;And change SLTTBL also
453	025B	08	EX	AF,AF'	;Restore possible returned values
454	025C	D9	EXX		
455	025D	C9	RET		

```

456 ;
457 ;
458 ; ----- ENASLT -----
459 ;
460 ; Selects the appropriate slot according to the value given
461 ; through registers, and permanently enables the slot.
462 ;
463 ; Input parameters:
464 ;
465 ; A - FxxxSSPP
466 ;      | | | |
467 ;      | | | | primary slot # (0-3)
468 ;      | | | | secondary slot # (0-3)
469 ;      +----- 1 if secondary slot # specified
470 ;
471 ; HL - address of target memory
472 ;
473 ; Note: Interrupts are disabled automatically but never enabled
474 ;       by this routine.
475 ;
476 ENASLT:
477     CD 027E      SELPRM      ;Calculate bit pattern and mask code
478     FA 026B      JP          ;Expanded slot specified
479     DB A8        IN          A,(PPI.AR)
480     A1           AND        C      ;Cancel current setting for target address
481     B0           OR         B      ;Add new setting
482     D3 A8        OUT        (PPI.AW),A
483     C9           RET
484
485 ENESLT:
486     E5           PUSH       HL      ;Save target address
487     CD 02A3      CALL       SELEXP ;Select secondary slot

```

487	026F	4F	LD	C,A	
488	0270	06 00	LD	B,0	;Move primary slot # to [BC]
489	0272	7D	LD	A,L	
490	0273	A4	AND	H	;Re-calculate what is currently output
491	0274	B2	OR	D	;to expansion slot register
492	0275	21 FCC5	LD	HL,SLTTBL	
493	0278	09	ADD	HL,BC	;Calculate address into SLTTBL
494	0279	77	LD	(HL),A	
495					;Set current value output to expansion
496	027A	E1	POP	HL	;slot register
497	027B	79	LD	A,C	;Restore target address
498	027C	18 E0	JR	ENASLT	;Restore primary slot # to [Acc]
					;Enable by primary slot register

```

( MSX ROM BASIC BIOS ) Macro-80      3.44      01-Jan-85      PAGE  14
-SLOT-

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      SELPRM:
027E      F3
027E      F5
027F      7C
0280      07
0281      07
0282      07
0283      E6 03
0285      5F
0286      3E C0
0288
0288      07
0289      07
028A      1D
028B      F2 0288
028E      5F

      SLPRM1:
0288      RLCA
0289      RLCA
028A      DEC
028B      JP
028E      LD

      CPL
028F      2F
0290      4F

      POP
0291      F1
0292      F5
0293      E6 03
0295      3C
0296      47

      DI
      PUSH
      LD
      RLCA
      RLCA
      AND
      LD
      LD

      AF
      A,H
      00000011B
      E,A
      A,0C0H

      E
      P,SLPRM1
      E,A

      00000011
      00001100
      00110000
      11000000

      0000-3FFF
      4000-7FFF
      8000-BFFF
      C000-FFFF

      ;Save mask pattern
      ;
      ;
      ;
      ;

      ;Save mask pattern
      ;
      ;
      ;
      ;
      ;Restore slot address

      AF
      AF
      00000011B
      A
      B,A

      ;Save slot address
      ;Extract upper 2 bits

      ;Format mask pat. correspond to address

      0000-3FFF
      4000-7FFF
      8000-BFFF
      C000-FFFF

      ;Save mask pattern
      ;
      ;
      ;
      ;
      ;Restore slot address

      ;Extract primary slot #

```

(MSX ROM BASIC BIOS)	Macro-80	3.44	01-Jan-85	PAGE	14-1
530	0297	3E AB			
531	0299		A,10101011B		;Convert slot # to proper bit pattern
532	0299	C6 55			
533	029B	10 FC	A,01010101B		
534	029D	57	SLPRM2		;Save bit pattern for primary slot #
535			D,A		; 00000000 slot #0
536					; 01010101 slot #1
537					; 10101010 slot #2
538					; 11111111 slot #3
539	029E	A3			;Extract significant bits
540	029F	47	E		;Set it to [B]
541	02A0	F1	B,A		;Expanded slot specified?
542	02A1	A7	AF		;Set sign flag if so
543	02A2	C9	A		
544	02A3				
545	02A3	F5			
546	02A4	7A	AF		;Save target slot
547	02A5	E6 C0	A,D		;Get bit pattern for primary slot
548	02A7	4F	11000000B		;Extract slot # for 0C000H..0FFFFH
549	02A8	F1	C,A		;Save it
550	02A9	F5	AF		;Restore target slot
551	02AA	57	AF		;Save target slot
552	02AB	DB A8	D,A		;Load [D] with specified slot address
553	02AD	47	A,(PPI.AR)		
554	02AE	E6 3F	B,A		;Save current setting
555	02B0	B1	00111111B		;Cancel current setting for 0C000H..0FFFFH
556	02B1	D3 A8	C		
557	02B3	7A	(PPI.AW),A		;Enable 0C000H..0FFFFH or target bank
558	02B4	0F	A,D		;Load slot information
559	02B5	0F			
560	02B6	E6 03	00000011B		;Extract secondary slot #

-SLOT-

```

( MSX ROM BASIC BIOS ) Macro-80      3.44      01-Jan-85      PAGE 14-2
-SLOT-

561      02B8      57      LD      D,A
562      02B9      3E AB      LD      A,10101011B
563      02BB      SLEXP1:
564      02BB      C6 55      ADD     A,01010101B
565      02BD      15      DEC     D
566      02BE      F2 02BB      JP      P,SLEXP1
567
568
569
570      02C1      A3      AND
571      02C2      57      LD      D,A
572      02C3      7B      LD      A,E
573      02C4      2F      CPL
574      02C5      67      LD      H,A
575      02C6      3A FFFF      LD      A,(0FFFFH)
576      02C9      2F      CPL
577      02CA      6F      LD      L,A
578      02CB      A4      AND     H
579      02CC      B2      OR      D
580      02CD      32 FFFF      LD      (0FFFFH),A
581      02D0      78      LD      A,B
582      02D1      D3 A8      OUT     (PPI.AW),A
583      02D3      F1      POP     AF
584      02D4      E6 03      AND     00000011B
585      02D6      C9      RET
586

SUBTTL - MSXIO - I/O Module

```

```

;Convert secondary slot # to proper
;bit pattern
;
; 00000000 slot #0
; 01010101 slot #1
; 10101010 slot #2
; 11111111 slot #3
;Make bit pattern to be added
;Save this
;Make bit pattern to strip off old value
;Save this
;Read expanded slot register
;Save current setting
;Strip off old bits
;And set new bits
;Set secondary slot register
;Restore original primary port
;Restore target slot
;Fake read from primary slot

```



```

( MSX ROM BASIC BIOS ) Macro-80      3.44  01-Jan-85      PAGE  15-1
- MSXIO - I/O Module

618      00AA      PPI.CW      EQU      10101010B      ;AAH      write to PPI Port C
619      00AB      PPI.CM      EQU      10101011B      ;ABH      write to PPI command register
620      ;
621      ;      Printer port definition
622      ;
623      0091      LPT.DW      EQU      10010001B      ;Data port
624      0090      LPT.SB      EQU      10010000B      ;Strobe output
625      0090      LPT.ST      EQU      10010000B      ;Printer status
626      ;
627      ;      Text mode (40*24)      SCREEN 0
628      ;
629      ;      TXTNAM,TXTCGP
630      ;
631      ;      Text mode (graphics 1)      SCREEN 1
632      ;
633      ;      T32NAM,T32COL,T32CGP,T32ATR,T32PAT
634      ;
635      ;      Hires mode      SCREEN 2
636      ;
637      ;      GRPNAM,GRPCOL,GRPCGP,GRPATR,GRPPAT
638      ;
639      ;      Multi-color mode      SCREEN 3
640      ;
641      ;      MLTNAM,MLTCGP,MLTATR,MLTPAT
642      ;
643      ;      Screen size
644      ;
645      ;      LINLEN,CRTCNT,LINL32,LINL40
646      ;
647      ;      External constants
648      ;

```

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649 ;
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671 ;
672 ;
673 ;
674 ;
675 ;
676 ;
677 ;
678 ;
679 ;

; CGTABL
;
; External variables
;
;   FORCLR      Foreground color
;   BAKCLR      Background color
;   BDRCLR      Border color for PAINT
;   SCRMOD      Current screen mode
;               ; 0 - 40*24 text
;               ; 1 - 32*24 text
;               ; 2 - hiresolution graphics
;               ; 3 - Multicolor graphics
;
;   OLDSCR      Base of current name table
;   NAMBAS      Base of current cgen table
;   CGPBAS      Base of current sprite pattern table
;   PATBAS      Base of current sprite attribute table
;   ATRBAS      Jiffy count
;   JIFFY       Click switch
;   CLIKSW      Click flag to suppress multiple key clicks
;   CLIKFL      VDP register #0 save area
;   RG0SAV      VDP register #1 save area
;   RGLSAV      VDP status register
;   STATFL      Work area for pattern converter
;   PATWRK
;
; External routines
;
;   GETQ
;   PUTQ
;   INITQ
;
SUBTTL - MSXIO - Find available RAM
```

- MSXIO - Find available RAM

```

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        CHKRAM:
        ;
        ; ----- CHKRAM -----
        ;
        ; Look into every slot from 0FFFFH to C000H, and set system work
        ; area. Note that we cannot use RAM as work area nor perform
        ; subroutine calls 'cause we do not yet know where the available
        ; RAM exits. Everything has to be done inside ROM and CPU's
        ; register until the RAM is found.
        ;
        ;
        ; LD      A,82H      ;Port A - output (mode 0)
        ; OUT    (PPI.CM),A ;Port B - input (mode 0)
        ; XOR     A         ;Port C - output (mode 0)
        ; OUT    (PPI.AW),A ;Select slot 0 for all addresses
        ; LD      A,'p'     ;Disable all cassette related outputs
        ; OUT    (PPI.CW),A ;Motor off
        ;
        ; Start searching
        ;
        ; Register usage:
        ; B - non 0 if we're now checking secondary slot
        ; SPH - slot # of the biggest RAM block
        ; SPL - secondary slot # of the biggest RAM block (if any)
        ; DE - lowest address of the biggest RAM block ever found
        ; C - 'slot-expanded' flag
        ;
        ; 0000xxxx
        ; ||||
        ; |||+- slot #3 expanded
        ; |||+- slot #2 expanded

```


742	0307	77	LD
743	0308	BE	CP
744	0309	2F	CPL
745	030A	77	LD
746	030B	20 07	JR
747	030D	2C	INC
748	030E	20 F5	JR
749	0310	25	DEC
750	0311	FA 0305	JP
751	0314		CKRM25:
752	0314	2E 00	
753	0316	24	
754	0317	7D	
755	0318	93	
756	0319	7C	
757	031A	9A	
758	031B	30 0A	
759	031D	EB	
760	031E	3A FFFF	
761	0321	2F	CPL
762	0322	6F	LD
763	0323	DB A8	IN
764	0325	67	LD
765	0326	F9	LD
766	0327		CKRM30:
767	0327	78	
768	0328	A7	
769	0329	28 0A	
770	032B	3A FFFF	
771	032E	2F	
772	032F	C6 10	
		.	ADD

(HL), A → y comparar con la de mismo
 (HL) → de para que haya auto. v.
 (HL), A |
 NZ, CKRM25 ; RAM not equipped in this page
 L ; Make sure it's not a coincidence
 NZ, CKRM20 ; Check more
 H ; Check next page
 M, CKRM20

$(HL)_A \rightarrow$ lo almuerzo
 $(HL) \rightarrow$ y comparen con el mismo
 $(HL)_A \rightarrow$ de lo que había antes.

[illegible]

773	0331	FE 40	CP	01000000B	
774	0333	38 CA	JR	C,CKRM10	
775	0335		CKRM35:		
776	0335	DB A8	IN	A,(PPI.AR)	
777	0337	C6 50	ADD	A,01010000B	
778	0339	30 AC	JR	NC,CKRM05	

;Continue if more secondary slots remain

;Prepare to select next slot

;Continue if more primary slots remain

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809

;
; Check is done, select the biggest one
;
      LD HL,0
      ADD HL,SP
      LD A,H
      OUT (PPI.AW),A
      LD A,L
      LD (0FFFFH),A
      ;Set primary slot register
      ;Set possible secondary slot register
;
; Next, check 0C000H..0FFFFH
;
      LD A,C
      RLCA
      RLCA
      RLCA
      RLCA
      LD C,A
      LD DE,0FFFFH
      IN A,(PPI.AR)
      AND 00111111B
      ;Initialize lowest address ever found
      ;Start from slot #0

      CKRM50:
      OUT (PPI.AW),A
      LD B,0
      RLC C
      JR NC,CKRM60
      INC B
      LD A,(0FFFFH)
      CPL
      AND 00111111B
      ;Select the slot
      ;Assume this slot is not expanded
      ;Shift bit pattern
      ;This slot is not expanded
      ;We're checking expanded slot
```

810	0362		CKRM55:	LD	(0FFFFH),A	;Select the expanded slot
811	0362	32 FFFF				
812	0365		CKRM60:	LD	HL,0FE00H	;Start checking from 0FE00H to 0C000H
813	0365	21 FE00				
814	0368		CKRM65:	LD	A,(HL)	
815	0368	7E		CPL		
816	0369	2F		LD	(HL),A	
817	036A	77		CP	(HL)	
818	036B	BE		CPL		
819	036C	2F		LD	(HL),A	
820	036D	77		JR	NZ,CKRM70	;RAM not equipped in this page
821	036E	20 09		INC	L	;Make sure it's not a coincidence
822	0370	2C		JR	NZ,CKRM65	;Check more
823	0371	20 F5		DEC	H	
824	0373	25		LD	A,H	
825	0374	7C		CP	0C0H	
826	0375	FE C0		JR	NC,CKRM65	;Check next page
827	0377	30 EF				
828	0379		CKRM70:	LD	L,0	
829	0379	2E 00		INC	H	
830	037B	24		LD	A,L	
831	037C	7D		SUB	E	
832	037D	93		LD	A,H	
833	037E	7C		SBC	A,D	
834	037F	9A		JR	NC,CKRM75	;No
835	0380	30 0A		EX	DE,HL	;Register this address as the lowest
836	0382	EB		LD	A,(0FFFFH)	;Set possible secondary slot #
837	0383	3A FFFF		CPL		
838	0386	2F		LD	L,A	
839	0387	6F		IN	A,(PPI.AR)	;Set primary slot #
840	0388	DB A8				

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Find available RAM

3.44 01-Jan-85 PAGE 17-2
841 038A 67 LD H,A
842 038B F9 LD SP,HL ;Register these slot #'s
843 038C CKRM75:
844 038C 78 LD A,B
845 038D A7 AND A ;Are we checking secondary slot
846 038E 28 08 JR Z,CKRM80 ;No
847 0390 3A FFFF LD A,(0FFFFH)
848 0393 2F CPL
849 0394 C6 40 ADD A,01000000B ;Prepare to select next secondary slot
850 0396 30 CA JR NC,CKRM55 ;Continue if more secondary slots remain
851 0398 CKRM80:
852 0398 DB A8 IN A,(PPI.AR)
853 039A C6 40 ADD A,01000000B ;Prepare to select next slot
854 039C 30 B5 JR NC,CKRM50 ;Continue if more primary slots remain
855 SUBTTTL - MSXIO - Slot attribute setup

```

```
856  
857  
858  
859  
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885  
886  
21 0000  
39  
7C  
D3 A8  
7D  
32 FFFF  
79  
01 0C49  
11 F381  
21 F380  
36 00  
ED B0  
4F  
06 04  
21 FCC4  
CB 19  
9F  
E6 80  
77  
2B  
039E  
03A1  
03A2  
03A3  
03A5  
03A6  
03A9  
03AA  
03AD  
03B0  
03B3  
03B5  
03B7  
03B8  
03BA  
03BD  
03BD  
03BF  
03C0  
03C2  
03C3  
21 0000  
39  
7C  
D3 A8  
7D  
32 FFFF  
79  
01 0C49  
11 F381  
21 F380  
36 00  
ED B0  
4F  
06 04  
21 FCC4  
CB 19  
9F  
E6 80  
77  
2B  
LD HL,0  
ADD HL,SP  
LD A,H  
OUT (PPI.AW),A  
LD A,L  
LD (0FFFFH),A  
LD A,C  
LD BC,0C49H  
LD DE,RAMLOW+1  
LD HL,RAMLOW  
LD (HL),0  
LDIR  
LD C,A  
LD B,4  
LD HL,EXPTBL+3  
RR C  
SBC A,A  
AND 80H  
LD (HL),A  
DEC HL  
; Check is done, select the biggest one  
;  
; Set primary slot register  
; Set possible secondary slot register  
; Set 'slot expanded' flag  
; Clear work area with zero  
;  
; length of work area  
; beginning of work  
; init first byte  
; transfer it to rest of area  
; Set 'slot-expanded' flag  
; Loop 4 times  
; Set carry if LSB is set  
; [Acc]=255 if expanded, 0 if not expanded  
; Affects only MSB  
; Set table for each slot
```

(MSX ROM BASIC BIOS) Macro-80
- MSXIO - Slot attribute setup

	03C4	10 F7	DJNZ	SSTLP	
887					
888					
889			; Set SLTTBL		
890					
891				A,(PPI.AR)	;Remember primary slot register's content
892	03C6	DB A8	IN	C,A	
893	03C8	4F	LD	A	;Read from slot #0
894	03C9	AF	XOR	(PPI.AW),A	
895	03CA	D3 A8	OUT	A,(0FFFFH)	
896	03CC	3A FFFF	LD		
897	03CF	2F	CPL	L,A	
898	03D0	6F	LD	A,0100000B	;Read from slot #1
899	03D1	3E 40	LD	(PPI.AW),A	
900	03D3	D3 A8	OUT	A,(0FFFFH)	
901	03D5	3A FFFF	LD		
902	03D8	2F	CPL	H,A	
903	03D9	67	LD	A,80H	;Read from slot #2
904	03DA	3E 80	LD	(PPI.AW),A	
905	03DC	D3 A8	OUT	A,(0FFFFH)	
906	03DE	3A FFFF	LD	E,A	
907	03E1	2F	CPL	A,0C0H	;Read from slot #3
908	03E2	5F	LD	(PPI.AW),A	
909	03E3	3E C0	LD	A,(0FFFFH)	
910	03E5	D3 A8	OUT		
911	03E7	3A FFFF	LD	D,A	
912	03EA	2F	CPL	A,C	;Restore primary slot register
913	03EB	57	LD	(PPI.AW),A	
914	03EC	79	LD	(SLTTBL),HL	;Set SLTTBL
915	03ED	D3 A8	OUT	DE,HL	
916	03EF	22 FCC5	LD	(SLTTBL+2),HL	
917	03F2	EB	EX		
918	03F3	22 FCC7	LD		

(MSX ROM BASIC BIOS) Macro-80
-- MSXIO - Slot attribute setup

3.44 01-Jan-85

PAGE 18-2

37

918 03F6 ED 56
919 03F8 C3 2680
920

IM 1 ;IM 1

JP INIT

SUBTTL - MSXIO - Control-[C] processing

(MSX ROM BASIC BIOS) Macro-80
 - MSXIO - Control-[C] processing

PAGE 19-1

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3.44

```

952 0421 F5          PUSH
953 0422 CD 0A27     CALL
954 0425 F1          POP
955 0426 C1          POP
956 0427 D1          POP
957 0428 E1          POP
958 0429 FE 03       CP
959 042B C0          RET
960 042C             EXCABO:
961 042C E5          PUSH
962 042D CD 0468     CALL
963 0430 CD 0454     CALL
964 0433 30 0A       JR
965 0435 21 FC6A     LD
966 0438 F3          DI
967 0439 CD 0EF1     CALL
968 043C FB          EI
969 043D E1          POP
970 043E C9          RET
971 043F             EXABOL:
972 ;
973 043F CD 083B     CALL
974 0442 3A FCC1     LD
975 0445 26 40       LD
976 0447 CD 025E     CALL
977 044A E1          POP
978 044B AF          XOR
979 044C ED 7B F6B1   LD
980 0450 C5          PUSH
981 0451 C3 63E6     JP
982
```

AF
 CKERC0
 AF
 BC
 DE
 HL
 3
 NZ
 HL
 KILBUF
 CKSTTP
 NC,EXABOL
 HL,REQSTP
 REQTRP
 HL
 TOTEXT
 A,(EXPTBL)
 H,01000000B
 ENASLT
 HL
 A
 SP,(SAVSTK)
 BC
 STOP

;Erase cursor if disabled
 ;Abort?
 ;No
 ;Save text pointer
 ;Cancel any input
 ;Is STOP trap ON
 ;No, accept this break
 ;Request STOP trap
 ;Since REQTRP does not change interrupt mask,
 ;this must be enclosed by 'DI' and 'EI'
 ;Restore text pointer
 ;Make sure we're in text mode
 ;Make sure BASIC is enabled
 ;Restore text pointer
 ;Must return with carry cleared, zero set
 ;LSPD

```

( MSX ROM BASIC BIOS ) Macro-80          3.44    01-Jan-85    PAGE    19-2
- MSXIO - Control-[C] processing

983      0454      CKSTTP:
984      ;
985      ; Check for STOP trap
986      ;
987      ;
988      0454      3A FC6A      LD      A,(REQSTP)      ;Is STOP trap ON
989      0457      0F          RRCA
990      0458      D0          RET      NC              ;No, accept this break
991      0459      2A FC6B      LD      HL,(REQSTP+1)    ;Is STOP trap specified
992      045C      7C          LD      A,H
993      045D      B5          OR      L
994      045E      C8          RET      Z              ;No, accept this break
995      045F      2A F41C      LD      HL,(CURLIN)      ;Are we in direct mode
996      0462      23          INC     HL
997      0463      7C          LD      A,H
998      0464      B5          OR      L
999      0465      C8          RET      Z              ;Yes, treat as break
1000     0466      37          SCF
1001     0467      C9          RET
1002     0468      KILBUF:
1003     ;
1004     0468      2A F3F8      LD      HL,(PUTPNT)      ;Empties ring buffer
1005     046B      22 F3FA      LD      (GETPNT),HL
1006     046E      C9          RET

```

(MSX ROM BASIC BIOS) Macro-80
- MSXIO - Control-[C] processing

3.44

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PAGE

20

```

1007
1008      046F      BREAKX:
1009
1010      ;
1011      ; Check if stop key pressed. If pressed, return with carry set.
1012      ;
1013
1014      DB AA      IN      A,(PPI.CR)
1015      E6 F0      AND     0F0H
1016      F6 07      OR      7
1017      D3 AA      OUT     (PPI.CW),A
1018      DB A9      IN      A,(PPI.BR)
1019      E6 10      AND     10H
1020      C0         RET     NZ
1021      DB AA      IN      A,(PPI.CR)
1022      3D         DEC     A
1023      D3 AA      OUT     (PPI.CW),A
1024      DB A9      IN      A,(PPI.BR)
1025      E6 02      AND     2
1026      C0         RET     NZ
1027      E5         PUSH    HL
1028      2A F3F8    LD      HL,(PUTPNT)
1029      22 F3FA    LD      (GETPNT),HL
1030      E1         POP     HL
1031      3A FBEL    LD      A,(OLDKEY+7)
1032      E6 EF      AND     0EFH
1033      32 FBEL    LD      (OLDKEY+7),A
1034      3E 0D      LD      A,0DH
1035      32 F3F7    LD      (REPCNT),A
1036      37         SCF
1037      C9         RET
1038
1039      SUBTTTL - MSXIO - PSG Initialization

```

;Leave others unaffected
;Select 6th row

;STOP key is assigned to bit 4
;0 when pressed

;Cancel any input

;STOP pressed, mark as pressed to prevent
; to be doubly recognized


```
1068 ; First, clear all static data
1069 ;
1070 LD HL,MUSICF
1071 LD B,71H
1072 XOR A
1073 ;=VCBC + VCBSIZ + MUSCIF
1074 MUSCLL: LD (HL),A
1075 INC HL
1076 DJNZ MUSCLL
1077 ;
1078 ; Then clear music dynamic queue
1079 ;
1080 LD DE,VOICAQ
1081 LD B,7FH
1082 LD HL,80H
1083 ;Address of music queue
1084 ;Mask pattern, 7F = Music queue len - 1
1085 ;Queue length
1086 GICINI: PUSH HL
1087 ;Save length of queue
1088 PUSH DE
1089 ;Save address of queue
1090 PUSH BC
1091 ;Save mask pattern
1092 PUSH AF
1093 ;Save queue ID
1094 CALL INITQ
1095 POP AF
1096 ;Initialize a queue by [Acc],[B],[DE]
1097 ADD A,8
1098 ;write to regs 8,9,10
1099 LD E,0
1100 WRTPSG
1101 ;0 out amplitude (turn voice off)
1102 ;Restore [Acc]
1103 SUB 8
1104 ;Save queue ID
1105 PUSH AF
1106 ;OctaveX
1107 LD L,0FH
1108 GETVC1
1109 CALL GETVC1
1110 EX DE,HL
1111 ;[HL] points to octave for voice [A]
1112 LD HL,MUSITB
1113 ;[HL] points to default value table
```

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - PSG Initialization

1099      04EE      01 0006      LD      BC,6      ;EMSTB - MUSITB
1100      04F1      ED B0      LDIR      ;default variables for this voice
1101      04F3      F1      POP      AF      ;Restore queue ID
1102      04F4      C1      POP      BC      ;Restore mask
1103      04F5      E1      POP      HL      ;Restore queue address
1104      04F6      D1      POP      DE      ;Restore queue length
1105      04F7      19      ADD      HL,DE      ;Update queue address
1106      04F8      EB      EX      DE,HL
1107      04F9      3C      INC      A      ;Next channel
1108      04FA      FE 03      CP      3      ;Loop till done all three voices
1109      04FC      38 D5      JR      C,GICIN1      ;write to reg 7 mixer control
1110      04FE      3E 07      LD      A,7      ;input port A, output port B
1111      0500      1E B8      LD      E,0B8H      ;disable noise, enable all 3 tones
1112      0502      CD 1102      CALL      WRTPSG      ;Restore environments
1113      0505      C3 08DA      JP      POPALL
1114      0508
1115
1116
1117
1118      0508      04      DB      04H      ;default octave
1119      0509      04      DB      04H      ;default note length
1120      050A      78      DB      78H      ;default tempo
1121      050B      88      DB      88H      ;default volume
1122      050C      FF      DB      0FFH      ;default envelope period
1123      050D      00      DB      00H
1124      050E
1125

      MUSITB:
      ;
      ; table of default values for music variables
      ;
      04      DB      04H      ;default octave
      04      DB      04H      ;default note length
      78      DB      78H      ;default tempo
      88      DB      88H      ;default volume
      FF      DB      0FFH      ;default envelope period
      00      DB      00H

      EMSTB:
      ;end of music table
SUBTTTL - MSXIO - Utility routines for VDP

```



```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Utility routines for VDP

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1157 054C 22 F922 LD (NAMBAS),HL
1158 054F 2A F3C1 LD HL,(T32CGP)
1159 0552 22 F924 LD (CGPBAS),HL
1160 0555 2A F3C5 LD HL,(T32PAT)
1161 0558 22 F926 LD (PATBAS),HL
1162 055B 2A F3C3 LD HL,(T32ATR)
1163 055E 22 F928 LD (ATRBAS),HL
1164 0561 CD 07F7 CALL CHGCLR
1165 0564 CD 077E CALL CLRTXT
1166 0567 CD 071E CALL INIPAT
1167 056A CD 06BB CALL ERASPR
1168 056D CD 05B4 CALL SETT32
1169 0570 ENASCR:
1170 ;
1171 ; Enable screen display
1172 ;
1173 0570 3A F3E0 LD A,(RGLSAV)
1174 0573 F6 40 OR 01000000B
1175 0575 18 05 JR DISSC1
1176 0577 DISSCR:
1177 ;
1178 ; Disable screen display
1179 ;
1180 0577 3A F3E0 LD A,(RGLSAV)
1181 057A E6 BF AND 0BFH
1182 057C DISSC1:
1183 057C 47 LD B,A
1184 057D 0E 01 LD C,1

;Set border foreground background color
;Initialize character pattern
;Clear sprites
;Actually set VDP registers

```

```

1185
1186      057F      WRTVDP:
1187      ;
1188      ; Write data to VDP
1189      ;
1190      ; C = register #
1191      ; B = value to be set
1192      ;
1193      ; Register save area for the register is properly set
1194      ;
1195      057F      78      LD      A,B      ;Get data to set
1196      0580      F3      DI
1197      0581      D3 99      OUT      (VDP.CW),A
1198      0583      79      LD      A,C      ;Get register #
1199      0584      F6 80      OR      80H
1200      0586      D3 99      OUT      (VDP.CW),A
1201      0588      FB      EI
1202      0589      E5      PUSH     HL
1203      058A      78      LD      A,B      ;Remember this value 'cause this is
1204      058B      06 00      LD      B,0      ;a write-only register
1205      058D      21 F3DF    LD      HL,RG0SAV
1206      0590      09      ADD      HL,BC
1207      0591      77      LD      (HL),A
1208      0592      E1      POP      HL
1209      0593      C9      RET
1210      0594
1211
1212      ;
1213      ; Set VDP for text mode (40 by 32)
1214
1215      0594      3A F3DF    LD      A,(RG0SAV)      ;Set register #0
1216      0597      E6 01      AND      1

```

```
1216 0599 47 LD B,A
1217 059A 0E 00 LD C,0
1218 059C CD 057F WRTVDP
1219 059F 3A F3E0 LD A,(RGLSAV) ;Set register #1
1220 05A2 E6 E7 AND 0E7H
1221 05A4 F6 10 OR 10H
1222 05A6 47 LD B,A
1223 05A7 0C INC C
1224 05A8 CD 057F WRTVDP
1225 05AB 21 F3B3 HL,TXTNAM
1226 05AE 11 0000 LD DE,0
1227 05B1 C3 0677 JP SETSCM ;Set mask pattern
1228 05B4 SETT32: ;Set screen mode
1229 ;
1230 ; Set VDP for text mode (graphics 1)
1231 ;
1232 05B4 3A F3DF LD A,(RG0SAV) ;Set register #0
1233 05B7 E6 01 AND 1
1234 05B9 47 LD B,A
1235 05BA 0E 00 LD C,0
1236 05BC CD 057F WRTVDP
1237 05BF 3A F3E0 LD A,(RGLSAV) ;Set register #1
1238 05C2 E6 E7 AND 0E7H
1239 05C4 47 LD B,A
1240 05C5 0C INC C
1241 05C6 CD 057F WRTVDP
1242 05C9 21 F3BD HL,T32NAM
1243 05CC 11 0000 LD DE,0
1244 05CF C3 0677 JP SETSCM ;Set mask pattern
1245 05D2 INIGRP: ;Set screen mode
1246 ;
```



```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Utility routines for VDP

3.44 01-Jan-85 PAGE 23-3

1278 060D 3A F3E0 LD A,(RGLSAV) ;Set register #1
1279 0610 E6 E7 AND 0E7H
1280 0612 47 LD B,A
1281 0613 0C INC C
1282 0614 CD 057F CALL WRTVDP
1283 0617 21 F3C7 LD HL,GRPNAM
1284 061A 11 7F03 LD DE,7F03H
1285 061D 18 58 JR SETSCM
1286 061F INIMLT:
1287 ;
1288 ; Initialize VDP for multi-color mode
1289 ;
1290 061F CD 0577 CALL DISSCR
1291 0622 3E 03 LD A,3
1292 0624 32 FCAF LD (SCRMOD),A
1293 0627 2A F3D9 LD HL,(MLTPAT)
1294 062A 22 F926 LD (PATBAS),HL
1295 062D 2A F3D7 LD HL,(MLTATR)
1296 0630 22 F928 LD (ATRBAS),HL
1297 0633 2A F3D1 LD HL,(MLTNAM)
1298 0636 CD 07DF CALL SETWRT
1299 0639 11 0006 LD DE,6
1300 063C INIML1:
1301 063C 0E 04 LD C,4
1302 063E INIML2:
1303 063E 7A LD A,D
1304 063F 06 20 LD B,' '
1305 0641 INIML3:
1306 0641 D3 98 OUT (VDP.DRW),A
1307 0643 3C INC A
1308 0644 10 FB DJNZ INIML3

```

;Initialize name table

```
1309 0646 0D DEC C
1310 0647 20 F5 JR NZ,INIML2
1311 0649 57 LD D,A
1312 064A 1D DEC E
1313 064B 20 EF JR NZ,INIML1
1314 064D CD 07B9 CALL CLSMLT
1315 0650 CD 06BB CALL ERASPR
1316 0653 CD 0659 CALL SETMLT
1317 0656 C3 0570 JP ENASCR
1318 0659
1319
1320
1321
1322 0659 3A F3DF LD A,(RG0SAV)
1323 065C E6 01 AND 1
1324 065E 47 LD B,A
1325 065F 0E 00 LD C,0
1326 0661 CD 057F CALL WRTVDP
1327 0664 3A F3E0 LD A,(RG1SAV)
1328 0667 E6 E7 AND 0E7H
1329 0669 F6 08 OR 8
1330 066B 47 LD B,A
1331 066C 0E 01 LD C,1
1332 066E CD 057F CALL WRTVDP
1333 0671 21 F3D1 LD HL,MLTNAM
1334 0674 11 0000 LD DE,0
1335 0677
1336 0677 01 0602 LD BC,SETGRP
1337 067A CD 0690 CALL SETREG
1338 067D 06 0A LD B,0AH
1339 067F 7A LD A,D

SETMLT:
;
; Set VDP for multicolor mode
;
;
; Set register #0
; Set register #1
; Set mask pattern
; Set name table
```

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Utility routines for VDP

1340 0680 CD 0691          SETRG1      ;Set color table
1341 0683 06 05          LD B,5
1342 0685 7B             LD A,E
1343 0686 CD 0691          SETRG1      ;Set pattern table
1344 0689 06 09          LD B,9
1345 068B CD 0690          SETRG      ;Set sprite attribute table
1346 068E 06 05          LD B,5      ;Set sprite pattern table
1347 0690
1348 0690 AF             SETREG:
1349 0691
1350 0691 E5             SETRG1:
1351 0692 F5             PUSH
1352 0693 7E             PUSH
1353 0694 23             LD A,(HL)
1354 0695 66             INC HL
1355 0696 6F             LD H,(HL)
1356 0697 AF             LD L,A
1357 0698
1358 0698 29             XOR A
1359 0699 8F             SETRG2:
1360 069A 10 FC          ADD HL,HL
1361 069C 6F             ADC A,A
1362 069D F1             DJNZ SETRG2
1363 069E B5             LD L,A
1364 069F 47             POP AF
1365 06A0 CD 057F        OR L
1366 06A3 E1             LD B,A
1367 06A4 23             WRTVDP
1368 06A5 23             HL
1369 06A6 0C             INC HL
1370 06A7 C9             INC C
RET

```

```
1371
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1391
1392
1393
1394
1395
1396
1397
1398
1399
1400
1401

06A8
06A8
06AB
06AC
06AE
06B1
06B4
06B7
06B8
06BB
06BB
06BE
06BF
06C2
06C5
06C5
06C7
06CA
06CB
06CC
06CD
06D0
06D1
06D2
06D5

3A F3E0
47
0E 01
CD 057F
2A F926
01 0800
AF
CD 0815
3A F3E9
5F
2A F928
01 2000
3E D1
CD 07CD
23
23
79
CD 07CD
23
0C
3A F3E0
0F

CLRSPR:
;
; Clear all sprites
;
LD A,(RGLSAV)
LD B,A
LD C,1
CALL WRTVDP
LD HL,(PATBAS)
LD BC,0800H
XOR A
CALL FILVRM
ERASPR:
LD A,(FORCLR)
LD E,A
LD HL,(ATRBAS)
LD BC,2000H
CLRSPR2:
; default sprite name to [C]
;
LD A,0D1H
CALL WRTVRM
INC HL
INC HL
LD A,C
CALL WRTVRM
INC HL
INC C
LD A,(RGLSAV)
RRCA

;Set register #1
;Clear sprite pattern table
;Length of sprite pattern table
;Load foreground color (default) to [E]
;Set number of sprite plane to [B]
;Erase code (i.e. vertical position)
;Set vertical position
;Load default sprite name
;Prepare for next
```

```
1402 06D6 0F RRCA
1403 06D7 30 03 JR NC,CLSPR3
1404 06D9 0C INC C
1405 06DA 0C INC C
1406 06DB 0C INC C
1407 06DC CLSPR3:
1408 06DC 7B LD A,E
1409 06DD CD 07CD WRTVRM
1410 06E0 23 INC HL
1411 06E1 10 E2 DJNZ CLSPR2
1412 06E3 C9 RET
1413 06E4 CALPAT:
1414 ;
1415 06E4 6F LD L,A
1416 06E5 26 00 LD H,0
1417 06E7 29 ADD HL,HL
1418 06E8 29 ADD HL,HL
1419 06E9 29 ADD HL,HL
1420 06EA CD 0704 GSPSIZ
1421 06ED FE 08 8
1422 06EF 28 02 JR Z,GSPAD1
1423 06F1 29 ADD HL,HL
1424 06F2 29 ADD HL,HL
1425 06F3 GSPAD1:
1426 06F3 EB DE,HL
1427 06F4 2A F926 HL,(PATBAS)
1428 06F7 19 ADD HL,DE
1429 06F8 C9 RET
1430 06F9 CALATR:
1431 ;
1432 06F9 6F LD L,A
;Get plane number to [L]
```

;16*16?
;No
;Yes, C=C+4

;Load default color

;Assume 8 byte long

;Check size of sprite

;Good assumption
;32 byte long sprite

;Get base address of sprite pattern table
;Form destination/source address

```

1433 06FA 26 00 LD H,0
1434 06FC 29 ADD HL,HL
1435 06FD 29 ADD HL,HL
1436 06FE EB DE,HL
1437 06FF 2A F928 LD HL,(ATRBAS)
1438 0702 19 ADD HL,DE
1439 0703 C9 RET
1440 0704 GSPSIZ:
1441 ;
1442 ; Get sprite size
1443 ;
1444 0704 3A F3E0 LD A,(RGLSAV)
1445 0707 0F RRCA
1446 0708 0F RRCA
1447 0709 3E 08 LD A,8
1448 070B D0 RET NC
1449 070C 3E 20 LD A,32
1450 070E C9 RET

;Sprite attribute consists of 4 bytes
;Load base address
;Calculate target address
;Assume 8 byte long
;Good assumption
;32 byte long sprite

```



```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Utility routines for VDP

3.44 01-Jan-85 PAGE 25-1

1482 0735 CD 01B6
1483 0738 FB
1484 0739 C1
1485 073A D3 98
1486 073C 23
1487 073D 0B
1488 073E 79
1489 073F B0
1490 0740 20 EF
1491 0742 F1
1492 0743 C9
1493 0744
1494
1495 0744 EB
1496 0745 CD 07DF
1497 0748
1498 0748 1A
1499 0749 D3 98
1500 074B 13
1501 074C 0B
1502 074D 79
1503 074E B0
1504 074F 20 F7
1505 0751 C9
1506 0752
1507
1508
1509
1510
1511
1512

CALL RDSLT ;Read from specified slot
EI
POP BC ;Restore counter
OUT (VDP.DRW),A
INC HL ;Bump character source pointer
DEC BC
LD A,C
OR B
JR NZ,INIPT1
POP AF ;Discard stack
RET

LDIRVM:
;
EX DE,HL
CALL SETWRT
LDIVM1:
LD A,(DE)
OUT (VDP.DRW),A
INC DE
DEC BC
LD A,C
OR B
JR NZ,LDIVM1
RET

GETPAT:
;
; Get pattern corresponding to ASCII code in [A]
;
; Pattern is returned to 8 byte work area (PATWRK). Entered
; by GRPPRT (print a character to graphic screen) subroutine.
;

```

(MSX ROM BASIC BIOS) Macro-80
- MSXIO - Utility routines for VDP

3.44 01-Jan-85 PAGE 25-2

```

1513
1514
1515      0752 26 00      LD H,0      ;Prepare for calculation
1516      0754 6F      LD L,A
1517      0755 29      ADD HL,HL
1518      0756 29      ADD HL,HL
1519      0757 29      ADD HL,HL
1520      0758 EB      EX DE,HL
1521      0759 2A F920   LD HL,(CGPNT+1)
1522      075C 19      ADD HL,DE
1523      075D 11 FC40   LD DE,PATWRK
1524      0760 06 C8      LD B,8
1525      0762 3A F91F   LD A,(CGPNT)
1526      0765      GTPAT1:
1527      0765 F5      PUSH AF      ;Save source slot
1528      0766 E5      PUSH HL      ;Save source address
1529      0767 D5      PUSH DE      ;Save destination address
1530      0768 C5      PUSH BC      ;Save counter
1531      0769 CD 01B6   CALL RDSLT  ;Read from specified slot
1532      076C FB      EI
1533      076D C1      POP BC        ;Restore counter
1534      076E D1      POP DE        ;Restore destination address
1535      076F E1      POP HL        ;Restore source address
1536      0770 12      LD (DE),A
1537      0771 13      INC DE
1538      0772 23      INC HL
1539      0773 F1      POP AF
1540      0774 10 EF     DJNZ GTPAT1 ;Bump destination pointer
1541      0776 C9      RET          ;Bump character source pointer
1542      0777      CLSSUB:         ;Restore source slot
1543
; All registers are completely destroyed
;

```

```

1544 0777 CD 0B9F          CALL      CHKSCR      ;Check current screen mode
1545 077A 28 25          JR          Z,CLSHRS    ;Hires
1546 077C 30 3B          JR          NC,CLSMLT    ;Multi-color
1547 077E                      CLRTXT:
1548                                ;
1549                                ; Clear screen (text mode)
1550                                ;
1551 077E 3A FCAF          LD          A,(SCRMOD)
1552 0781 A7              AND          A
1553 0782 2A F922          LD          HL,(NAMBAS)
1554 0785 01 03C0          LD          BC,03C0H
1555 0788 28 03          JR          Z,CLRTX1
1556 078A 01 0300          LD          BC,0300H
1557 078D                      CLRTX1:
1558 078D 3E 20          LD          A,' '
1559 078F          CALL      FILVRM
1560 0792 CD 0A7F          CALL      CSHOME
1561 0795 21 FBB2          LD          HL,LINTTB
1562 0798 06 18          LD          B,18H
1563 079A                      CLRTX2:
1564 079A 70              LD          (HL),B
1565 079B 23              INC          HL
1566 079C 10 FC          DJNZ        CLRTX2
1567 079E C3 0B26          JP          FNKSB
1568 07A1                      CLSHRS:
1569                                ;
1570 07A1 CD 0832          CALL      CHGBDR
1571 07A4 01 1800          LD          BC,1800H
1572 07A7 C5              PUSH        BC
1573 07A8 2A F3C9          LD          HL,(GRPCOL)
1574 07AB 3A F3EA          LD          A,(BAKCLR)

```

			3.44	01-Jan-85	PAGE 25-4
(MSX ROM BASIC BIOS) Macro-80					
- MSXIO - Utility routines for VDP					
1575	07AE	CD 0815	CALL	FILVRM	
1576	07B1	2A F3CB	LD	HL, (GRPCGP)	
1577	07B4	C1	POP	BC	;Load 6144
1578	07B5	AF	XOR	A	
1579	07B6		JFLVRM:		
1580	07B6	C3 0815	JP	FILVRM	
1581	07B9		CLSMLT:		
1582			;		
1583	07B9	CD 0832	CALL	CHGBDR	;Set border color
1584	07BC	21 F3EA	LD	HL, BAKCLR	;Set all pixels to background color
1585	07BF	7E	LD	A, (HL)	
1586	07C0	87	ADD	A, A	
1587	07C1	87	ADD	A, A	
1588	07C2	87	ADD	A, A	
1589	07C3	87	ADD	A, A	
1590	07C4	B6	OR	(HL)	
1591	07C5	2A F3D5	LD	HL, (MLTCGP)	;Set up address for write
1592	07C8	01 0600	LD	BC, 0600H	
1593	07CB	18 E9	JR	JFLVRM	;Clear sprites (except sprite pattern)

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1623
1624

07CD
07CD
07CE
07D1
07D2
07D3
07D4
07D6
07D7

F5
CD 07DF
E3
E3
F1
D3 98
C9

PUSH AF
CALL SETWRT
EX (SP),HL
EX (SP),HL
POP AF
OUT (VDP.DRW),A
RET

RDVRM:
;
; Read a byte from VRAM
;
07D7
07DA
07DB
07DC
07DE
07DF

CD 07EC
E3
E3
DB 98
C9

CALL SETRD
EX (SP),HL
EX (SP),HL
IN A,(VDP.DRW)
RET

SETWRT:
;
; Set address for write to VDP
;
; Address is passed to HL
;
07DF
07E0
07E1
07E3

7D
F3
D3 99
7C

LD A,L
DI
OUT (VDP.CW),A
LD A,H
```

;Save data to be written

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Utility routines for VDP

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1625 07E4 E6 3F
1626 07E6 F6 40
1627 07E8 D3 99
1628 07EA FB
1629 07EB C9
1630 07EC
1631
1632
1633
1634
1635
1636 07EC 7D
1637 07ED F3
1638 07EE D3 99
1639 07F0 7C
1640 07F1 E6 3F
1641 07F3 D3 99
1642 07F5 FB
1643 07F6 C9
1644 07F7
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00111111B
01000000B
(VDP.CW),A
EI
RET
SETRD:
;
; Set address for read from VDP
;
; Address is passed to HL
;
LD A,L
DI
OUT (VDP.CW),A
LD A,H
AND 00111111B
OUT (VDP.CW),A
EI
RET
CHGCLR:
;
; CHGCLR - changes foreground, background, and border color
;
LD A,(SCRMOD)
DEC A
JP M,CHCLTX
PUSH AF
CALL CHGBDR
POP AF
RET NZ
LD A,(FORCLR)
;Are we in text mode
;Yes, change color in 40*24 text mode
;Change border color for all
;No
;We're in 32*24 text mode

```

1656	0807	87	ADD	A,A
1657	0808	87	ADD	A,A
1658	0809	87	ADD	A,A
1659	080A	87	ADD	A,A
1660	080B	21 F3EA	LD	HL,BAKCLR
1661	080E	B6	OR	(HL)
1662	080F	2A F3BF	LD	HL,(T32COL)
1663	0812	01 0020	LD	BC,20H
1664	0815			
1665	0815	F5	PUSH	AF
1666	0816	CD 07DF	CALL	SETWRT
1667	0819			
1668	0819	F1	POP	AF
1669	081A	D3 98	OUT	(VDP.DRW),A
1670	081C	F5	PUSH	AF
1671	081D	0B	DEC	BC
1672	081E	79	LD	A,C
1673	081F	B0	OR	B
1674	0820	20 F7	JR	NZ,FLVRM1
1675	0822	F1	POP	AF
1676	0823	C9	RET	
1677	0824			
1678			CHCLTX:	
1679	0824	3A F3E9	LD	A,(FORCLR)
1680	0827	87	ADD	A,A
1681	0828	87	ADD	A,A
1682	0829	87	ADD	A,A
1683	082A	87	ADD	A,A
1684	082B	21 F3EA	LD	HL,BAKCLR
1685	082E	B6	OR	(HL)
1686	082F	47	LD	B,A

(MSX ROM BASIC BIOS) Macro-80
 - MSXIO - Utility routines for VDP

1687	0830	18 03	JR	CHGBD1	
1688	0832		CHGBDR:		
1689			;		
1690	0832	3A F3EB	LD	A, (BDRCLR)	;Get border color
1691	0835				
1692	0835	47	LD	B,A	
1693	0836	0E 07	LD	C,7	
1694	0838	C3 057F	JP	WRTVDP	

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083B
;
; TOTEXT - Force screen to text mode
;
;
083B CD 0B9F
083E D8
083F 3A FCB0
0842 CD FDBD
0845 C3 084F
0848
;
; CLS - clears screen
;
;
0848 C0
0849 E5
084A CD 0777
084D E1
084E C9
084F
;
; CHGMOD - changes mode of screen
;
;
084F 3D
0850 FA 050E
0853 CA 0538
0856 3D
0857 CA 05D2
085A C3 061F

TOTEXT:
;
; TOTEXT - Force screen to text mode
;
;
CALL CHKSCR
RET C
LD A,(OLDSCR)
CALL H.TOTE
JP CHGMOD
;No, change to text mode then
CLS:
;
; CLS - clears screen
;
;
RET NZ
PUSH HL
CALL CLSSUB
POP HL
RET
;Statement not ending
;Save text pointer
;Restore text pointer
CHGMOD:
;
; CHGMOD - changes mode of screen
;
;
DEC A
JP M,INITXT
JP Z,INIT32
DEC A
JP Z,INIGRP
JP INIMLT
;Change to what mode
;To text mode
;To hires mode
;To multicolor mode
SUBTTTL - MSXIO - Some entry points
```

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085D
085D
0860
0861
0861
0864
0866
0869
086B
086C
086C
086D
086F
0870
0872
0873
0875
0876
0877
0878
0878
0879
087C
087E
0881
0882

CD FFB6
F5
CD 046F
38 12
CD 0884
28 F6
F1
F5
D3 91
AF
D3 90
3D
D3 90
F1
A7
C9
AF
32 F415
3E 0D
CD 086C
F1
37

LPTOUT:
;
; Output a character to printer
;
CALL H.LPTO
PUSH AF
; Save character to output

CHPLP1:
CALL BREAKX
JR C,LPTABO
CALL LPTSTT
JR Z,CHPLP1
POP AF
; No
; Restore character

CHPLP2:
PUSH AF
OUT (LPT.DW),A
XOR A
OUT (LPT.SB),A
DEC A
OUT (LPT.SB),A
POP AF
AND A
RET
; Restore data output

LPTABO:
;
XOR A
LD (LPTPOS),A
LD A,0DH
CALL CHPLP2
POP AF
SCF
; Reset carriage position
; Send CR even if LPT not active
```

```
1756 0883 C9 RET
1757 0884 LPTSTT:
1758 ;
1759 0884 CD FFBB CALL H.LPTS
1760 0887 DB 90 IN A,(90H) ;LSB is 0 if ready
1761 0889 0F RRCA
1762 088A 0F RRCA
1763 088B 3F CCF
1764 088C 9F SBC A,A
1765 088D C9 RET
1766 088E POSIT:
1767 ;
1768 ; Position cursor to specified position
1769 ;
1770 088E 3E 1B LD A,1BH
1771 0890 DF RST 18H
1772 0891 3E 59 LD A,'Y'
1773 0893 DF RST 18H
1774 0894 7D LD A,L
1775 0895 C6 1F ADD A,1FH
1776 0897 DF RST 18H
1777 0898 7C LD A,H
1778 0899 C6 1F ADD A,1FH
1779 089B DF RST 18H
1780 089C C9 RET
1781 089D CNVCHR:
1782 ;
1783 ; Convert character code
1784 ;
1785 089D E5 PUSH HL
1786 089E F5 PUSH AF
;= ' ' - 1
```

(MSX ROM BASIC BIOS) Macro-80
- MSXIO - Some entry points

PAGE 28-2

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1787	089F	21 FCA6	LD	HL,GRPHED	;Preceded by a header byte
1788	08A2	AF	XOR	A	
1789	08A3	BE	CP	(HL)	
1790	08A4	77	LD	(HL),A	;Clear this since seen
1791	08A5	28 0D	JR	Z,CNVCH3	;No
1792	08A7	F1	POP	AF	
1793	08A8	D6 40	SUB	01000000B	;Get rid of offset
1794	08AA	FE 20	CP	' '	;Valid range
1795	08AC	38 04	JR	C,CNVCH2	;Yes
1796	08AE	C6 40	ADD	A,01000000B	;Compensate value
1797	08B0			CNVCH1:	
1798	08B0	BF	CP	A	;Set Z flag
1799	08B1	37	SCF		;Make sure carry is cleared
1800	08B2			CNVCH2:	
1801	08B2	E1	POP	HL	
1802	08B3	C9	RET		
1803	08B4			CNVCH3:	
1804				;	
1805	08B4	F1	POP	AF	
1806	08B5	FE 01	CP	1	;Graphic header
1807	08B7	20 F7	JR	NZ,CNVCH1	;No, do not modify
1808	08B9	77	LD	(HL),A	;Set GRPHED flag
1809	08BA	E1	POP	HL	;Carry is clear indicating one more byte is
1810	08BB	C9	RET		;required
1811				SUBTTTL - MSXIO -	Output a character to CRT


```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Output a character to CRT

1843 08E6 21 FCA7 LD HL,ESCCNT
1844 08E9 7E LD A,(HL)
1845 08EA A7 AND A
1846 08EB C2 098F JP NZ,INESC
1847 08EE 79 LD A,C
1848 08EF FE 20 CP ' '
1849 08F1 38 21 JR C,CNTPUT
1850 08F3
1851 08F3 2A F3DC LD HL,(CSRY)
1852 08F6 FE 7F CP 7FH
1853 08F8 CA 0AE3 JP Z,RUBOUT
1854 08FB CD 0BE6 CALL PUTVRM
1855 08FE CD 0A44 CALL RIGHT
1856 0901 C0 RET NZ
1857 0902 AF XOR A
1858 0903 CD 0C2B CALL SETTRM
1859 0906 26 01 LD H,l
1860 0908
1861 LF:
1862 ;
1863 ; Line feed
1864 ;
1865 0908 CD 0A61 CALL DOWN
1866 090B C0 RET NZ
1867 090C CD 0A69 CALL STOCsr
1868 090F 2E 01 LD L,l
1869 0911 C3 0A88 JP DELLNO
1870 0914
1871 CNTPUT:
1872 ;
1873 ; Following control codes are supported
1874 ;
1875 ; 7 Bell

```

```
1874      ; 8 Back space
1875      ; 9 Tab
1876      ; 10 Line feed
1877      ; 11 Cursor home
1878      ; 12 Clear screen
1879      ; 13 Carriage return
1880      ;
1881      ; 27 Enter escape sequence
1882      ; 28 Cursor right
1883      ; 29 Cursor left
1884      ; 30 Cursor up
1885      ; 31 Cursor down
1886      ;
1887      21 092D      LD      HL,JMPBC
1888      0E 0C      LD      C,0CH
1889
1890      23          INC     HL
1891      23          INC     HL
1892      A7          AND     A
1893      0D          DEC     C
1894      F8          RET     M
1895      BE          CP      (HL)
1896      23          INC     HL
1897      20 F7      JR      NZ,INDJMP
1898      4E          LD      C,(HL)
1899      23          INC     HL
1900      46          LD      B,(HL)
1901      2A F3DC     LD      HL,(CSRY)
1902      CD 092D     CALL    JMPBC
1903      AF          XOR     A
1904      C9          RET

;Make sure carry is cleared
;Undefined function
;Found?
;No
;Get routine address in BC
;
;Jump to each routine with cursor pos
;Tell screen editor not to echo this character
```

```

1905      092D      JMPBC:
1906      ;
1907      092D      C5      PUSH BC
1908      092E      C9      RET
1909      ;
1910      ;
1911      ;
1912      092F      ;
1913      092F      07      DB 7 ;Beep
1914      0930      1113    DW BEEP
1915      0932      08      DB 8 ;Back space
1916      0933      0A4C    DW BS
1917      0935      09      DB 9 ;Tabulation
1918      0936      0A71    DW TAB
1919      0938      0A      DB 10 ;Line feed
1920      0939      0908    DW LF
1921      093B      0B      DB 11 ;Home
1922      093C      0A7F    DW CSHOME
1923      093E      0C      DB 12 ;Clear
1924      093F      077E    DW CLRTXT
1925      0941      0D      DB 13 ;Carriage return
1926      0942      0A81    DW CR
1927      0944      1B      DB 27 ;Enter escape sequence
1928      0945      0989    DW ENTESC
1929      0947      1C      DB 28 ;Cursor right
1930      0948      0A5B    DW ADVCUR
1931      094A      1D      DB 29 ;Cursor left
1932      094B      0A4C    DW BS
1933      094D      1E      DB 30 ;Cursor up
1934      094E      0A57    DW UP
1935      0950      1F      DB 31 ;Cursor down
  
```

Function dispatch table

CNTTBL:

(MSX ROM BASIC BIOS) Macro-80
- MSXIO - Output a character to CRT

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73

1936 0951 0A61
1937

DW DOWN

SUBTTL - MSXIO - Escape sequence handler

Year	Code	Function
1938		
1939		
1940	0953	ESC TBL :
1941	6A	DB
1942	0953	DW
1943	077E	DB
1944	45	DW
1945	0956	DB
1946	077E	DW
1947	4B	DB
1948	0959	DW
1949	0AEE	DB
1950	4A	DW
1951	095C	DB
1952	0B05	DW
1953	095D	DB
1954	6C	DW
1955	0AFC	DB
1956	4C	DW
1957	0962	DB
1958	0AB4	DW
1959	0963	DB
1960	4D	DW
1961	0965	DB
1962	0A85	DW
1963	59	DB
1964	0968	DW
1965	0969	DB
1966	096B	DW
1967	096C	DB
1968	0A57	DW
	42	DB
	0A61	DW
	43	DB
	0A44	DW
	44	DB
	0A55	DW
	48	DB
	0A7F	DW
	78	DB
	097A	DW
	097B	DB
	097D	DW
	79	DB

```
1969 097E 0983          SETMOD:          RSTMOD
1970 0980
1971
1972 ;
1973 ; Function dispatch table
1974
1975 0980 0983 3E 01          LD      A,1
1976 0982 01          DB      1
1977
1978 0983 3E 02          RSTMOD:          LD      A,2
1979 0985 01          DB      1
1980
1981 0986          LOC:
1982 0986 3E 04          LD      A,4
1983 0988 01          DB      1
1984
1985 0989          ENTESC:
1986 0989 3E FF          LD      A,0FFH
1987 098B 32 FCA7          LD      (ESCCNT),A
1988 098E C9          RET
1989
```

;Say row is expected next
;LXI B' instruction
;Tell him we're in escape sequence

```
1986
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1990
1991
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2007
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2012
2013
2014
2015
2016

098F
098F
0992
0994
0995
0998
099A
099D
099D
099E
09A0
09A1
09A3
09A4
09A5
09A8
09AB
09AD
09AF
09B2
09B3
09B3
09B4
09B5
09B7
09B8
09B9
09BA

F2 099D
36 00
79
21 0951
0E 0F
C3 0919
3D
28 1E
3D
28 25
3D
77
3A F3B0
11 F3DD
28 06
36 03
CD 0C32
1B
47
79
D6 20
B8
3C
12
D8

INESC:
;
JP
LD
LD
LD
LD
JP
INESC1:
;
DEC
JR
DEC
JR
DEC
LD
LD
LD
JR
LD
CALL
DEC
INESC2:
LD
LD
SUB
CP
INC
LD
RET

P,INESC1
(HL),0
A,C
HL,ESCTBL-2
C,0FH
INDJMP
A
Z,GOSSET
A
Z,GORSET
A
(HL),A
A,(LINLEN)
DE,CSRX
Z,INESC2
(HL),3
GETLEN
DE
B,A
A,C
, ,
B
A
(DE),A
C

;Arguments expected
;Exit from escape sequence
;Restore character
;Number of ESC handler entries
;Set modes?
;Yes
;Reset modes?
;Update ESCCNT
;Assume column expected
;
;Column expected
;Row expected
;Point CSRY
;Get max limit in B
;Restore character
;0-xx
;Legal value
```

2017	09BB	78		LD	A,B		
2018	09BC	12		LD	(DE),A		
2019	09BD	C9		RET			
2020	09BE			GOSSET:			
2021				;			
2022				;	Set various modes		
2023				;			
2024	09BE	77		LD	(HL),A		
2025	09BF	79		LD	A,C		
2026	09C0	D6 34		SUB	'4'		
2027	09C2	28 0B		JR	Z,STSTYL		
2028	09C4	3D		DEC	A		
2029	09C5	28 0F		JR	Z,STCSSW		
2030	09C7	C9		RET			
2031	09C8			GORSET:			
2032				;			
2033				;	Reset various modes		
2034				;			
2035	09C8	77		LD	(HL),A		
2036	09C9	79		LD	A,C		
2037	09CA	D6 34		SUB	'4'		
2038	09CC	20 05		JR	NZ,RSETl0		
2039	09CE	3C		INC	A		
2040	09CF			STSTYL:			
2041	09CF	32 FCAA		LD	(CSTYLE),A		
2042	09D2	C9		RET			
2043	09D3			RSETl0:			
2044				;			
2045	09D3	3D		DEC	A		
2046	09D4	C0		RET	NZ		
2047	09D5	3C		INC	A		

(MSX ROM BASIC BIOS) Macro-80
 - MSXIO - Escape sequence handler

```

2048      09D6      STCSSW:
2049      09D6      32 FCA9      LD      (CSRSW),A
2050      09D9      C9          RET
2051      09DA      CKDPC0:
2052      09DA      ;
2053      09DA      ; Display cursor if disabled
2054      09DA      ;
2055      09DA      3A FCA9      LD      A,(CSRSW)
2056      09DD      A7          AND     A
2057      09DE      C0          RET     NZ
2058      09DF      18 05      JR      DPCSR
2059      09E1      CKDPCS:
2060      09E1      ;
2061      09E1      ; Display cursor if enabled
2062      09E1      ;
2063      09E1      3A FCA9      LD      A,(CSRSW)
2064      09E4      A7          AND     A
2065      09E5      C8          RET     Z
2066      09E6      DPCCSR:
2067      09E6      ;
2068      09E6      ; Display a cursor
2069      09E6      ;
2070      09E6      CD FDA9      CALL    H.DSPC
2071      09E9      CD 0B9F      CALL    CHKSCR
2072      09EC      D0          RET     NC
2073      09ED      2A F3DC      LD      HL,(CSRY)
2074      09F0      E5          PUSH    HL
2075      09F1      CD 0BD8      CALL    GETVRM
2076      09F4      32 FBCC      LD      (CODSAV),A
2077      09F7      6F          LD      L,A
2078      09F8      26 00      LD      H,0

```

;Get current cursor position
 ;Save it for future use
 ;Get a raw character at cursor
 ;Remember this code
 ;Then read pattern for this code

```

2079 09FA 29          ADD HL,HL
2080 09FB 29          ADD HL,HL
2081 09FC 29          ADD HL,HL
2082 09FD EB          EX DE,HL
2083 09FE 2A F924     LD HL,(CGPBAS)
2084 0A01 E5          PUSH HL
2085 0A02 19          ADD HL,DE
2086 0A03 CD 0BA5     CALL GET8B
2087 0A06 21 FC1F     LD HL,BUFEND+7
2088 0A09 06 08      LD B,8
2089 0A0B 3A FCAA     LD A,(CSTYLE)
2090 0A0E A7          AND A
2091 0A0F 28 02      JR Z,DSPCS1
2092 0A11 06 03      LD B,3
2093 0A13           DSPCS1:
2094 0A13 7E          LD A,(HL)
2095 0A14 2F          CPL
2096 0A15 77          LD (HL),A
2097 0A16 2B          DEC HL
2098 0A17 10 FA      DJNZ DSPCS1
2099 0A19 E1          POP HL
2100 0A1A 01 07F8    LD BC,07F8H
2101 0A1D 09          ADD HL,BC
2102 0A1E CD 0BBE     CALL PUT8B
2103 0A21 E1          POP HL
2104 0A22 0E FF      LD C,0FFH
2105 0A24 C3 0BE6    JP PUTVRM
2106 0A27           CKERC0:
2107           ;
2108           ; Erase cursor if disabled
2109           ;

```

; [A] * 8

;Make a complement of this pattern
 ;Assume full reverse cursor

;Good assumption
 ;No, reverse bottom 3 lines only

;Assign this pattern to 255

;Restore cursor position
 ;Get code for cursor
 ;Set it at cursor position

```
2110 0A27 3A FCA9 LD A,(CSRSW)
2111 0A2A A7 AND A
2112 0A2B C0 RET NZ
2113 0A2C 18 05 JR ERACSR
2114 0A2E CKERCS:
2115 ;
2116 ; Erase a cursor if enabled
2117 ;
2118 0A2E 3A FCA9 LD A,(CSRSW)
2119 0A31 A7 AND A
2120 0A32 C8 RET Z
2121 0A33 ERACSR:
2122 ;
2123 ; Erase cursor
2124 ;
2125 0A33 CD FDAE CALL H.ERAC
2126 0A36 CD 0B9F CALL CHKSCR
2127 0A39 D0 RET NC
2128 0A3A 2A F3DC LD HL,(CSRY)
2129 0A3D 3A FBCC LD A,(CODSAV)
2130 0A40 4F LD C,A
2131 0A41 C3 0BE6 JP PUTVRM
2132 ;
2133 SUBTTL - MSXIO - Cursor movements
; Restore old code
```

Address	Instruction	Comments
2134	0A44	
2135		RIGHT:
2136		;
2137		; Cursor right
2138		;
2139	0A44 3A F3B0	LD A, (LINLEN)
2140	0A47 BC	CP H
2141	0A48 C8	RET Z
2142	0A49 24	INC H
2143	0A4A 18 1D	JR STOCSR
2144	0A4C	BS:
2145		;
2146		; Back space
2147		;
2148	0A4C CD 0A55	CALL LEFT
2149	0A4F C0	RET NZ
2150	0A50 3A F3B0	LD A, (LINLEN)
2151	0A53 67	LD H, A
2152	0A54 11	DB 11H
2153	0A55	LEFT:
2154		;
2155		; Cursor left
2156		;
2157	0A55 25	DEC H
2158	0A56 3E	DB 3EH
2159	0A57	UP:
2160		;
2161		; Cursor up
2162		;
2163	0A57 2D	DEC L
2164	0A58 C8	RET Z

- MSXIO - Cursor movements

```

2165 0A59 18 0E          JR      STOCSR
2166 0A5B          ADVCUR:
2167          ;
2168          ; Advance cursor
2169          ;
2170 0A5B CD 0A44          CALL  RIGHT
2171 0A5E C0              RET    NZ
2172 0A5F 26 01          LD     H,1
2173 0A61          DOWN:
2174          ;
2175          ; Cursor down
2176          ;
2177 0A61 CD 0C32          CALL  GETLEN
2178 0A64 BD              CP     L
2179 0A65 C8              RET    Z
2180 0A66 38 05          JR     C,DOWNL
2181 0A68 2C              INC    L
2182 0A69          STOCSR:
2183 0A69 22 F3DC          LD     (CSRY),HL
2184 0A6C C9              RET
2185 0A6D          DOWNL:
2186          ;
2187 0A6D 2D              DEC    L
2188 0A6E AF              XOR    A
2189 0A6F 18 F8          JR     STOCSR
2190 0A71          TAB:
2191          ;
2192          ; Tabulation
2193          ;
2194 0A71 3E 20          LD     A,' '
2195 0A73 CD 08DF          CALL  CHPUT1

```

```

;Get an actual bottom of screen
;Are we at the bottom of screen?
;Yes, return with Z flag
;We're below screen bottom
;Go to next line

```

```

( MSX ROM BASIC BIOS ) Macro-80      3.44    01-Jan-85    PAGE    32-2
- MSXIO - Cursor movements

2196 0A76 3A F3DD          LD      A,(CSRX)
2197 0A79 3D  DEC          A
2198 0A7A E6 07  AND          7
2199 0A7C 20 F3  JR      NZ,TAB
2200 0A7E C9  RET
2201 0A7F
2202
2203 ;
2204 ; Cursor home
2205
2206 0A7F 2E 01  LD      L,1
2207 0A81  CR:
2208 ;
2209 ; Carriage return
2210
2211 0A81 26 01  LD      H,1
2212 0A83 18 E4  JR      STOCSR
2213 ;
SUBTTTL - MSXIO - Line insert and delete of CRT

```

```
2214
2215 0A85
2216
2217 ;
2218 ; Delete a line specified by [L]
2219 ;
2220 ; Cursor should be set at the top of line
2221 ;
2222 CD 0A81 CALL CR
2223 DELLN0:
2224 0A88
2225 0A88 CD 0C32 CALL GETLEN ;Get an actual height of screen
2226 0A8B 95 SUB L
2227 0A8C D8 RET C
2228 0A8D CA 0AEC JP Z,ELN ;Something is wrong
2229 0A90 E5 PUSH HL ;Delete the bottom line only
2230 0A91 F5 PUSH AF ;Save row
2231 0A92 4F LD C,A ;Save counter (# of lines to be moved upward)
2232 0A93 06 00 LD B,0
2233 0A95 CD 0C1D CALL GETTRM ;Get address of [LINTTB] in [DE]
2234 0A98 6B LD L,E
2235 0A99 62 LD H,D
2236 0A9A 23 INC HL
2237 0A9B ED B0 LDIR
2238 0A9D 21 FBCA LD HL,FSTPOS
2239 0AA0 35 DEC (HL)
2240 0AA1 F1 POP AF
2241 0AA2 E1 POP HL
2242 0AA3
2243 DELLN1:
2244 0AA3 F5 PUSH AF ;Save counter
2245 0AA4 2C INC L
2246 0AA5 CD 0BAA CALL GETLIN ;Get 1 line specified by L
2247 0AA8 2D DEC L
```

2245	0AA9	CD 0BC3	CALL	PUTLN		
2246	0AAC	2C	INC	L		;Put 1 line specified by L
2247	0AAD	F1	POP	AF		;Restore counter
2248	0AAE	3D	DEC	A		
2249	0AAF	20 F2	JR	NZ,DELLN1		
2250	0AB1	C3 0AEC	JP	ELN		;Blank bottom line
2251	0AB4					
2252						
2253						
2254						
2255						
2256						
2257	0AB4	CD 0A81	CALL	CR		
2258	0AB7					
2259	0AB7	CD 0C32	CALL	GETLEN		;Get an actual height of screen
2260	0ABA	67	LD	H,A		
2261	0ABB	95	SUB	L		
2262	0ABC	D8	RET	C		;Something is wrong!!
2263	0ABD	CA 0AEC	JP	Z,ELN		
2264	0AC0	6C	LD	L,H		
2265	0AC1	E5	PUSH	HL		;Save row to be inserted
2266	0AC2	F5	PUSH	AF		;Save # of lines to be moved downward
2267	0AC3	4F	LD	C,A		
2268	0AC4	06 00	LD	B,0		
2269	0AC6	CD 0C1D	CALL	GETTRM		
2270	0AC9	6B	LD	L,E		
2271	0ACA	62	LD	H,D		
2272	0ACB	E5	PUSH	HL		;Save pointer to [LINTTB] for the bottom line
2273	0ACC	2B	DEC	HL		;Form source address
2274	0ACD	ED B8	LDDR			
2275	0ACF	E1	POP	HL		

```

( MSX ROM BASIC BIOS ) Macro-80      3.44      01-Jan-85      PAGE      33-2
- MSXIO - Line insert and delete of CRT

2276      0AD0      74      LD      (HL),H      ;Make sure the bottom line is terminated
2277      0AD1      F1      POP      AF
2278      0AD2      E1      POP      HL
2279      0AD3      INSLN1:      PUSH      AF
2280      0AD3      F5      DEC      L
2281      0AD4      2D      CALL      GETLN
2282      0AD5      CD 0BAA      INC      L
2283      0AD8      2C      CALL      PUTLN
2284      0AD9      CD 0BC3      DEC      L
2285      0ADC      2D      POP      AF
2286      0ADD      F1      A
2287      0ADE      3D      NZ,INSLN1
2288      0ADF      20 F2      JR      ELN
2289      0AE1      18 09      JR
2290
2291      ;
      SUBTTTL - MSXIO - Character(s) erase

```

```

( MSX ROM BASIC BIOS ) Macro-80      3.44    01-Jan-85      PAGE    34
- MSXIO - Character(s) erase

2292                                RUBOUT:
2293                                ;
2294                                ; Erase previous character
2295                                ;
2296                                ;
2297                                CALL BS          ;Back space
2298                                RET Z           ;We're at the top of screen
2299                                LD C,' '       ;Overstrike with a space
2300                                JP PUTVRM
2301                                ELN:
2302                                ;
2303                                ; Erase entire line
2304                                ;
2305                                ; Cursor should remain unchanged
2306                                ;
2307                                LD H,1
2308                                EOL:
2309                                ;
2310                                ; Erase to end-of-line
2311                                ;
2312                                ; Cursor should remain unchanged
2313                                ;
2314                                CD 0C29        CALL TERMIN
2315                                E5             PUSH HL          ;Save current position (column)
2316                                CD 0BF2        CALL VADDR
2317                                CD 07DF        CALL SETWRT
2318                                AF8           POP HL           ;Restore current position
2319                                AFB           EREOL1:
2320                                AF9           LD A,' '          ;Overstrike with a space
2321                                AFB           OUT (VDP.DRW),A
2322                                AFD           INC H

```

```

2323 0AFE 3A F3B0
2324 0B01 BC
2325 0B02 30 F5
2326 0B04 C9
2327 0B05
      EOP:
2328 ;
2329 ; Erase to end-of-page
2330 ;
2331 ; Cursor should remain unchanged
2332 ;
2333 0B05 E5
2334 0B06 CD 0AEE
2335 0B09 E1
2336 0B0A CD 0C32
2337 0B0D BD
2338 0B0E D8
2339 0B0F C8
2340 0B10 26 01
2341 0B12 2C
2342 0B13 18 F0
2343
2344 ;
      SUBTTL - MSXIO - Function keys display/erase.
; Save current position
; Erase to end-of-line
; Restore current position
; Get an actual height of CRT
; Something is wrong
; All done

```

```
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2371
2372
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2374
2375

0B15
0B15
0B18
0B19
0B1C
0B1D
0B1E
0B21
0B24
0B25
0B26

CD FDB8
AF
CD 0B9C
D0
E5
2A F3B1
CD 0AEC
E1
C9

ERAFNK:
;
; Erase function key
;
CALL H.ERAF
XOR A
CALL SETCHK
RET NC
PUSH HL
LD HL,(CRTCNT)
CALL ELN
POP HL
RET

FNKSB:
;
; Display function key if enabled
;
LD A,(CNSDFG)
AND A
RET Z
;No

DSPFNK:
;
; Display function key
;
CALL H.DSPF
LD A,0FFH
CALL SETCHK
RET NC
PUSH HL
LD A,(CSRY)

;Say no function key is displayed
;We're not in text mode, just set flag
;Save possible text pointer
;Erase last line
;Restore possible text pointer

;Say function key is displayed
;Say function key is displayed
;We're not in text mode, just set flag
;Save possible text pointer
```

2376	0E38	21 F3B1	LD	HL,CRTCNT	
2377	0E3B	BE	CP	(HL)	
2378	0E3C	3E 0A	LD	A,0AH	;Scroll up if we're at the bottom of screen
2379	0E3E	20 01	JR	NZ,NTBOTM	
2380	0B40	DF	RST	18H	
2381	0B41				
2382	0B41	3A FBEB	LD	A,(SFTKEY)	;Get current shift status
2383	0B44	0F	RRCA		
2384	0B45	21 F87F	LD	HL,FNKSTR	;Assume shift not pressed
2385	0B48	3E 01	LD	A,1	
2386	0B4A	38 04	JR	C,DSPFK1	;Good assumption
2387	0B4C	21 F8CF	LD	HL,FNKSTR+80	;Shift is being pressed
2388	0B4F	AF	XOR	A	
2389	0B50				
2390	0B50	32 FBCE	LD	(FNKSWI),A	;Mark which part of function key is displayed
2391	0B53	11 FC18	LD	DE,BUFEND	
2392	0B56	D5	PUSH	DE	
2393	0B57	06 28	LD	B,'('	;=40
2394	0B59	3E 20	LD	A,' '	
2395	0B5B				
2396	0B5B	12	LD	(DE),A	
2397	0B5C	13	INC	DE	
2398	0B5D	10 FC	DJNZ	DSPFKCL	
2399	0B5F	D1	POP	DE	
2400	0B60	0E 05	LD	C,5	;Restore temporary destination in [DE] ;Total number of keys ;Calculate (LINLEN-4) / 5
2401	0B62	3A F3B0	LD	A,(LINLEN)	
2402	0B65	D6 04	SUB	4	
2403	0B67	38 2B	JR	C,DSPFKE	;Not enough room for function keys
2404	0B69	06 FF	LD	B,OFFH	
2405	0B6B				
2406	0B6B	04	INC	B	

2407	0B6C	D6 05	SUB	5		
2408	0B6E	30 FB	JR	NC,DSPFK4		
2409	0B70	78	LD	A,B		
2410	0B71	A7	AND	A		
2411	0B72	28 20	JR	Z,DSPFKE		;No enough room
2412	0B74	3E	DB	3EH		;Skip next byte
2413	0B75					
2414	0B75	13	INC	DE		;Put separator space
2415	0B76	C5	PUSH	BC		;Save key counter
2416	0B77	0E 00	LD	C,0		;Reset # of characters actually fetched
2417	0B79					
2418	0B79	7E	LD	A,(HL)		;Get from function key string
2419	0B7A	23	INC	HL		;Prepare for next fetch
2420	0B7B	0C	INC	C		
2421	0B7C	CD 089D	CALL	CNVCHR		
2422	0B7F	30 F8	JR	NC,DSPFK5		;This is a graphic header, fetch more
2423	0B81	20 04	JR	NZ,DSPFK8		;Converted graphics character, store this
2424	0B83	FE 20	CP	,		;Printable?
2425	0B85	38 01	JR	C,DSPFK6		;No, ignore this
2426	0B87					
2427	0B87	12	LD	(DE),A		
2428	0B88					
2429	0B88	13	INC	DE		
2430	0B89	10 EE	DJNZ	DSPFK5		
2431	0B8B	3E 10	LD	A,10H		
2432	0B8D	91	SUB	C		
2433	0B8E	4F	LD	C,A		;Skip rest
2434	0B8F	09	ADD	HL,BC		
2435	0B90	C1	POP	BC		;Restore counter
2436	0B91	0D	DEC	C		
2437	0B92	20 E1	JR	NZ,DSPFK2		

- MSXIO - Function keys display/erase.

```

2438      0B94      DSPFKE:
2439      0B94      2A F3B1      LD      HL,(CRTCNT)      ;Display at the lowest line
2440      0B97      CD 0BC3      CALL   PUTLN
2441      0B9A      E1          POP     HL      ;Restore possible text pointer
2442      0B9B      C9          RET
2443      ;
2444      SUBTTL - MSXIO - Low level routines

```

```
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2471
2472
2473
2474
2475

0B9C
0B9C 32 F3DE
0B9F

0B9F 3A FCAF
0BA2 FE 02
0BA4 C9
0BA5

0BA5 E5
0BA6 0E 08
0BA8 18 0A
0BAA

0BAA E5
0BAB 26 01
0BAD CD 0BF2
0BB0 3A F3B0
0BB3 4F

        SETCHK:
        ;
        ; Set CNSDFG and check current screen mode
        ;
        LD      (CNSDFG),A
CHKSCR:
        ;
        ; Check current screen mode
        ;
        LD      A,(SCRMOD)
        CP      2
        RET
        ;Return with the status

GET8B:
        ;
        ; Get 8 bytes from HL
        ;
        PUSH    HL
        LD      C,8
        JR      GET11
GET11:
        ;
        ; Get character and attribute of position specified by H,L
        ;
        ; Character returned in C
        ;
        PUSH    HL
        LD      H,1
        CALL    VADDR
        LD      A,(LINLEN)
        LD      C,A
```

```

( MSX ROM BASIC BIOS ) Macro-80      3.44      01-Jan-85      PAGE      36-1
- MSXIO - Low level routines

2476      0BB4      GETL1:      LD      B,0
2477      0BB4      06 00      LD      DE,BUFEND
2478      0BB6      11 FC18      CALL   LDIRMV
2479      0BB9      CD 070F      POP     HL
2480      0BBC      E1          RET
2481      0BBD      C9
2482      0BBE      PUT8B:
2483      ;
2484      0BBE      E5          PUSH    HL
2485      0BBF      0E 08      LD      C,8
2486      0BC1      18 0A      JR       PUTL1
2487      0BC3      PUTL1:
2488      ;
2489      0BC3      E5          PUSH    HL
2490      0BC4      26 01      LD      H,1
2491      0BC6      CD 0BF2      CALL   VADDR
2492      0BC9      3A F3B      LD      A,(LINLEN)
2493      0BCC      4F          LD      C,A
2494      0BCD      PUTL1:
2495      0BCD      06 00      LD      B,0
2496      0BCF      EB          DE,HL
2497      0BD0      21 FC18      LD      HL,BUFEND
2498      0BD3      CD 0744      CALL   LDIRVM
2499      0BD6      E1          POP     HL
2500      0BD7      C9          RET
2501      0BD8      GETVRM:
2502      ;
2503      0BD8      E5          PUSH    HL
2504      0BD9      CD 0BF2      CALL   VADDR
2505      0BDC      CD 07EC      CALL   SETRD
2506      0BDF      E3          EX      (SP),HL
;Save coordinate
;Calculate VRAM address
;Set up VDP for read

```

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Low level routines

3.44 01-Jan-85 PAGE 36-2

2507 0BE0 E3 EX (SP),HL
2508 0BE1 DB 98 IN A,(VDP.DRW) ;Get character code in C
2509 0BE3 4F LD C,A
2510 0BE4 E1 POP HL ;Restore coordinate
2511 0BE5 C9 RET
2512 0BE6 PUTVRM:
2513 ;
2514 0BE6 E5 PUSH HL
2515 0BE7 CD 0BF2 CALL VADDR
2516 0BEA CD 07DF CALL SETWRT
2517 0BED 79 LD A,C
2518 0BEE D3 98 OUT (VDP.DRW),A
2519 0BF0 E1 POP HL
2520 0BF1 C9 RET
2521 0BF2 VADDR:
2522 ;
2523 ; Calculate buffer address out of H,L (column,row)
2524 ;
2525 ; address returned in HL
2526 ;
2527 0BF2 C5 PUSH BC
2528 0BF3 5C LD E,H ;Get column in L
2529 0BF4 26 00 LD H,0
2530 0BF6 54 LD D,H
2531 0BF7 2D DEC L
2532 0BF8 29 ADD HL,HL
2533 0BF9 29 ADD HL,HL
2534 0BFA 29 ADD HL,HL
2535 0BFB 4D LD C,L
2536 0BFC 44 LD B,H
2537 0BFD 29 ADD HL,HL

```

(MSX ROM BASIC BIOS) Macro-80
 - MSXIO - Low level routines

```

2538 0BFE 29          HL,HL
2539 0BFF 19          HL,DE
2540 0C00 3A FCAF     A,(SCRMOD)
2541 0C03 A7          A
2542 0C04 3A F3B0     A,(LINLEN)
2543 0C07 28 04      Z,VADDR1
2544 0C09 D6 22      SUB
2545 0C0B 18 03      JR
2546 0C0D            VADDR2
2547 ;
2548 0C0D 09          HL,BC
2549 0C0E D6 2A      SUB
2550 0C10            VADDR2:
2551 0C10 2F          CPL
2552 0C11 A7          AND
2553 0C12 1F          RRA
2554 0C13 5F          LD
2555 0C14 19          E,A
2556 0C15 EB          HL,DE
2557 0C16 2A F922     DE,HL
2558 0C19 19          HL,(NAMBAS)
2559 0C1A 2B          HL,DE
2560 0C1B C1          HL
2561 0C1C C9          POP
2562 0C1D            BC
2563 ;
2564 ; Get value of line-terminator-table and affect flags
2565 ;
2566 ; Entry: L has the line #
2567 ; Exit: DE has the address of corresponding terminator byte.
2568 ; Z flag is affected.

```

```
2569 ;
2570 0C1D E5          PUSH HL          ; Save HL
2571 0C1E 11 FB11    LD DE,BASROM
2572 0C21 26 00      LD H,0
2573 0C23 19        ADD HL,DE
2574 0C24 7E        LD A,(HL)
2575 0C25 EB        DE,HL
2576 0C26 E1        POP HL
2577 0C27 A7        AND A
2578 0C28 C9        RET
2579 0C29
2580
2581 0C29 3E        DB 3EH          ; Load non 0 value in Acc
2582 0C2A
2583 0C2A AF        XOR A
2584 0C2B
2585 0C2B F5        PUSH AF
2586 0C2C CD 0C1D    CALL GETTRM
2587 0C2F F1        POP AF
2588 0C30 12        LD (DE),A
2589 0C31 C9        RET
2590 0C32
2591
2592 ;
2593 ; Get an actual height of screen
2594 0C32 3A F3DE    LD A,(CNSDFG)    ; 0 or -1
2595 0C35 E5        PUSH HL
2596 0C36 21 FB11    LD HL,CRTCNT
2597 0C39 86        ADD A,(HL)
2598 0C3A E1        POP HL
2599 0C3B C9        RET
```

(MSX ROM BASIC BIOS) Macro-80	3.44	01-Jan-85	PAGE	36-5
- MSXIO - Low level routines				
2600				
2601				
; SUBTTL - MSXIO - Keyboard encoding routines				

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                                OC3C
                                KEYINT:
                                ;
                                ; Encode keyboard
                                ;
                                ; Timer interrupt routine
                                ;

                                PUSH HL
                                PUSH DE
                                PUSH BC
                                PUSH AF
                                EXX
                                EX AF,AF'
                                PUSH HL
                                PUSH DE
                                PUSH BC
                                PUSH AF
                                PUSH IY
                                PUSH IX
                                CALL H.KEYI
                                IN A,(VDP.SR)
                                AND A
                                JP P,INTRET
                                CALL H.TIMI
                                EI

                                FB
                                32 F3E7
                                E6 20
                                LD (STATFL),A
                                AND ' ,
                                ;Save all registers

                                ;To allow other interrupts than 60Hz timer
                                ;Clear possible interrupt request
                                ;Interrupt requested by VDP?
                                ;No, skip the rest
                                ;To allow timer interrupt to be
                                ;used elsewhere.
                                ;Now that it became obvious that VDP
                                ;generated the interrupt, we re-enable
                                ;interrupt here to allow RS232C's
                                ;interrupt or something like that.
                                ;Store this new status
                                ;Collision detected?
```

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Keyboard encoding routines

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2633 0C5C 21 FC6D LD HL,TRPTBL+33 ;Assume so
2634 0C5F C4 0EF1 CALL NZ,REQTRP ;Request trap if so
2635 ;
2636 ; Check interval trap
2637 ;
2638 0C62 2A FCA2 LD HL,(INTCNT) ;Count down interval count
2639 0C65 2B DEC HL
2640 0C66 7C LD A,H
2641 0C67 B5 OR L
2642 0C68 20 09 JR NZ,NTINTT ;Not yet reached 0
2643 0C6A 21 FC7F LD HL,TRPTBL+3*17 ;Request trap
2644 0C6D CD 0EF1 CALL REQTRP
2645 0C70 2A FCA0 LD HL,(INTVAL) ;Load initial value
2646 0C73 NTINTT:
2647 0C73 LD (INTCNT),HL ;Update interval count
2648 ;
2649 ; Increment jiffy count
2650 ;
2651 0C76 2A FC9E LD HL,(JIFFY)
2652 0C79 23 INC HL
2653 0C7A 22 FC9E LD (JIFFY),HL
2654 ;
2655 ; Check music queue
2656 ;
2657 0C7D 3A FB3F LD A,(MUSICF) ;Check music flag
2658 0C80 4F LD C,A
2659 0C81 AF XOR A ;Start with queue 0
2660 ;
2661 0C82 CB 19 RR C ;C7=carry, carry=C0, [C]=[C]/2
2662 0C84 F5 PUSH AF ;Save queue ID
2663 0C85 C5 PUSH BC ;Save MUSICF

```

2664	0C86	DC 113B	CALL	C,ACTION	
2665	0C89	C1	POP	BC	
2666	0C8A	F1	POP	AF	
2667	0C8B	3C	INC	A	;Next queue
2668	0C8C	FE 03	CP	3	;All done?
2669	0C8E	38 F2	JR	C,MUSINT	;Not yet
2670	0C90	21 F3F6	LD	HL,SCNCNT	
2671	0C93	35	DEC	(HL)	;Need to scan?
2672	0C94	20 6C	JR	NZ,INTRET	;No, return soon
2673	0C96	36 03	LD	(HL),3	;Time delay of first repeat
2674					
2675					
2676					
2677	0C98	AF	XOR	A	
2678	0C99	CD 120C	CALL	SLSTCK	
2679	0C9C	E6 30	AND	00110000B	;Read joystick A
2680	0C9E	F5	PUSH	AF	
2681	0C9F	3E 01	LD	A,1	
2682	0CA1	CD 120C	CALL	SLSTCK	
2683	0CA4	E6 30	AND	'0'	
2684	0CA6	07	RLCA		
2685	0CA7	07	RLCA		
2686	0CA8	C1	POP	BC	
2687	0CA9	B0	OR	B	
2688	0CAA	F5	PUSH	AF	
2689	0CAB	CD 1226	CALL	GTROW8	
2690	0CAE	E6 01	AND	1	
2691	0CB0	C1	POP	BC	
2692	0CB1	B0	OR	B	
2693	0CB2	4F	LD	C,A	
2694	0CB3	21 F3E8	LD	HL,TRGFLG	;Save this

(MSX ROM BASIC BIOS) Macro-80
 - MSXIO - Keyboard encoding routines

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```

2695 0CB6 AE
2696 0CB7 A6
2697 0CB8 71
2698 0CB9 4F
2699 0CBA 0F
2700 0CBB 21 FC70
2701 0CBE DC 0EF1
2702 0CC1 CB 11
2703 0CC3 21 FC7C
2704 0CC6 DC 0EF1
2705 0CC9 CB 11
2706 0CCB 21 FC76
2707 0CCE DC 0EF1
2708 0CD1 CB 11
2709 0CD3 21 FC79
2710 0CD6 DC 0EF1
2711 0CD9 CB 11
2712 0CDB 21 FC73
2713 0CDE DC 0EF1
2714
2715
2716
2717 0CE1 AF
2718 0CE2 32 FBD9
2719 0CE5 CD 0D12
2720 0CE8 20 18
2721 0CEA 21 F3F7
2722 0CED 35
2723 0CEE 20 12
2724 0CF0 36 01
2725 0CF2 21 FBDA

; Scan keyboard
;
;
; Enable first key click
; Detect valid key transition and check buffer
; Some characters still remain, don't repeat
; Need to enter repeat mode
; No
; Set short time repeat
; Clear OLDKEY status

```

```

(HL)
(HL)
(HL),C
C,A

```

```

XOR
AND
LD
LD
RRCA
LD
CALL
RL
LD
CALL
RL
LD
CALL
RL
LD
CALL
RL
LD
CALL

```

```

HL,TRPTBL+3*12
C,REQTRP
C
HL,TRPTBL+3*16
C,REQTRP
C
HL,TRPTBL+3*14
C,REQTRP
C
HL,TRPTBL+3*15
C,REQTRP
C
HL,TRPTBL+3*13
C,REQTRP

```

```

;Any transition?
;Is this transition negative
;Update trigger status
;Check space key trigger
;Check trigger 4
;Check trigger 2
;Check trigger 3
;Check trigger 1

```

```

A
(CLKFL),A
KEYCHK
NZ,INTRET
HL,REPCNT
(HL)
NZ,INTRET
(HL),1
HL,OLDKEY

```

```

;Enable first key click
;Detect valid key transition and check buffer
;Some characters still remain, don't repeat
;Need to enter repeat mode
;No
;Set short time repeat
;Clear OLDKEY status

```

2726	0CF5	11 FBDB	LD	DE,OLDKEY+1	
2727	0CF8	01 000A	LD	BC,0AH	
2728	0CFB	36 FF	LD	(HL),0FFH	
2729	0CFD	ED B0	LDIR		
2730	0CFF	CD 0D4E	CALL	KEYCK4	;Check if currently pressed key is valid
2731	0D02				
INTRET:					
2732	0D02	DD E1	POP	IX	;Restore all registers
2733	0D04	FD E1	POP	IY	
2734	0D06	F1	POP	AF	
2735	0D07	C1	POP	BC	
2736	0D08	D1	POP	DE	
2737	0D09	E1	POP	HL	
2738	0D0A	08	EX	AF,AF'	
2739	0D0B	D9	EXX		
2740	0D0C	F1	POP	AF	
2741	0D0D	C1	POP	BC	
2742	0D0E	D1	POP	DE	
2743	0D0F	E1	POP	HL	
2744	0D10	FB	EI		
2745	0D11	C9	RET		
2746	0D12				
KEYCHK:					
2747					
2748	0D12	DB AA	IN	A,(PPI CR)	;Get what is currently output to Port C
2749	0D14	E6 F0	AND	0F0H	;Leave higher 4 bits unaffected
2750	0D16	4F	LD	C,A	
2751	0D17	06 0B	LD	B,0BH	
2752	0D19	21 FBE5	LF	HL,NEWKEY	;Move current key status to NEWKEY
2753	0D1C				
2754	0D1C	79	LD	A,C	
2755	0D1D	D3 AA	OUT	(PPI.CW),A	;Select row
2756	0D1F	DB A9	IN	A,(PPI.BR)	;Get column information of selected row

2757	0D21	77	LD	(HL),A	
2758	0D22	0C	INC	C	;Move it
2759	0D23	23	INC	HL	;Select next row
2760	0D24	10 F6	DJNZ	KEYCK1	
2761	0D26	3A FBB0	LD	A,(ENSTOP)	;Loop until all rows are sensed
2762	0D29	A7	AND	A	;Warm start enabled?
2763	0D2A	28 0E	JR	Z,NOSTOP	;No
2764	0D2C	3A FBEB	LD	A,(SFTKEY)	;Get current status of the 6th row
2765	0D2F	FE E8	CP	0E8H	;Check if KANA, GRAPH, CTRL and SHIFT
2766	0D31	20 07	JR	NZ,NOSTOP	;are pressed simultaneously
2767	0D33	DD 21 409B	LD	IX,READYR	
2768	0D37	C3 01FF	JP	CALBAS	
2769	0D3A				
2770			NOSTOP:		
2771	0D3A	11 FBE5	LD	DE,NEWKEY	
2772	0D3D	06 0B	LD	B,0BH	;[OLDKEY] + 11
2773	0D3F		KEYCK2:		
2774	0D3F	1B	DEC	DE	
2775	0D40	2B	DEC	HL	
2776	0D41	1A	LD	A,(DE)	;Get OLDKEY status
2777	0D42	BE	CP	(HL)	;Compare with NEWKEY status
2778	0D43	20 04	JR	NZ,KEYCK3	;Changed, set long repeat interval
2779	0D45	10 F8	DJNZ	KEYCK2	
2780	0D47	18 05	JR	KEYCK4	
2781	0D49		KEYCK3:		;No change
2782					
2783	0D49	3E 0D	LD	A,0DH	
2784	0D4B	32 F3F7	LD	(REPCNT),A	
2785	0D4E		KEYCK4:		
2786	0D4E	06 0B	LD	B,0BH	;Set number of rows
2787	0D50	21 FBDA	LD	HL,OLDKEY	

- MSXIO - Keyboard encoding routines

```

2788 0D53 11 FBE5          LD      DE,NEWKEY
2789 0D56          KEYCK5:
2790 0D56 1A             LD      A,(DE)
2791 0D57 4F             LD      C,A
2792 0D58 AE             XOR     (HL)
2793 0D59 A6             AND     (HL)
2794 0D5A 71             LD      (HL),C
2795 0D5B C4 0D89        CALL    NZ,KEYANY
2796 0D5E 13             INC     DE
2797 0D5F 23             INC     HL
2798 0D60 10 F4          DJNZ    KEYCK5
2799 0D62          CHKBUF:
2800          ;
2801          ; Check if buffer is empty or not
2802          ;
2803 0D62 2A F3FA        LD      HL,(GETPNT)
2804 0D65 3A F3F8        LD      A,(PUTPNT)
2805 0D68 95             SUB     L
2806 0D69 C9             RET
2807 0D6A          CHSNS:
2808          ;
2809 0D6A FB             EI
2810 0D6B E5             PUSH    HL
2811 0D6C D5             PUSH    DE
2812 0D6D C5             PUSH    BC
2813 0D6E CD 0B9F        CALL    CHKSCR
2814 0D71 30 0F          JR      NC,CHSNS1
2815 0D73 3A FBCE        LD      A,(FNKSWI)
2816 0D76 21 FBEB        LD      HL,SFTKEY
2817 0D79 AE             XOR     (HL)
2818 0D7A 21 F3DE        LD      HL,CNSDFG

```

;Get current key status
 ;See if any bit changed
 ;See if this change is negative transition
 ;Update old status
 ;Active transition, go find it
 ;Load GETPNT
 ;Load lower 8 bit of PUTPNT
 ;Check if same
 ;Make sure interrupts are enabled
 ;Save environments
 ;Are we in text mode?
 ;No, do not flip function keys
 ;Get current shift status
 ;Get current function key display
 ;Are they different
 ;Function key displayed at all?

```
2819 0D7D A6 AND (HL)
2820 0D7E 0F RRCA
2821 0D7F DC 0B2B CALL C,DSPFNK ;Update display
2822 0D82 CHSNS1:
2823 0D82 CD 0D62 CALL CHKBUF
2824 0D85 C1 POP BC ;Restore environments
2825 0D86 D1 POP DE
2826 0D87 E1 POP HL
2827 0D88 C9 RET
2828 0D89 KEYANY:
2829
2830 ;
2831 ; [[[ SUBROUTINE 'KEYANY' ]]]
2832
2832 0D89 E5 PUSH HL ;Save environments
2833 0D8A D5 PUSH DE
2834 0D8B C5 PUSH BC
2835 0D8C F5 PUSH AF ;Save pressed bit
2836 0D8D 3E 0B LD A,0BH ;Calculate base code
2837 0D8F 90 SUB B
2838 0D90 87 ADD A,A
2839 0D91 87 ADD A,A
2840 0D92 87 ADD A,A
2841 0D93 4F LD C,A
2842 0D94 06 08 LD B,8 ;Set up counter for 8 bit
2843 0D96 F1 POP AF ;Restore pressed bit
2844 0D97 KYANY1:
2845 0D97 1F RRA
2846 0D98 C5 PUSH BC
2847 0D99 F5 PUSH AF
2848 0D9A DC 0E3B CALL C,KEYCOD ;If pressed bit, call key coder.
2849 0D9D F1 POP AF
```


2881	0DC4	0F06		DW	KYCLS	;CLS/HOME key
2882	0DC6	FF		DB	255	
2883	0DC7	0F10		DW	KYEASY	
2884						
2885	0DC9			NMSFTB:		
2886	0DC9	FF		DB	255	
2887	ODCA	21		DB	" "	
2888	0DCB	22		DB	34	
2889	0DCC	23 24 25 26		DB	"#\$%&'()"	;Double quote
2890	0DD0	27 28 29				
2891						
2892	0DD3			ALPJMP:		
2893	0DD3	0F55		DW	PUTCHR	;CTRL+shift
2894	0DD5	0F55		DW	PUTCHR	;CTRL
2895	0DD7	0E93		DW	KEYSFT	;SHIFT
2896	0DD9	0E95		DW	KEYNOM	;
2897						
2898	0ddb			KYCLTB:		
2899	0ddb	0DFD		DW	KYL SFC-10	;CTRL+SHIFT
2900	0DDD	0DF1		DW	KYL CNT-10	;CTRL
2901	0DDF	0DE5		DW	KYL SFT-10	;SHIFT
2902	0DE1	0DD9		DW	KYL NOM-10	;
2903	0DE3			KYINOM:		
2904	0DE3	2D 5E 5C 40		DB	"_^\ @[::],./"	
2905	0DE7	5B 3B 3A 5D				
2906	0DEB	2C 2E 2F				
2907	0DEE	FF		DB	255	
2908	0DEF			KYLSFT:		
2909	0DEF	3D 7E 7C 60		DB	"= ~ \ { + * } "	
2910	0DF3	7B 2B 2A 7D				
2911	0DF7	3C		DB	00111100B	;Less than sign

2912	0DF8	3E	DB	00111110B	
2913	0DF9	3F 5F	DB	"? " _	
2914	0DFB				;Greater than sign
2915	0DFB	2D	DB	" _	
2916	0DFC	1E	DB	" ^ " " @ "	
2917	0DFD	1C	DB	" \ " " @ "	
2918	0DFE	00	DB	" @ " " @ "	
2919	0DFF	1B	DB	" [" " @ "	
2920	0E00	3B 3A	DB	" ; " _	
2921	0E02	1D	DB	"] " " @ "	
2922	0E03	2C 2E 2F	DB	" , . / "	
2923	0E06	FF	DB	255	
2924	0E07				
2925	0E07	3D	DB	" = "	
2926	0E08	1E	DB	" ^ " " @ "	
2927	0E09	1C	DB	" \ " " @ "	
2928	0E0A	00	DB	" @ " " @ "	
2929	0E0B	1B	DB	" [" " @ "	
2930	0E0C	2B 2A	DB	" + * "	
2931	0E0E	1D	DB	"] " " @ "	
2932	0E0F	3C	DB	00111100B	
2933	0E10	3E	DB	00111110B	
2934	0E11	3F	DB	" ? "	
2935	0E12	1F	DB	" _ " " @ "	
2936					
2937	0E13				
2938	0E13	00	DB	0	;Shift (48)
2939	0E14	00	DB	0	;Control (49)
2940	0E15	00	DB	0	;Graph (50)
2941	0E16	00	DB	0	;Cap lock (51)
2942	0E17	00	DB	0	;Kana lock (52)

; EASYTB:

2943	0E18	00	DB	0	;F1	(53)
2944	0E19	00	DB	0	;F2	(54)
2945	0E1A	00	DB	0	;F3	(55)
2946	0E1B	00	DB	0	;F4	(56)
2947	0E1C	00	DB	0	;F5	(57)
2948	0E1D	1B	DB	27	;Escape	(58)
2949	0E1E	09	DB	9	;Tab	(59)
2950	0E1F	00	DB	0	;Stop	(60)
2951	0E20	08	DB	8	;Back space	(61)
2952	0E21	18	DB	"X"-"@"	;Select	(62)
2953	0E22	0D	DB	13	;Enter	(63)
2954	0E23	20	DB	32	;Space	(64)
2955	0E24	0C	DB	12	;Clear	(65)
2956	0E25	12	DB	"R"-"@"	;Insert	(66)
2957	0E26	7F	DB	127	;Rubout	(67)
2958	0E27	1D	DB	29	;Left	(68)
2959	0E28	1E	DB	30	;Up	(69)
2960	0E29	1F	DB	31	;Down	(70)
2961	0E2A	1C	DB	28	;Right	(71)
2962						
2963						
2964					For additional key matrix	
2965	0E2B	01	DB	"A"-"@"	; ;	(72)
2966	0E2C	04	DB	"D"-"@"	; ;	(73)
2967	0E2D	0F	DB	"O"-"@"	; ;	(74)
2968	0E2E	10	DB	"P"-"@"	; ;	(75)
2969	0E2F	11	DB	"Q"-"@"	; ;	(76)
2970	0E30	12	DB	"R"-"@"	; ;	(77)
2971	0E31	13	DB	"S"-"@"	; ;	(78)
2972	0E32	14	DB	"T"-"@"	; ;	(79)
2973	0E33	00	DB	0	; ;	(80)

(MSX ROM BASIC BIOS) Macro-80
 - MSXIO - Keyboard encoding routines

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2974	0E34	00	DB	0	;	(81)
2975	0E35	00	DB	0	;	(82)
2976	0E36	00	DB	0	;	(83)
2977	0E37	00	DB	0	;	(84)
2978	0E38	00	DB	0	;	(85)
2979	0E39	00	DB	0	;	(86)
2980	0E3A	00	DB	0	;	(87)

```
2981
2982
2983
2984
2985
2986
2987
2988
2989
2990
2991
2992
2993
2994
2995
2996
2997
2998
2999
3000
3001
3002
3003
3004
3005
3006
3007
3008
3009
3010
3011

0E3B
;
KEYCOD:
;
; [[[ SUBROUTINE 'KEYCOD' ]]]
;
; Return key-code in buffer if valid
;
;
0E3B 79 LD A,C ;Get raw code
0E3C FE FF CP OFFH ;Just for fail safe
0E3E C8 RET Z
0E3F 21 0DA5 LD HL,KYJTAB
0E42 CD FDCC CALL H.KEYC
0E45 FE 30 CP 48
0E47 30 13 JR NC,KYCLAS ;Possibly a KANA or graphic character
0E49 3A FBEB LD A,(SFTKEY) ;No
0E4C 0F RRCA ;Get shift key status
0E4D 0F RRCA ;Control pressed?
0E4E 30 0B JR NC,KYCLA0 ;Yes, this supersedes everything
0E50 0F RRCA ;How about graphic shift
0E51 D2 107D JP NC,KYGRAP ;Yes, this has the 2nd priority
0E54 3A FCAC LD A,(KANAST) ;KANA lock active
0E57 A7 AND A
0E58 C2 0F83 JP NZ,KYKANA ;Yes
;
KYCLA0:
0E5B 79 LD A,C
0E5C BE CP (HL) ;Compare range
0E5D 23 INC HL
0E5E 5E LD E,(HL) ;Get jump address in [DE]
0E5F 23 INC HL
```

```
3012 0E60 56 LD D,(HL)
3013 0E61 23 INC HL
3014 0E62 D5 PUSH DE
3015 0E63 D8 RET C
3016 0E64 D1 POP DE
3017 0E65 18 F5 JR KYCLAS
3018 0E67 KYNUM:
3019 ;
3020 0E67 C6 30 ADD A,'0'
3021 0E69 47 LD B,A
3022 0E6A 3A FBEB LD A,(SFTKEY)
3023 0E6D 0F RRCA
3024 0E6E 78 LD A,B
3025 0E6F 38 0A JR C,JPUTCH
3026 0E71 06 00 LD B,0
3027 0E73 21 0DC9 LD HL,NMSFTB
3028 0E76 09 LD HL,BC
3029 0E77 7E 7E LD A,(HL)
3030 0E78 FE FF CP OFFH
3031 0E7A C8 RET Z
3032 0E7B JPUTCH:
3033 0E7B C3 0F55 JP PUTCHR
3034 0E7E KYALP:
3035 ;
3036 0E7E 3A FBEB LD A,(SFTKEY)
3037 0E81 E6 03 AND 3
3038 0E83 87 ADD A,A
3039 0E84 5F E,A
3040 0E85 16 00 LD D,0
3041 0E87 21 0DD3 LD HL,ALPJMP
3042 0E8A 19 ADD HL,DE

;Assume matched
;Good assumption
;Discard stack
;Check next possibility

;Assume no shift
;Save code
;Check shift status

;Restore code
;Good assumption

;This must not be 'DADF'
;Get code for shift-number
;Shift '0'?
;Yes, ignore this

;Put this in buffer
```

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXIO - Keyboard encoding routines

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3043 0E8B 7E LD A,(HL) ;Get jump address
3044 0E8C 23 INC HL
3045 0E8D 66 LD H,(HL)
3046 0E8E 6F LD L,A
3047 0E8F 79 LD A,C
3048 0E90 D6 15 SUB 15H ;Get code
3049 0E92 E9 JP (HL) ;Make it a control character (1 - 26)
3050 0E93 KEYSFT:
3051 ;
3052 0E93 C6 20 ADD A,' '
3053 0E95 KEYNOM:
3054 0E95 47 LD B,A
3055 0E96 3A FCAB LD A,(CAPST) ;Save code
3056 0E99 2F CPL
3057 0E9A E6 20 AND 00100000B
3058 0E9C A8 XOR B
3059 0E9D C6 40 ADD A,01000000B
3060 0E9F 18 DA JR JPUTCH
3061 0EAl KYCOD1:
3062 ;
3063 0EAl 21 0DDB LD HL,KYCLTB
3064 0EA4 3A FBEB LD A,(SFTKEY)
3065 0EA7 E6 03 AND 3
3066 0EA9 87 ADD A,A
3067 0EAA 5F E,A
3068 0EAB 16 00 LD D,0
3069 0EAD 19 LD HL,DE
3070 0EAE 7E LD A,(HL)
3071 0EAF 23 INC HL
3072 0EB0 66 LD H,(HL)
3073 0EB1 6F LD L,A ;Extract shift and control status

```

```
3074 0EB2 59          LD      E,C
3075 0EB3 19          ADD     HL,DE
3076 0EB4 7E          LD      A,(HL)
3077 0EB5 FE FF       CP      0FFH
3078 0EB7 C2 0F55     JP      NZ,PUTCHR
3079 0EBA C9          RET
3080 0EBB             KYFUNC:
3081             ;
3082             ; Function keys
3083             ;
3084 0EBB 3A FBEB       LD      A,(SFTKEY)
3085 0EBE 0F           RRCA
3086 0EBF 38 04       JR      C,KYFNC1
3087 0EC1 79          LD      A,C
3088 0EC2 C6 05       ADD     A,5
3089 0EC4 4F          LD      C,A
3090 0EC5             KYFNC1:
3091 0EC5 59          LD      E,C
3092 0EC6 16 00       LD      D,0
3093 0EC8 21 FB99     LD      HL,FNKFLG-53
3094 0ECB 19          ADD     HL,DE
3095 0ECC 7E          LD      A,(HL)
3096 0ECD A7          AND     A
3097 0ECE 20 13       JR      NZ,FNKINT
3098 0ED0             KYFNC2:
3099 0ED0 EB          EX
3100 0ED1 29          ADD     HL,HL
3101 0ED2 29          ADD     HL,HL
3102 0ED3 29          ADD     HL,HL
3103 0ED4 29          ADD     HL,HL
3104 0ED5 11 F52F     LD      DE,FNKSTR-53*16

;Should generate some code?
;Yes
;No code should be generated

;Is shift pressed?
;No

;[DE] is (56..65)
;Check if this function key is an event device

;Request trap if not in direct mode
```

3105	0ED8	19		ADD	HL,DE	
3106	0ED9	EB		EX	DE,HL	;Get function key string address
3107	0EDA					;Move address to DE
3108	0EDA	1A		LD	A,(DE)	
3109	0EDB	A7		AND	A	;Get from function key string
3110	0EDC	C8		RET	Z	;End of string
3111	0EDD	CD 0F55		CALL	PUTCHR	;Yes
3112	0EE0	13		INC	DE	;Put this character in buffer
3113	0EE1	18 F7		JR	KYFNC3	;Check next character
3114	0EE3					
3115				FNKINT:		
3116	0EE3	2A F41C		LD	HL,(CURLIN)	
3117	0EE6	23		INC	HL	;Are we in direct mode (CURLIN=65535)
3118	0EE7	7C		LD	A,H	
3119	0EE8	B5		OR	L	
3120	0EE9	28 E5		JR	Z,KYFNC2	
3121	0EEB	21 FEAD		LD	HL,TRPTBL-53*3	
3122	0EEE	19		ADD	HL,DE	
3123	0EEF	19		ADD	HL,DE	
3124	0EF0	19		ADD	HL,DE	

```
3125
3126
3127 0EF1
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3141
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3143
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3155

;
REQTRP:
;
; Request trap (called to request trap for event devices)
;
;
; Since REQTRP is mostly called from within an interrupt routine,
; don't touch the interrupt mask through DI or EI.
;
LD A,(HL)
AND 1
RET Z
LD A,(HL)
OR 4
CP (HL)
RET Z
LD (HL),A
XOR 5
RET NZ
LD A,(ONGSBF)
INC A
LD (ONGSBF),A
RET

;
KYCLS:
LD 7E
RRCA
LD 3A FBEB
LD 0F
LD 3E 0C
LD DE 00
LD 18 45
LD A,(SFTKEY)
;Set carry if shift not pressed
RRCA
LD A,0CH
SBC A,0
JR PUTCHR
;Load code for CLS
;Change to HOME if shift not pressed
```

(MSX ROM BASIC BIOS) Macro-80

- MSXIO - Keyboard encoding routines

```

3156 0F10
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3183
3184
3185
3186

      CD FDD1
      5F
      16 00
      21 0DE3
      19
      7E
      A7
      C8
      18 36

      CALL H.KYEA
      LD E,A
      LD D,0
      LD HL,EASYTB-48
      ADD HL,DE
      LD A,(HL)
      AND A
      RET Z
      JR PUTCHR

      KYKLOK:
      ;
      ; Kana lock key
      ;

      21 FCAC
      7E
      2F
      77
      3E 0F
      D3 A0
      DB A2
      E6 7F
      47
      7E
      2F
      E6 80
      B0
      D3 A1

      LD HL,KANAST
      LD A,(HL)
      CPL
      LD (HL),A
      LD A,0FH
      (PSG.LW),A
      IN A,(PSG.DR)
      AND 7FH
      LD B,A
      LD A,(HL)
      CPL
      AND 80H
      OR B
      OUT (PSG.DW),A

```

;For CCP (Cut, copy, paste) editor rom
;These character are simply taken from table

;Should this key generate some code
;No
;Yes

```
3187 0F35          NOKEY:
3188 0F35          C9          RET
3189 0F36          KYLOCK:
3190          ;
3191          ; Capital lock key
3192          ;
3193 0F36 21 FCAB      LD      HL,CAPST
3194 0F39 7E          LD      A,(HL)
3195 0F3A 2F          CPL
3196 0F3B 77          LD      (HL),A
3197 0F3C 2F          CPL
3198 0F3D          CHGCAP:
3199 0F3D A7          AND      A
3200 0F3E 3E 0C       LD      A,0CH
3201 0F40 28 01       JR      Z,CGCAP1
3202 0F42 3C          INC      A
3203 0F43          CGCAP1:
3204 0F43 D3 AB       OUT      (PPI.CM),A
3205 0F45 C9          RET
3206 0F46          KYSTOP:
3207          ;
3208          ; STOP key
3209          ;
3210 0F46 3A FBEB      LD      A,(SFTKEY)
3211 0F49 0F          RRCA
3212 0F4A 0F          RRCA
3213 0F4B 3E 03       LD      A,3
3214 0F4D 30 01       JR      NC,KYSTP1
3215 0F4F 3C          INC      A
3216 0F50          KYSTP1:
3217 0F50 32 FC9B      LD      (INTFLG),A

          ;Toggle capital status
          ;Update capital status
          ;Assume 'turn off'
          ;Good assumption
          ;Change to 'turn on'
          ;Move CTRL status to carry
          ;Assume CTRL pressed also
          ;Good assumption
          ;CTRL not pressed, just treat as pause
```

```
3218 0F53 38 0F          JR      C,GENCLK      ;Only generate click if pause
3219 0F55                PUTCHR:
3220                      ;
3221                      ; Put one character in key buffer .
3222                      ;
3223 0F55 2A F3F8          LD      HL,(PUTPNT)      ;Load PUTPNT in [HL]
3224 0F58 77              LD      (HL),A          ;Save the character to buffer
3225 0F59 CD 10C2          CALL  UPDATE          ;Increment PUTPNT
3226 0F5C 3A F3FA          LD      A,(GETPNT)      ;Load lower 8bit of GETPNT
3227 0F5F BD              CP      L              ;Compare it with new PUTPNT
3228 0F60 C8              RET                     ;If same skip next step
3229 0F61 22 F3F8          LD      (PUTPNT),HL     ;Save HL in PUTPNT
3230 0F64                GENCLK:
3231 0F64 3A F3DB          LD      A,(CLIKSW)      ;Key click enabled?
3232 0F67 A7              AND      A              ;
3233 0F68 C8              RET                     ;No
3234 0F69 3A FBD9          LD      A,(CLIKFL)      ;Already generated?
3235 0F6C A7              AND      A              ;
3236 0F6D C0              RET                     ;Yes, don't click any more
3237 0F6E 3E 0F          LD      A,0FH          ;
3238 0F70 32 FBD9          LD      (CLIKFL),A      ;Set flag to disable more clicks
3239 0F73 D3 AB          OUT     (PPI.CM),A
3240 0F75 3E 0A          LD      A,0AH
3241 0F77                CLICKW:
3242 0F77 3D              DEC      A
3243 0F78 20 FD          JR      NZ,CLICKW
3244 0F7A                CHGSND:
3245 0F7A A7              AND      A
3246 0F7B 3E 0E          LD      A,0EH          ;Assume 'turn off'
3247 0F7D 28 01          JR      Z,CGSND1        ;Good assumption
3248 0F7F 3C              INC      A              ;Change to 'turn on'
```

```
3249 0F80 CGSND1: OUT (PPI.CM),A
3250 0F80 D3 AB RET
3251 0F82 C9
3252 0F83 KYKANA:
3253 ;
3254 ; KANA key pressed while KANA lock is active
3255 ;
3256 0F83 3A FCAD LD A,(KANAMD) ;JIS or AIUEO?
3257 0F86 A7 AND A ;Affect Z flag
3258 0F87 3A FBEB LD A,(SFTKEY) ;Check shift key
3259 0F8A 0F RRCA ;Affect Carry flag
3260 0F8B 28 0A JR Z,KAIUEO ;AIUEO order
3261 0F8D 21 101D LD HL,KANJNO
3262 0F90 38 0D JR C,KYKAN1
3263 0F92 21 104D LD HL,KANJSF
3264 0F95 18 08 JR KYKAN1
3265 0F97 KAIUEO:
3266 ;
3267 0F97 21 0FBD LD HL,KANANO ;Assume shift not pressed
3268 0F9A 38 03 JR C,KYKAN1 ;Good assumption
3269 0F9C 21 0FED LD HL,KANASF
3270 0F9F KYKAN1:
3271 0F9F 06 00 LD B,0
3272 0FA1 09 HL,BC
3273 0FA2 01 0F55 LD BC,PUTCHR
3274 0FA5 C5 BC ;Push jump address
3275 0FA6 3A FCAB LD A,(CAPST) ;Capital lock (katakana) active?
3276 0FA9 A7 AND A
3277 0FAA 7E A,(HL)
3278 0FAB C0 LD NZ
3279 0FAC FE A6 CP 165+1 ;active
;Special characters?
```

3280	0FAE	D8	RET	C	
3281	0FAF	FE B0	CP	0B0H	
3282	0FB1	C8	RET	Z	
3283	0FB2	FE DE	CP	0DEH	
3284	0FB4	D0	RET	NC	
3285	0FB5	D6 20	SUB	' '	;Assume first half
3286	0FB7	FE A0	CP	191-32+1	;Really first half
3287	0FB9	D8	RET	C	;Good assumption
3288	0FBA	C6 40	ADD	A,32+32	;Compensate
3289	0FBC	C9	RET		
3290	0FBD				
3291					
3292					
3293	0FBD	C9 B1 B2 B3	DB	0C9H,0B1H,0B2H,0B3H,0B4H,0B5H,0C5H	
3294	0FC1	B4 B5 C5			
3295	0FC4	C6 C7 C8 D7	DB	0C6H,0C7H,0C8H,0D7H,0D8H,0D9H,0DAH	
3296	0FC8	D8 D9 DA			
3297	0FCB	DB D3 DE DF	DB	0DBH,0D3H,0DEH,0DFH,0D6H,0DCH,0A6H	
3298	0FCF	D6 DC A6			
3299	0FD2	DD BB C4 C2	DB	0DDH,0BBH,0C4H,0C2H,0BDH,0B8H,0BEH	
3300	0FD6	BD B8 BE			
3301	0FD9	BF CF CC D0	DB	0BFH,0CFH,0CCH,0D0H,0D1H,0D2H,0D5H	
3302	0FDD	D1 D2 D5			
3303	0FE0	D4 CD CE B6	DB	0D4H,0CDH,0CEH,0B6H,0B9H,0BCH,0BAH	
3304	0FE4	B9 BC BA			
3305	0FE7	CB C3 B7 C1	DB	0CBH,0C3H,0B7H,0C1H,0CAH,0C0H	
3306	0FEB	CA C0			
3307	0FED				
3308					
3309					
3310	0FED	C9 A7 A8 A9	DB	0C9H,0A7H,0A8H,0A9H,0AAH,0ABH,0C5H	

KANANO:
;
;
Kana table (AIUEO order, un-shifted)

KANASF:
;
;
Shifted

3311	OFF1	AA AB C5		
3312	OFF4	C6 C7 C8 D7	DB	0C6H,0C7H,0C8H,0D7H,0D8H,0D9H,0DAH
3313	OFF8	D8 D9 DA		
3314	OFFB	A2 D3 B0 A3	DB	0A2H,0D3H,0B0H,0A3H,0AEH,0A4H,0A1H
3315	OFFF	AE A4 A1		
3316	1002	A5 BB C4 AF	DB	0A5H,0BBH,0C4H,0AFH,0BDH,0B8H,0BEH
3317	1006	BD B8 BE		
3318	1009	BF CF CC D0	DB	0BFH,0CFH,0CCH,0D0H,0D1H,0D2H,0ADH
3319	100D	D1 D2 AD		
3320	1010	AC CD CE B6	DB	0ACH,0CDH,0CEH,0B6H,0B9H,0BCH,0BAH
3321	1014	B9 BC BA		
3322	1017	CB C3 B7 C1	DB	0CBH,0C3H,0B7H,0C1H,0CAH,0C0H
3323	101B	CA C0		
3324	101D			

KANJNO:

; Kana table JIS order, un-shifted

;

3327	101D	DC C7 CC B1	DB	0DCH,0C7H,0CCH,0B1H,0B3H,0B4H,0B5H
3328	1021	B3 B4 B5		
3329	1024	D4 D5 D6 CE	DB	0D4H,0D5H,0D6H,0CEH,0CDH,0B0H,0DEH
3330	1028	CD B0 DE		
3331	102B	DF DA B9 D1	DB	0DFH,0DAH,0B9H,0D1H,0C8H,0D9H,0D2H
3332	102F	C8 D9 D2		
3333	1032	DB C1 BA BF	DB	0DBH,0C1H,0BAH,0BFH,0BCH,0B2H,0CAH
3334	1036	BC B2 CA		
3335	1039	B7 B8 C6 CF	DB	0B7H,0B8H,0C6H,0CFH,0C9H,0D8H,0D3H
3336	103D	C9 D8 D3		
3337	1040	D0 D7 BE C0	DB	0D0H,0D7H,0BEH,0C0H,0BDH,0C4H,0B6H
3338	1044	BD C4 B6		
3339	1047	C5 CB C3 BB	DB	0C5H,0CBH,0C3H,0BBH,0DDH,0C2H
3340	104B	DD C2		
3341	104D			

KANJSF:

[illegible]

```
3358
3359
3360 107D
3361
3362
3363
3364 107D 06 00 LD B,0
3365 107F 21 1092 LD HL,GRPTAB
3366 1082 09 ADD HL,BC
3367 1083 7E LD A,(HL)
3368 1084 A7 AND A
3369 1085 C8 RET Z
3370 1086 FE 80 CP 80H
3371 1088 F5 PUSH AF
3372 1089 3E 01 LD A,1
3373 108B DC 0F55 CALL C,PUTCHR
3374 108E F1 POP AF
3375 108F C3 0F55 JP PUTCHR
3376
3377 1092
3378 1092 4F 47 41 42 DB 4FH,47H,41H,42H,43H,44H,45H
3379 1096 43 44 45 DB 46H,4DH,4EH,57H,00H,49H,00H
3380 1099 46 4D 4E 57 DB 84H,82H,81H,85H,5FH,5DH,80H
3381 109D 00 49 00 DB 83H,00H,5BH,5AH,54H,58H,55H
3382 10A0 84 82 81 85 DB 53H,4AH,56H,00H,00H,5EH,4BH
3383 10A4 5F 5D 80 DB 00H,00H,50H,00H,52H,4CH,59H
3384 10A7 83 00 5B 5A
3385 10AB 54 58 55
3386 10AE 53 4A 56 00
3387 10B2 00 5E 4B
3388 10B5 00 00 50 00
```

; KYGRAP:
;
; Graphic characters
;
LD B,0
HL,GRPTAB
ADD HL,BC
LD A,(HL)
AND A
RET Z
CP 80H
PUSH AF
LD A,1
CALL C,PUTCHR
POP AF
JP PUTCHR

;Get from graphic key table
;Should generate some code
;No
;1 byte code?
;Assume not
;Was 2 byte code, put header byte

```
3389 10B9 52 4C 59
3390 10BC 00 51 00 5C DB 00H,51H,00H,5CH,48H,00H
3391 10C0 48 00
3392
3393 10C2 ; UPDATE:
3394 ;
3395 ; Update pointer
3396 ;
3397 10C2 23 INC HL
3398 10C3 7D LD A,L
3399 10C4 FE 18 CP 18H ;Check buffer boundary
3400 10C6 C0 RET NZ
3401 10C7 21 FBF0 LD HL,KEYBUF
3402 10CA C9 RET
3403 10CB CHGET:
3404 ;
3405 ; Get one character from keyboard
3406 ;
3407 10CB E5 PUSH HL
3408 10CC D5 PUSH DE
3409 10CD C5 PUSH BC
3410 10CE CD FDC2 CALL H,CHGE
3411 10D1 CD OD6A CALL CHSNS
3412 10D4 20 0B JR NZ,CHGET2 ;Character already there?
3413 10D6 CD 09DA CALL CKDPC0 ;Yes, do not touch cursor
3414 10D9 CHGET1: ;Display cursor if disabled
3415 10D9 CD OD6A CALL CHSNS ;Any character in buffer?
3416 10DC 28 FB JR Z,CHGET1 ;No, wait
3417 10DE CD 0A27 CALL CKERC0 ;Erase cursor if disabled
3418 10E1 CHGET2:
3419 10E1 21 FC9B LD HL,INTFLG
```

```
3420 10E4 7E          LD      A,(HL)
3421 10E5 FE 04       CP      4
3422 10E7 20 02       JR      NZ,CHGET3
3423 10E9 36 00       LD      (HL),0
3424 10EB             CHGET3:
3425 10EB 2A F3FA     LD      HL,(GETPNT)
3426 10EE 4E          LD      C,(HL)
3427 10EF CD 10C2     CALL   UPDATE
3428 10F2 22 F3FA     LD      (GETPNT),HL
3429 10F5 79          LD      A,C
3430 10F6 C3 08DB     JP      PBDHRT
3431 10F9             CKCNTC:
3432 ;
3433 ; Check ctl-C
3434 ;
3435 10F9 E5          PUSH   HL
3436 10FA 21 0000     LD      HL,0
3437 10FD CD 03FB     CALL   ISCNTC
3438 1100 E1          POP    HL
3439 1101 C9          RET
3440 ;
3441 SUBTTL - MSXIO - Music routines
```

;Code for pause?

;No

;Clear this

;Save pressed key

;Update [GETPNT]

;Set new [GETPNT]

;Pass result to Acc

;To disable CONTINUING

(MSX ROM BASIC BIOS)
 - MSXIO - Music routines

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      1102
      WRTPSG:
      ;
      ; Write data to specified register of GI sound chip
      ; Entry - (E)=data,(A)=register number
      ; Exit - All regs preserved
      ;
      ; GI Reg# - usage
      ;
      ; 0 voice A fine tune
      ; 1 voice A coarse tune
      ; 2 voice B fine tune
      ; 3 voice B coarse tune
      ; 4 voice C fine tune
      ; 5 voice C coarse tune
      ; 7 B7,B6 = Reg 14,15 Input Output flags
      ; B5,B4,B3 = voice C,B,A noise enable (0=enabled)
      ; B2,B1,B0 = voice C,B,A tone enable (0=enabled)
      ; 8 voice A volume (0..15 = volume, 16=use envelope)
      ; 9 voice B volume (0..15 = volume, 16=use envelope)
      ; 10 voice C volume (0..15 = volume, 16=use envelope)
      ; 11-12 envelope period
      ; 13 envelope shape (0..15)
      ; 14 joystick 1 port
      ; 15 joystick 2 port
      ;
      F3 DI
      D3 A0 OUT (PSG.LW),A ;LATCH ADDRESS
      F5 PUSH AF
      7B LD A,E
      D3 A1 OUT (PSG.DW),A ;OUTPUT DATA

```

```
3473 1109 FB EI
3474 110A F1 POP AF
3475 110B C9 RET
3476 110C INGI:
3477 ;
3478 ; Input data from PAD
3479 ;
3480 110C 3E 0E LD A,PSG.PA
3481 110E RDPSPG:
3482 110E D3 A0 OUT (PSG.LW),A
3483 1110 DB A2 IN A,(PSG.DR)
3484 1112 C9 RET
3485 1113 BEEP:
3486 ;
3487 ; BEEP causes a 'bell' sound
3488 ;
3489 ; Exit - all registers are destroyed
3490 ;
3491 1113 AF XOR A
3492 1114 1E 55 LD E,01010101B
3493 1116 CD 1102 CALL WRTPSG
3494 1119 5F LD E,A
3495 111A 3C INC A
3496 111B CD 1102 CALL WRTPSG
3497 111E 1E BE LD E,10111110B
3498 1120 3E 07 LD A,7
3499 1122 CD 1102 CALL WRTPSG
3500 1125 5F LD E,A
3501 1126 3C INC A
3502 1127 CD 1102 CALL WRTPSG
3503 112A 01 07D0 LD BC,07D0H

;[A]=fine tune register for voice A
;data to be written on R0
;0 to coarse tune register
;R1 coarse
;enable voice [A] tone
;[A]=voice enable register
;R7
;set volume to 7
;[A]=voice A volume register
;R8
```



```
3535 ;
3536 ; IF HO 2 BITS = 0 then this is the HO byte of the tone period.
3537 ; IF HO 2 BITS = 2 then this is just a volume control byte.
3538 ; IF BIT 4 IS ON, envelope control is in effect, and bits
3539 ; 0-3 give shape number of envelope.
3540 ; IF BIT 4 IS OFF, BITS 0-3 give amplitude number.
3541 ; IF HO 2 BITS = 3 THEN this byte will be followed by a 2 byte
3542 ; envelope period, HO first.
3543 ;
3544 ; ENTRY - (A)=Channel count number (0..2)
3545 ;
3546 LD B,A ;Save channel number
3547 CALL GETVCP ;Get pointer into vcb of channel
3548 DEC HL
3549 LD D,(HL)
3550 DEC HL
3551 LD E,(HL)
3552 DEC DE ;[DE]=countdown timer for voice
3553 LD (HL),E ;Decrement timer
3554 INC HL ;Put it back to first
3555 LD (HL),D
3556 LD A,D
3557 OR E
3558 RET NZ ;No action if not zero
3559 LD A,B ;Voice 0 uses queue 0
3560 LD (QUEUEN),A ;Set queue ID for further 'CALL XGETQ'
3561 CALL XGETQ
3562 CP OFFH
3563 JR Z,VOICOF ;branch if EOF marker
3564 LD D,A ;SAVE IN [D]
3565 AND OE0H ;Get number of following items
```

[illegible]

```
3597 1181          XVOL:
3598          ;
3599 1181          ;
3600 1182          LD      H,A
3601 1184          AND     80H
3602          JR      Z,XEPER
3603          ;
3604          ; Set volume
3605          ;
3606 1186          LD      E,D
3607 1187          LD      A,B
3608 1188          ADD     A,8
3609 118A          CALL    WRTPSG
3610 118D          LD      A,E
3611 118E          AND     10H
3612 1190          LD      A,0DH
3613 1192          CALL    NZ,WRTPSG
3614 1195          XEPER:
3615          ;
3616          ; Set envelope period
3617          ;
3618 1195          LD      A,H
3619 1196          AND     01000000B
3620 1198          JR      Z,MORACT
3621 119A          CALL    XGETQ
3622 119D          LD      D,A
3623 119E          CALL    XGETQ
3624 11A1          LD      E,A
3625 11A2          LD      A,0BH
3626 11A4          CALL    WRTPSG
3627 11A7          INC     A
3628 11A8          LD      E,D

;[A] has junk in ho which shouldn't matter
;Get back voice number
;Regs 8,9,10
;Output amplitude reg
;Check envelope generate bit
;Reg 13 for shape
;Set envelope shape if enabled

;See if set envelope period
;No
;Get ho byte of envelope period
;Get low byte of envelope period
;Register 11 for fine tune
;Point to coarse tune
```

```
3628 11A9 CD 1102          CALL WRTPSG
3629 11AC OD              DEC C
3630 11AD OD              DEC C
3631 11AE 18 B6          JR MORACT
3632 11B0
3633
3634
3635
3636
3637
3638
3639
3640
3641
3642
3643
3644
3645
3646
3647
3648
3649
3650
3651
3652
3653
3654
3655
3656
3657
3658

11B0
11B1
11B3
11B5
11B8
11B9
11BC
11BD
11BE
11BE
11BF
11C1
11C2
11C3
11C4

78
C6 08
1E 00
CD 1102
04
21 FB3F
AF
37
17
10 FD
A6
AE
77

          LD A,B
          ADD A,8
          LD E,0
          CALL WRTPSG
          INC B
          LD HL,MUSICF
          XOR A
          SCF

RSTFL1:
          RLA
          DJNZ
          AND (HL)
          XOR (HL)
          LD (HL),A

STRTMS:
;
; STRTMS starts the background music task if:
; 1) - it is currently idle (MUSICF=0) and
; 2) - there is work queued for it (PLYCNT.GTR. 0)
;
          LD A,(MUSICF)
          OR A
```

```

3659 11C8 C0 RET
3660 11C9 21 FB40 LD
3661 11CC 7E LD
3662 11CD B7 OR
3663 11CE C8 RET
3664 11CF 35 DEC
3665 11D0 21 0001 LD
3666 11D3 22 FB41 LD
3667 11D6 22 FB66 LD
3668 11D9 22 FB8B LD
3669 11DC 3E 07 LD
3670 11DE 32 FB3F LD
3671 11E1 C9 RET
3672 11E2
3673
3674 11E2 3A FB3E LD
3675 11E5 E5 PUSH
3676 11E6 D5 PUSH
3677 11E7 C5 PUSH
3678 11E8 CD 14AD CALL
3679 11EB C3 08DB JP
3680
3681
  XGETQ:
  ;
  LD A,(QUEUEEN) ;Get queue ID
  HL DE
  DE DE
  BC BC
  GETQ
  PBDHRT
  JP
  ;
  SUBTTL - MSXIO - Joystick and Paddle interface
  ;return if background task is active
  ;return if nothing for it to do
  ;1 less thing for it to do
  ;start it playing now
  ;Trigger!

```

```
3682
3683      11EE
3684
3685      3D
3686      FA 1200
3687      CD 120C
3688      11F5 21 1233
3689      11F8
3690      11F8 E6 0F
3691      11FA 5F
3692      11FB 16 00
3693      11FD 19
3694      11FE 7E
3695      11FF C9
3696      1200
3697
3698      1200 CD 1226
3699      1203 0F
3700      1204 0F
3701      1205 0F
3702      1206 0F
3703      1207 21 1243
3704      120A 18 EC
3705      120C
3706
3707
3708
3709      120C 47
3710      120D 3E 0F
3711      120F F3
3712      1210 CD 110E

      GTSTCK:
      ;
      DEC
      JP
      CALL
      LD
      STICK1:
      AND
      LD
      LD
      ADD
      LD
      RET
      KYSTCK:
      ;
      GTROW8
      CALL
      RRCA
      RRCA
      RRCA
      RRCA
      LD
      JR
      SLSTCK:
      ;
      ; Select proper joystick and read from it
      ;
      LD
      LD
      DI
      CALL
      RDPSG
      ;Read what is currently output to port B

      A
      M,KYSTCK
      SLSTCK
      HL,STKTBL
      0FH
      E,A
      D,0
      HL,DE
      A,(HL)
      ;STICK(0) - read cursor keys
      ;Read joystick
```

```

3713 1213 10 06          DJNZ      SLSTC1
3714 1215 E6 DF          AND       0DFH
3715 1217 F6 4C          OR        4CH
3716 1219 18 04          JR        SLSTC2
3717 121B          SLSTC1:
3718 ;
3719 121B E6 AF          AND       0AFH
3720 121D F6 03          OR        3
3721 121F          SLSTC2:
3722 121F D3 A1          OUT      (PSG.DW),A
3723 1221 CD 110C        CALL     INGI
3724 1224 FB           EI
3725 1225 C9           RET
3726 1226          GTROW8:
3727 ;
3728 ; Get keyboard's 8th row, bit assignments are as follows.
3729 ;
3730 ; RDULxxxxS
3731 ; ||||| |
3732 ; ||||| +- space
3733 ; |||+----- left
3734 ; ||+----- up
3735 ; |+----- down
3736 ; +----- right
3737 ;
3738 F3          DI
3739 1226 DB AA          IN      A,(PPI.CR)
3740 1227 E6 F0          AND     0F0H
3741 1229 C6 08          ADD     A,8
3742 122B D3 AA          OUT     (PPI.CW),A
3743 122D DB A9          IN      A,(PPI.BR)

```

```

3744 1231 FB EI
3745 1232 C9 RET
3746
3747 ;
3748 STKTBL:
3749 1233 00 DB 0
3750 1234 05 DB 5
3751 1235 01 DB 1
3752 1236 00 DB 0
3753 1237 03 DB 3
3754 1238 04 DB 4
3755 1239 02 DB 2
3756 123A 03 DB 3
3757 123B 07 DB 7
3758 123C 06 DB 6
3759 123D 08 DB 8
3760 123E 07 DB 7
3761 123F 00 DB 0
3762 1240 05 DB 5
3763 1241 01 DB 1
3764 1242 00 DB 0
3765 ;
3766 KSTKTBL:
3767 1243 00 DB 0
3768 1244 03 DB 3
3769 1245 05 DB 5
3770 1246 04 DB 4
3771 1247 01 DB 1
3772 1248 02 DB 2
3773 1249 00 DB 0
3774 124A 03 DB 3
3775 124B 07 DB 7
3776 ;RLBF
3777 ;RLB
3778 ;RL F
3779 ;RL
3780 ;R BF
3781 ;R B
3782 ;R F
3783 ;R
3784 ; LBF
3785 ; LB
3786 ; L F
3787 ; L
3788 ; BF
3789 ; B
3790 ; F
3791 ;
3792 ;RBFL
3793 ;RBF
3794 ;RB L
3795 ;RB
3796 ;R FL
3797 ;R F
3798 ;R L
3799 ;R
3800 ; BFL

```

```
3775 124C 00 DB
3776 124D 06 DB
3777 124E 05 DB
3778 124F 08 DB
3779 1250 01 DB
3780 1251 07 DB
3781 1252 00 DB
3782
3783 GTTRIG:
3784 ;
3785 1253 3D DEC
3786 1254 FA 126C JP M,KEYTRG
3787 1257 F5 PUSH AF
3788 1258 E6 01 AND 1
3789 125A CD 120C CALL SLSTCK
3790 125D C1 POP BC
3791 125E 05 DEC B
3792 125F 05 DEC B
3793 1260 06 10 LD B,10H
3794 1262 FA 1267 JP M,TRIG1
3795 1265 06 20 LD B,' '
3796 1267 TRIG1:
3797 1267 A0 AND B
3798 1268 TRIG2:
3799 1268 D6 01 SUB 1
3800 126A 9F SBC A,A
3801 126B C9 RET
3802 126C KEYTRG:
3803 ;
3804 126C CD 1226 CALL GTROW8
3805 126F E6 01 AND 1

; BF
; B L
; B
; FL
; F
; L
;
;STRIG(0), use keyboard
;Read joystick
;Prepare mask pattern for trigger A
;Prepare mask pattern for trigger B
;Extract trigger status
;Return 255 if [Acc]=0, 0 if non-0
;Read keyboard
;Extract space status
```

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85 PAGE 42-4
 - MSXIO - Joystick and Paddle interface

```

3806      1271      18 F5      JR      TRIG2
3807      1273      GTPDL:
3808      ;
3809      ; Get value of paddle
3810      ;
3811      ; Input parameters (passed via [Acc])
3812      ;
3813      ; 1 - Paddle A connected to joystick port 1
3814      ; 2 - Paddle A connected to joystick port 2
3815      ; 3 - Paddle B connected to joystick port 1
3816      ; 4 - Paddle B connected to joystick port 2
3817      ; 5 - Paddle C connected to joystick port 1
3818      ; 6 - Paddle C connected to joystick port 2
3819      ; 7 - Paddle D connected to joystick port 1
3820      ; 8 - Paddle D connected to joystick port 2
3821      ; 9 - Paddle E connected to joystick port 1
3822      ; 10 - Paddle E connected to joystick port 2
3823      ; 11 - Paddle F connected to joystick port 1
3824      ; 12 - Paddle F connected to joystick port 2
3825      ;
3826      1273      3C      INC      A      ;Force parameter 2 based
3827      1274      A7      AND      A
3828      1275      1F      RRA
3829      1276      F5      PUSH     AF
3830      1277      47      LD       B,A
3831      1278      AF      XOR      A
3832      1279      37      SCF
3833      127A      PDL1:
3834      127A      17      RLA
3835      127B      10 FD   DJNZ     PDL1      ;Form mask pattern
3836      127D      47      LD       B,A      ;Set mask pattern

```

3837	127E	F1	POP	A,F	
3838	127F	0E 10	LD	C,10H	
3839	1281	11 03AF	LD	DE,03AFH	;Assume port 1
3840	1284	30 05	JR	NC,PDLPl	
3841	1286	0E 20	LD	C,' '	;Good assumption
3842	1288	11 4C9F	LD	DE,4C9FH	
3843	128B				
			PDLPl:		
3844	128B	3E 0F	LD	A,PSG.PB	
3845	128D	F3	DI		
3846	128E	CD 110E	CALL	RDPSPG	
3847	1291	A3	AND	E	;Get current port B content
3848	1292	B2	OR	D	
3849	1293	B1	OR	C	
3850	1294	D3 A1	OUT	(PSG.DW),A	;Set trigger high
3851	1296	A9	XOR	C	
3852	1297	D3 A1	OUT	(PSG.DW),A	;Set trigger low again
3853	1299	3E 0E	LD	A,0EH	
3854	129B	D3 A0	OUT	(PSG.LW),A	
3855	129D	0E 00	LD	C,0	;Initialize counter
3856	129F		PDL2:		
3857	129F	DB A2	IN	A,(PSG.DR)	
3858	12A1	A0	AND	B	;End of pulse?
3859	12A2	28 05	JR	Z,PDL3	;Yes
3860	12A4	0C	INC	C	;Bump counter
3861	12A5	C2 129F	JP	NZ,PDL2	;No overflow yet
3862	12A8	0D	DEC	C	;Make it 255
3863	12A9				
			PDL3:		
3864	12A9	FB	EI		
3865	12AA	79	LD	A,C	;Return counted value
3866	12AB	C9	RET		
3867	12AC				
			GTPAD:		

```

3868 ;
3869 ; Read touch pad (NEC PC-6051 compatible)
3870 ;
3871 ; Input parameter (passed via [Acc])
3872 ;
3873 ; 0 - sense touch pad status ---
3874 ; 1 - return X coordinate      |for touch pad connected
3875 ; 2 - return Y coordinate      |to joystick port 1
3876 ; 3 - return switch status ----
3877 ;
3878 ; 4 - sense touch pad status ---
3879 ; 5 - return X coordinate      |for touch pad connected
3880 ; 6 - return Y coordinate      |to joystick port 2
3881 ; 7 - return switch status ----
3882 ;
3883 ; Result is returned via [Acc]. As for status, 255 is returned
3884 ; if true, 0 if false.
3885 ;
3886 FE 04      CP      4      ;Read pad connected to port 1
3887 12AC      LD      DE,0CECH ;Assume so
3888 12AE      JR      C,GTPDP1  ;Good assumption
3889 38 05      LD      DE,03D3H ;Connected to port 2
3890 12B1      SUB     4
3891 12B6      GTPDP1:
3892 12B8      DEC     A      ;Argument=0?
3893 12B9      JP      M,GTPAD0 ;If so, read pad and return status
3894 12BC      DEC     A
3895 12BD      LD      A,(PADX) ;Assume PAD(1) - X coordinate
3896 12C0      RET     M      ;Good assumption
3897 12C1      LD      A,(PADY) ;Return Y coordinate
3898 12C4      RET     Z

```

```
3899 12C5      GTPAD0:      PUSH      AF
3900 12C5      F5           EX        DE,HL
3901 12C6      EB           LD        (RUNFLG),HL
3902 12C7      22 F866      SBC       A,A
3903 12CA      9F          CPL
3904 12CB      2F          AND
3905 12CC      E6 40       LD
3906 12CE      4F          LD
3907 12CF      3E 0F       DI
3908 12D1      F3          CALL
3909 12D2      CD 110E     AND
3910 12D5      E6 BF      OR
3911 12D7      B1          OUT
3912 12D8      D3 A1       POP
3913 12DA      F1          JP
3914 12DB      FA 12E8     CALL
3915 12DE      CD 110C     EI
3916 12E1      FB          AND
3917 12E2      E6 08       SUB
3918 12E4      D6 01       SBC
3919 12E6      9F          RET
3920 12E7      C9
3921 12E8
3922          TRYAGN:
3923          ;
3924          LD            C,0
3925          12EA          REDPAD
3926          12ED          REDPAD
3927          12F0          C,PADX1
3928          12F2          REDCOD
3929          12F5          C,PADX1
3930          12F7          DE
3931          PUSH
3932          ;
3933          ;inz
3934          ;sense Panel input and select X
3935          ;branch if no input
3936          ;read first coordinate
3937          ;branch if input released
3938          ;save for comparison
```

	(MSX ROM BASIC BIOS)	Macro-80	3.44	01-Jan-85	PAGE	42-8
- MSXIO - Joystick and Paddle interface						
3930	12F8	CD 1320	CALL	REDCOD		
3931	12FB	C1	POP	BC		;read another input
3932	12FC	38 1C	JR	C,PADXL		;restore previous coord
3933	12FE	78	LD	A,B		;branch if input released
3934	12FF	92	SUB	D		
3935	1300	30 02	JR	NC,NONEGL		;[A]=ABS(X0-X1)
3936	1302	2F	CPL			
3937	1303	3C	INC	A		
3938	1304					
3939	1304	FE 05	CP	5		;less than 5?
3940	1306	30 E0	JR	NC,TRYAGN		;no, try again
3941	1308	79	LD	A,C		
3942	1309	93	SUB	E		;[A]=ABS(Y0-Y1)
3943	130A	30 02	JR	NC,NONEG2		
3944	130C	2F	CPL			
3945	130D	3C	INC	A		
3946	130E					
3947	130E	FE 05	CP	5		;less than 5
3948	1310	30 D6	JR	NC,TRYAGN		;no, try again
3949	1312	7A	LD	A,D		
3950	1313	32 FC9D	LD	(PADX),A		;update coordinate [X]
3951	1316	7B	LD	A,E		
3952	1317	32 FC9C	LD	(PADY),A		;update coordinate [Y]
3953	131A					
3954	131A	FB	EI			;finally enable interrupt
3955	131B	7C	LD	A,H		;get SENSE input value
3956	131C	D6 01	SUB	1		
3957	131E	9F	SBC	A,A		
3958	131F	C9	RET			;return value
3959	1320					
3960						

```
3961      ; Read X,Y coordinate into [D,E]
3962      ;
3963      LD      0E 0A      C,0AH      ;change to channel to [Y] when done
3964      CD 1332      REDPAD      ;read [X]
3965      D8      RET      C      ;return if input released
3966      55      LD      D,L
3967      D5      PUSH      DE
3968      0E 00      LD      C,0      ;change to [X] after read
3969      CD 1332      CALL      REDPAD      ;read [Y]
3970      D1      POP      DE
3971      5D      LD      E,L      ;store Y read out
3972      AF      XOR      A      ;clear carry
3973      67      LD      H,A      ;force input is OK
3974      C9      RET
3975      REDPAD:
3976      ;
3977      ; Read touch panel input into [L]
3978      ; Carry set if input released during read
3979      ;
3980      CD 135B      CALL      CHKEOC      ;make sure AD completed
3981      06 08      LD      B,8      ;input 8 bits
3982      51      LD      D,C      ;input channel# after done
3983      REDLOP:
3984      CB 82      RES      0,D      ;serial clock(SCK)=1
3985      CB 92      RES      2,D
3986      CD 136D      CALL      OUTGI
3987      CD 110C      CALL      INGI
3988      67      LD      H,A      ;read PAD
3989      1F      RRA      ;save SENSE status
3990      1F      RRA
3991      1F      RRA
```

```

( MSX ROM BASIC BIOS ) Macro-80      3.44      01-Jan-85      PAGE      42-10
- MSXIO - Joystick and Paddle interface

3992      1346      CB 15      RL      L
3993      1348      CB C2      SET      0,D
3994      134A      CB D2      SET      2,D
3995      134C      CD 136D      CALL     OUTGI
3996      134F      10 E7      DJNZ     REDLOP
3997      1351      CB E2      SET      4,D
3998      1353      CB EA      SET      5,D
3999      1355      CD 136D      CALL     OUTGI
4000      1358      7C          LD       A,H
4001      1359      1F          RRA
4002      135A      C9          RET
4003      135B
4004
4005
4006
4007      135B      3E 35      LD       A,00110101B
4008      135D      B1          OR       C
4009      135E      57          LD       D,A
4010      135F      CD 136D      CALL     OUTGI
4011      1362
4012      1362      CD 110C      CALL     INGI
4013      1365      E6 02      AND      2
4014      1367      28 F9      JR       Z,EOCHK
4015      1369      CB A2      RES      4,D
4016      136B      CB AA      RES      5,D
4017      136D
4018
4019
4020
4021      136D      E5          PUSH     HL
4022      136E      D5          PUSH     DE

;
; Output [D] to PAD
;
;
; initiate another AD
; LSB=SENSE status
; SENSE status to carry
; OK if no carry

CHKEOC:
;
; Check and wait for EOC
;
;
EOCHK:
CALL 110C
AND 2
JR Z,EOCHK
RES 4,D
RES 5,D
OUTGI:
;
;
; test EOC
; set CS and return

```

4023	136F	2A F866	LD	HL, (RUNFLG)	
4024	1372	7D	LD	A,L	
4025	1373	2F	CPL		
4026	1374	A2	AND		
4027	1375	57	LD	D	
4028	1376	3E 0F	LD	D,A	
4029	1378	D3 A0	LD	A,PSG.PB	
4030	137A	DB A2	OUT	(PSG.LW),A	
4031	137C	A5	IN	A,(PSG.DR)	
4032	137D	B2	AND	L	
4033	137E	B4	OR	D	
4034	137F	D3 A1	OR	H	
4035	1381	D1	OUT	(PSG.DW),A	
4036	1382	E1	POP	DE	
4037	1383	C9	POP	HL	
4038			RET		
4039					

; SUBTTL - MSXIO - Misc. routines for MSXIO

(MSX ROM BASIC BIOS) Macro-80
- MSXIO - Misc. routines for MSXIO

[illegible]

```
4063
4064
4065      139D      ;
4066      ; INIFNK:
4067      ;
4068      ; Initialize function key strings
4069      ;
4070      139D      01 00A0      LD      BC,0A0H
4071      13A0      11 F87F      LD      DE,FNKSTR
4072      13A3      21 13A9      LD      HL,FKTABL
4073      13A6      ED B0      LDIR
4074      13A8      C9      RET
4075      ;
4076      13A9      63 6F 6C 6F      FNKDEF: DB      "color "
4077      13AD      72 20
4078      13AF      61 75 74 6F      DS      10
4079      13B9      20      DB      "auto "
4080      13BD      67 6F 74 6F      DS      11
4081      13BE      20      DB      "goto "
4082      13C9      6C 69 73 74      DS      11
4083      13CD      20      DB      "list "
4084      13CE      72 75 6E      DS      11
4085      13D9      0D      DB      "run"
4086      13DD      63 6F 6C 6F      DS      13
4087      13DE      72 20 31 35      DS      12
4088      13E9      2C 34 2C 37      DB      "color 15,4,7"
4089      13EC
4090      13ED
4091      13F9
4092      13FD
4093      1401
```

4094	1405	0D		DB	13
4095	1406			DS	3
4096	1409	63 6C 6F 61		DB	"cload"
4097	140D	64			
4098	140E	22		DB	34
4099	140F			DS	10
4100	1419	63 6F 6E 74		DB	"cont"
4101	141D	0D		DB	13
4102	141E			DS	11
4103	1429	6C 69 73 74		DB	"list."
4104	142D	2E			
4105	142E	0D 1E 1E		DB	13,30,30
4106	1431			DS	8
4107	1439	0C		DB	12
4108	143A	72 75 6E		DB	"run"
4109	143D	0D		DB	13
4110	143E			DS	11
4111			;		
4112	1449		RDVDP:		
4113			;		
4114	1449	DB 99		IN	A,(VDP.SR)
4115	144B	C9		RET	
4116	144C		RSLREG:		
4117			;		
4118	144C	DB A8		IN	A,(PPI.AR)
4119	144E	C9		RET	
4120	144F		WSLREG:		
4121			;		
4122	144F	D3 A8		OUT	(PPI.AW),A
4123	1451	C9		RET	
4124	1452		SNSMAT:		

```
4125 ;
4126 1452 4F LD C,A
4127 1453 F3 DI
4128 1454 DB AA IN A,(PPI.CR) ;Get what is currently output to Port C
4129 1456 E6 F0 AND 0F0H ;Leave higher 4 bits unaffected
4130 1458 81 ADD A,C
4131 1459 D3 AA OUT (PPI.CW),A ;Select row
4132 145B DB A9 IN A,(PPI.BR) ;Get column information of selected row
4133 145D FB EI
4134 145E C9 RET
4135 145F
4136
4137
4138 ;
4139 145F CD FEDF CALL H.ISFL
4140 1462 E5 PUSH HL ;Save [H,L]
4141 1463 2A F864 LD HL,(PTRFIL) ;Get file pointer
4142 1466 7D LD A,L
4143 1467 B4 OR H ;No zero?
4144 1468 E1 POP HL ;Restore [H,L]
4145 1469 C9 RET
4146 146A
4147
4148 ;
4149 ; COMPAR compares [H,L] with [D,E] unsigned
4150 ;
4151 ; [H,L] less than [D,E] set carry
4152 ; [H,L] = [D,E] set zero
4153 ;
4154 ; [A] is the only register used
4155 ;
4156 146A 7C LD A,H
```


(MSX ROM BASIC BIOS) Macro-80		3.44	01-Jan-85	PAGE	44-4
- MSXIO - Misc. routines for MSXIO					
4187	147E B7	OR	A		
4188	147F 28 07	JR	Z,GETVCX		
4189	1481 11 0025	LD	DE,25H		;VCB size
4190	1484	GETVCL:			
4191	1484 19	ADD	HL,DE		
4192	1485 3D	DEC	A		
4193	1486 20 FC	JR	NZ,GETVCL		
4194	1488	GETVCX:			
4195	1488 D1	POP	DE		
4196	1489 C9	RET			
4197	148A	PHYDIO:			
4198		;			
4199	148A CD FFA7	CALL	H.PHYD		
4200	148D C9	RET			
4201	148E	FORMAT:			
4202		;			
4203	148E CD FFAC	CALL	H.FORM		
4204	1491 C9	RET			
4205		SUBTTL - QUEUTL - Queue utility routines			

```
4206 ; Copyright (C) 1980 by Microsoft Corporation
4207 ; Written by Marc Wilson
4208 ;
4209 ; This utility provides for multiple queues with the following
4210 ; capabilities:
4211 ;
4212 ; Queues of varying length - 1,3,7,15,31,63,127,255
4213 ;
4214 ; Each queue can be any of the possible lengths
4215 ; The queues can be initialized at any time and be
4216 ; located anywhere a single pointer (QUEUES) provides
4217 ; the address of the queue table.
4218 ;
4219 ; The queue table has all information for each queue,
4220 ; 6 bytes per queue. A single non-zero character can
4221 ; be pushed back on top of the queue.
4222 ;
4223 ; The entry for each queue is as follows:
4224 ; +0 PUT OFFSET
4225 ; +1 GET OFFSET
4226 ; +2 BACK CHARACTER
4227 ; +3 QUEUE LENGTH
4228 ; +4,+5 QUEUE ADDRESS
4229 ;
4230 ; The utility assumes that the queue table is
4231 ; valid for all queue numbers passed to the routines
4232 ;
4233 ; ROUTINES:
4234 ; All routines assume that [A] equals the queue number,
4235 ; [QUEUES] contains the address of the queue table.
4236
```

```
4237 ; Other requirements follow.
4238 ; GETQ - Returns current top of queue in [A],
4239 ;       zero flag set if queue empty
4240 ; PUTQ - Puts byte in [E] reg on end of queue,
4241 ;       zero set if queue is full
4242 ;
4243 ;NOTE:
4244 ; The routines are designed to be reentrant, however
4245 ; there are some restrictions for cases involving a
4246 ; single queue (in any case operating on different
4247 ; queues is alright). The first restriction is that
4248 ; the same routine cannot be reentered. The second
4249 ; is that INITQ and POPQ do not allow PUTQ,
4250 ; GETQ or BCKQ to be entered.
4251 ;
4252 ; LFTQ - Returns unused number of bytes in queue in [A] reg
4253 ; INITQ - Initialize queue to empty state,
4254 ;       B reg=length, (DE)=ADDR
4255 ; *** All routines destroy the registers ***
4256 ;
4257 SUBTTL - QUEUTL - Queue routines
```

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149A
149B
149C
149D
149E
149F
14A0
14A1
14A2
14A3
14A4
14A5
14A6
14A8
14A9
14AA
14AB
14AC
14AD

CD 14FA
78
3C
23
A6
B9
C8
E5
2B
2B
2B
E3
23
4F
7E
23
66
6F
06 00
09
73
E1
71
C9

PUTQ:
;
; Put data on queue
;
CALL GETPTR
LD A,B
INC A
INC HL
AND (HL)
CP C
RET Z
PUSH HL
DEC HL
DEC HL
DEC HL
EX (SP),HL
INC HL
LD C,A
LD A,(HL)
INC HL
LD H,(HL)
LD L,A
LD B,0
ADD HL,BC
LD (HL),E
POP HL
LD (HL),C
RET

;Get queue pointers
;Bump PUT
;Wrap around
;QUEUE full

;Save place to put new pointer
;Pointer in C

;(HL) = QUEUE address
;(HL) = Address to put char

;set new pointer

GETQ:
;
```

```

4289      ; Get data from QUEUE
4290      ;
4291      14AD      CD 14FA      CALL      GETPTR
4292      14B0      36 00      LD      (HL),0
4293      14B2      20 1D      JR      NZ,GETBAK
4294      14B4      79      LD      A,C
4295      14B5      B8      CP      B
4296      14B6      C8      RET      Z
4297      14B7      23      INC     HL
4298      14B8      3C      INC     A
4299      14B9      A6      AND     (HL)
4300      14BA      2B      DEC     HL
4301      14BB      2B      DEC     HL
4302      14BC      E5      PUSH    HL
4303      14BD      23      INC     HL
4304      14BE      23      INC     HL
4305      14BF      23      INC     HL
4306      14C0      4F      LD      C,A
4307      14C1      7E      LD      A,(HL)
4308      14C2      23      INC     HL
4309      14C3      66      LD      H,(HL)
4310      14C4      6F      LD      L,A
4311      14C5      06 00      LD      B,0
4312      14C7      09      ADD     HL,BC
4313      14C8      7E      LD      A,(HL)
4314      14C9      E1      POP     HL
4315      14CA      71      LD      (HL),C
4316      14CB      B7      OR      A
4317      14CC      C0      RET     NZ
4318      14CD      3C      INC     A
4319      14CE      3E 00      LD      A,0

```

;Get queue pointers
 ;zero back character

 ;QUEUE empty!
 ;Bump GET offset
 ;wrap around

 ;Save place to store pointer

 ;offset in C

 ;[HL] = QUEUE address

 ;get char from QUEUE

- QUEUTL - Queue routines

```

4320 14D0 C9
4321 14D1
4322 14D1 4F
4323 14D2 06 00
4324 14D4 21 F970
4325 14D7 09
4326 14D8 7E
4327 14D9 C9
4328 14DA
4329
4330
4331
4332
4333
4334
4335
4336
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4338
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4349
4350

      GETBAK:
      LD C,A
      LD B,0
      LD HL,QUEBAK-1
      ADD HL,BC
      LD A,(HL)
      RET

      INITQ:
      ;
      ; INITQ - Initialize QUEUE
      ;
      PUSH BC
      CALL QSTART
      LD (HL),B
      INC HL
      LD (HL),B
      INC HL
      LD (HL),B
      INC HL
      POP AF
      LD (HL),A
      INC HL
      LD (HL),E
      INC HL
      LD (HL),D
      RET

      LFTQ:
      ;
      ; LFTQ - Returns number of bytes remaining in QUEUE
      ;

```

;Save queue length

;Get addr of start of QUEUE table entry

;Clear PUT offset

;Clear GET offset

;Clear back character

;Set QUEUE length

;Set QUEUE address

; LFTQ - Returns number of bytes remaining in QUEUE

```
4351 14EB CD 14FA
4352 14EE 78
4353 14EF 3C
4354 14F0 23
4355 14F1 A6
4356 14F2 47
4357 14F3 79
4358 14F4 90
4359 14F5 A6
4360 14F6 6F
4361 14F7 26 00
4362 14F9 C9
4363
4364 14FA
4365
4366
4367
4368 14FA CD 1504
4369 14FD 46
4370 14FE 23
4371 14FF 4E
4372 1500 23
4373 1501 7E
4374 1502 B7
4375 1503 C9
4376
4377 1504
4378 1504 07
4379 1505 47
4380 1506 07
4381 1507 80

;Get QUEUE ptrs
;B=PUT PTR+1
;subtract PUT from GET
;make it positive UNSIGNED INTEGER

GETPTR:
;
; QUEUE general routines
;
CALL QSTART
LD B,(HL)
INC HL
LD C,(HL)
INC HL
LD A,(HL)
OR A
RET

;
QSTART:
RLCA
LD B,A
RLCA
ADD A,B

;*2
;*4
;*6
```

(MSX ROM BASIC BIOS) Macro-80	3.44	01-Jan-85	PAGE	46-4
- QUEUTL - Queue routines				
4382 1508 4F	LD	C,A		
4383 1509 06 00	LD	B,0		
4384 150B 2A F3F3	LD	HL,(QUEUES)		
4385 150E 09	ADD	HL,BC		
4386 150F C9	RET			
4387	SUBTTL - MSXGRP - Graphic driver (Print a character on GRP screen)			


```

( MSX ROM BASIC BIOS ) Macro-80      3.44      01-Jan-85      PAGE      47-1
- MSXGRP - Graphic driver (Print a character on GRP screen)

4419      1546      E5      PUSH      HL      ;Save these
4420      1547      F5      PUSH      AF
4421      1548      1A      LD        A,(DE)  ;Get pattern for a row
4422      1549      87      ADD       A,A     ;Check each bit
4423      1549      F5      PUSH      AF
4424      154A      DC 167E  CALL     C,SETC ;Set it if 1
4425      154B      CD 16AC  CALL     TRIGHT ;Move 1 pixel right
4426      154E      E1      POP       HL      ;Assume out of screen
4427      1551      38 04   JR        C,GPRT30 ;Good assumption, skip the rest
4428      1552      E5      PUSH      HL
4429      1554      F1      POP       AF
4430      1555      10 F1   DJNZ     GPRT20 ;Loop till done all columns
4431      1556      F1      POP       AF
4432      1558      F1      POP       HL      ;Restore CLOC and CMASK
4433      1558      E1      CALL     STOREC ;Set these
4434      1559      CD 1640  CALL     TDOWNC ;Move 1 pixel down
4435      155A      CD 170A  CALL     C,GPRT40 ;Out of screen, skip rest and return
4436      155D      38 04   JR        DE     ;Point to next row
4437      1560      13      DEC       C
4438      1562      0D      JR        NZ,GPRT10 ;Loop till done all rows
4439      1563      20 DB   CALL     CHKMOD ;Check current screen mode
4440      1564      CD 15D9  LD        A,(GRPACX) ;We're in high-resolution mode
4441      1566      3A FCB7  JR        Z,GPRT50 ;We're going out of screen
4442      1569      28 06   ADD       A,' '
4443      156C      C6 20   JR        C,GRPCR
4444      156E      38 0C   JR        GPRT60
4445      1570      18 04   JR
4446      1572      1574
4447      1574
4448      1574
4449      1574

```

```

4450 1574 C6 08      ADD      A,8
4451 1576 38 06      JR        C,GRPCR
4452 1578                      GPRT60:
4453 1578 32 FCB7      LD        (GRPACX),A
4454 157B                      JPPPAL:
4455 157B C3 08DA      JP        POPALL
4456 157E                      GRPCR:
4457 157E                      ;
4458 157E AF          XOR        A
4459 157F 32 FCB7      LD        (GRPACX),A
4460 1582 CD 15D9      CALL      CHKMOD
4461 1585 3A FCB9      LD        A,(GRPACY)
4462 1588 28 03      JR        Z,GPRT70
4463 158A C6 20      ADD        A,4*8
4464 158C 01          DB        1
4465 158D                      GPRT70:
4466 158D C6 08      ADD        A,8
4467 158F FE C0      CP        0C0H
4468 1591 38 01      JR        C,GPRT80
4469 1593 AF          XOR        A
4470 1594                      GPRT80:
4471 1594 32 FCB9      LD        (GRPACY),A
4472 1597 18 E2      JR        JPPPAL
4473                      SUBTTL - MSXGRP - (Routines for general graphics)
;Update cursor position
;Reset X position
;Reset Y position also

```

```

( MSX ROM BASIC BIOS ) Macro-80      3.44    01-Jan-85      PAGE    48
- MSXGRP - (Routines for general graphics)

4474
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4500
4501
4502
4503
4504

1599
1599
159A
159B
159D
159E
159F
15A0
15A2
15A5
15A7
15A7
15AA
15AB
15AD
15AE
15AF
15AF
15B1
15B1
15B2

SCALXY:
;
; SCALXY - Clips X,Y to max values in physical size and flags out
; of range values.
;
; ENTRY [BC] = X (0 ... max X), [DE] = Y (0 ... max Y)
; EXIT [BC] = X clipped, [DE] = Y clipped
; CARRY is reset if one of the value was out of bound
;
;
; PUSH HL
; PUSH BC
; LD B,1
; EX DE,HL
; LD A,H
; ADD A,A
; JR NC,YPOSTV
; LD HL,0
; JR YNEGTV
;
; YPOSTV:
;
; LD DE,0C0H
; RST 20H
; JR C,SCLYOK
; EX DE,HL
; DEC HL
;
; YNEGTV:
; LD B,0
;
; SCLYOK:
; EX (SP),HL
; LD A,H
;
; save [HL]
; save [BC] - X coordinate
; no-error flag
; Y coordinate to [HL]
; Is Y coordinate negative?
; No, positive
; Substitute by 0 is negative
; And set out of bound flag
; Maximum Y+1
; Test [HL] with [DE]
; if carry, not out of bound
; [HL] = 192
; Y = 191, maximum Y coordinate
; set out of bound flag
; save Y and get X to [HL]
; Is X coordinate negative?

```

```

( MSX ROM BASIC BIOS ) Macro-80      3.44      01-Jan-85      PAGE      48-1
- MSXGRP - (Routines for general graphics)

4505 15B3 87      ADD      A,A
4506 15B4 30 05      JR      NC,XPOSTV
4507 15B6 21 0000      LD      HL,0
4508 15B9 18 08      JR      XNEGTV
4509 15BB      XPOSTV:
4510      ;
4511 15BB 11 0100      LD      DE,0100H
4512 15BE E7      RST      20H
4513 15BF 38 04      JR      C,SCLXOK
4514 15C1 EB      EX      DE,HL
4515 15C2 2B      DEC      HL
4516 15C3      XNEGTV:
4517 15C3 06 00      LD      B,0
4518 15C5      SCLXOK:
4519 15C5 D1      POP      DE
4520 15C6 CD 15D9      CALL  CHKMOD
4521 15C9 28 08      JR      Z,HRSSCL
4522 15CB CB 3D      SRL      L
4523 15CD CB 3D      SRL      L
4524 15CF CB 3B      SRL      E
4525 15D1 CB 3B      SRL      E
4526 15D3      HRSSCL:
4527 15D3 78      LD      A,B
4528 15D4 0F      RRCA
4529 15D5 44      LD      B,H
4530 15D6 4D      LD      C,L
4531 15D7 E1      POP      HL
4532 15D8 C9      RET
4533 15D9      CHKMOD:
4534      ;
4535      ; Check current screen mode
;No, positive
;Substitute by 0 if negative
;And set out of bound flag

;max X +1
;Test [HL] with [DE]

;[HL] = 256
;[HL] = 255 - max X coordinate

;error flag

;restore [DE] = Y

;We're in high-resolution mode
;Divide both X and Y by 4 because we're
;in multi-color mode

;set carry if no error
;[BC] = X

;restore [HL]

```

(MSX ROM BASIC BIOS) Macro-80
 - MSXGRP - (Routines for general graphics)

```

4536 ;
4537 15D9 3A FCAF LD A,(SCRMOD)
4538 15DC D6 02 SUB 2
4539 15DE C9 RET
4540 15DF MAPXYC:
4541 ;
4542 ; MAPXYC - Maps X,Y coordinates to "C" (address, mask)
4543 ;
4544 ; Entry: [BC] = X, [DE] = Y
4545 ;
4546 ; Exit: CLOC = [HL] -- Video Ram address
4547 ; CMASK = [A] -- Bit Mask
4548 ;
4549 ; [ High-resolution mode ]
4550 ;
4551 ; X coord - XXXXXXXX ( 8 bits, max=255)
4552 ; 76543210
4553 ;
4554 ; Y coord - YYYYYYYY ( 8 bits, max=191)
4555 ; 76543210
4556 ;
4557 ; CLOC = YYYYYXXXXXXYY
4558 ; 7654376543210
4559 ; XXX
4560 ; 210
4561 ; -----
4562 ; CMASK = 10000000 000
4563 ; 01000000 001
4564 ; 00100000 010
4565 ; 00010000 011
4566 ; 00001000 100

```

;In what mode are we now?
 ;Return with the condition flag

```

4567 ;
4568 ;
4569 ;
4570 ;
4571 ; [ Multi-color mode ]
4572 ;
4573 ; X coord - XXXXXX ( 6 bits, max=63 )
4574 ; 543210
4575 ;
4576 ; Y coord - YYYYYY ( 6 bits, max=47 )
4577 ; 543210
4578 ;
4579 ; CLOC = YYYYYYYYYY
4580 ; 54354321210
4581 ;
4582 ; CMASK = 11110000 if X0=0 (even)
4583 ; CMASK = 00001111 if X0=1 (odd)
4584 ;
4585 ; Note: The boundary check has already been done by a call
4586 ; to SCALXY, so no range checking is needed.
4587 ;
4588 ;
4589 ;
4590 ;
4591 ;
4592 ;
4593 ;
4594 ;
4595 ;
4596 ;
4597 ;

```

15DF	C5	PUSH	BC	;Save X
15E0	CD 15D9	CALL	CHKMOD	;Check current screen mode
15E3	20 2E	JR	NZ,MMPXYC	;Multi-color mode
15E5	51	LD	D,C	;Save X to D also
15E6	79	LD	A,C	
15E7	E6 07	AND	7	
15E9	4F	LD	C,A	
15EA	21 160B	LD	HL,TWOPWR	;Table of power of two
15ED	09	ADD	HL,BC	
15EE	7E	LD	A,(HL)	;read bit mask CMASK

```

( MSX ROM BASIC BIOS ) Macro-80      3.44    01-Jan-85    PAGE    48-4
- MSXGRP - (Routines for general graphics)

4598      15EF      32 F92C      LD      (CMASK),A
4599      15F2      7B          LD      A,E
4600      15F3      0F          RRCA
4601      15F4      0F          RRCA
4602      15F5      0F          RRCA
4603      15F6      E6 1F      AND     00011111B
4604      15F8      47          LD      B,A
4605      15F9      7A          LD      A,D
4606      15FA      E6 F8      AND     11111000B
4607      15FC      4F          LD      C,A
4608      15FD      7B          LD      A,E
4609      15FE      E6 07      AND     00000111B
4610      1600      B1          OR      C
4611      1601      4F          LD      C,A
4612      1602      2A F3CB      LD      HL,(GRPCGP)
4613      1605      09          ADD     HL,BC
4614      1606      22 F92A      LD      (CLOC),HL
4615      1609      C1          POP     BC
4616      160A      C9          RET
4617      160B      TWOPWR:
4618      ;
4619      ; Table of power of two
4620      ;
4621      80 40 20 10      DB      80H,40H,20H,10H
4622      08 04 02 01      DB      08H,04H,02H,01H
4623      ;
4624      1613      MMPXYC:
4625      ;
4626      ; Map XY for multi-color mode
4627      ;
4628      1613      79          LD      A,C
                                ;Get X position

```


(MSX ROM BASIC BIOS) Macro-80 3.44
 - MSXGRP - (Routines for general graphics)

4686	1655	A0	AND	B		
4687	1656	F5	PUSH	AF		;Extract specified pixel
4688	1657	01 2000	LD	BC,GRPDIF		;Save whether the pixel is on or off
4689	165A	09	ADD	HL,BC		
4690	165B	CD 07D7	CALL	RDVRM		;Read VDP's VRAM (color)
4691	165E	47	LD	B,A		;Save this to B
4692	165F	F1	POP	AF		;Restore condition
4693	1660	78	LD	A,B		;Restore color
4694	1661	28 04	JR	Z,READC1		;Specified dot is off, return
4695						;background color
4696	1663		READC0:			
4697	1663	0F	RRCA			;Specified dot is on, return foreground color
4698	1664	0F	RRCA			
4699	1665	0F	RRCA			
4700	1666	0F	RRCA			
4701	1667		READC1:			
4702	1667	E6 0F	AND	0FH		
4703	1669	E1	POP	HL		;Make it a legal value
4704	166A	C1	POP	BC		
4705	166B	C9	RET			
4706	166C		MREADC:			
4707			;			
4708	166C	CD 07D7	CALL	RDVRM		;Read VRAM
4709	166F	04	INC	B		;Check if specified pixel is even or odd
4710	1670	05	DEC	B		
4711	1671	F2 1667	JP	P,READC1		;Odd, return lower nibble
4712	1674	18 ED	JR	READC0		;Even, return upper nibble

- MSXGRP - (Routines for general graphics)

```

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4742
4743

1676
1677
1678
1679
167A
167B
167C
167D
167E

FE 10
3F
D8
32 F3F2
C9

CP 16
CCF
RET C
LD (ATTRBYT),A
RET

SETC:
;
; SETC - Sets the point indicated by the graphics accumulator
; to ATTRBYT
;
; All registers except AF must be preserved.
;
PUSH HL
PUSH BC
CALL CHKMOD
CALL FETCHC
JR NZ,MSETC
PUSH DE
CALL PATWRT
POP DE
POP BC
POP HL

167E
167F
1680
1683
1686
1688
1689
168C
168D
168E

E5
C5
CD 15D9
CD 1639
20 08
D5
CD 186C
D1
C1
E1

SETATR:
;
; SETATR - Sets the attribute (color, reverse, etc..) to be
; used in future actions.
;
; Entry - [A] = Attribute
; Exit - carry set if illegal value
;
FE 10
3F
D8
32 F3F2
C9

CP 16
CCF
RET C
LD (ATTRBYT),A
RET

SETC:
;
; SETC - Sets the point indicated by the graphics accumulator
; to ATTRBYT
;
; All registers except AF must be preserved.
;
PUSH HL
PUSH BC
CALL CHKMOD
CALL FETCHC
JR NZ,MSETC
PUSH DE
CALL PATWRT
POP DE
POP BC
POP HL

```

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85 PAGE 50-1
 - MSXGRP - (Routines for general graphics)

```

4744 168F C9 RET
4745 1690 MSETC:
4746 ;
4747 ; Set a pixel in multi-color mode
4748 ;
4749 1690 47 LD B,A ;Save CMASK in [B]
4750 1691 CD 07D7 RDVRM ;Read VRAM
4751 1694 4F LD C,A
4752 1695 78 LD A,B
4753 1696 2F CPL ;Leave another unaffected
4754 1697 A1 AND C
4755 1698 4F LD C,A
4756 1699 3A F3F2 LD A,(ATRYT) ;Get specified color
4757 169C 04 INC B ;Check if even or odd
4758 169D 05 DEC B
4759 169E F2 16A5 JP P,MSETC1 ;Odd
4760 16A1 87 ADD A,A
4761 16A2 87 ADD A,A
4762 16A3 87 ADD A,A
4763 16A4 87 ADD A,A
4764 16A5 MSETC1:
4765 16A5 B1 OR C ;Form new color
4766 16A6 CD 07CD CALL WRTVRM ;Write new pattern
4767 16A9 C1 POP BC
4768 16AA E1 POP HL
4769 16AB C9 RET
4770
SUBTTL - MSXGRP - (Graphic cursor movements)

```

(MSX ROM BASIC BIOS) Macro-80
 - MSXGRP - (Graphic cursor movements)

```

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;
; UPC, DOWNC, RIGHTC, LEFTC
;
; These are the C relative movement routines. They
; adjust the current graphics accumulator in the indicated
; direction without checking boundary conditions.
;
; -----
;
;
; TRIGHT:
;
; TRIGHT - move 1 pixel right
; Return carry set if already on border
;
;
; PUSH HL
; CALL CHKMOD
; JP NZ,MTRGT
; CALL FETCHC
; RRCA
; JR NC,HRZMV1
; LD A,L
; AND 0F8H
; CP 0F8H
; LD A,80H
; JR NZ,RGHTC1
; JP ONBRDI
;
; RIGHTC:
;
; RIGHTC - move 1 pixel right
;
;
;Get CLOC,CMASK
;Move 1 pixel right
;Within byte, just change CMASK
;Get low byte of CLOC
;On right edge?
;Assume not
;Goot assumption
;On border, set carry and return

```

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXGRP - (Graphic cursor movements)

3.44 01-Jan-85 PAGE 51-1

4802 16C5 E5          PUSH      HL
4803 16C6 CD 15D9     CALL      CHKMOD
4804 16C9 C2 178B     JP        NZ,MRGTC
4805 16CC CD 1639     CALL      FETCHC
4806 16CF 0F          RRCA
4807 16D0 30 32      JR        NC,HRZMV1
4808 16D2             RGHTCL:
4809 16D2 D5          PUSH      DE
4810 16D3 11 0008     LD        DE,8
4811 16D6 18 27      JR        HRZMOV
4812 16D8             TLEFT:
4813             ;
4814             ; TLEFT - move 1 pixel left
4815             ; Return carry set if already on border
4816             ;
4817 16D8 E5          PUSH      HL
4818 16D9 CD 15D9     CALL      CHKMOD
4819 16DC C2 179C     JP        NZ,MTLFT
4820 16DF CD 1639     CALL      FETCHC
4821 16E2 07          RLCA
4822 16E3 30 1F      JR        NC,HRZMV1
4823 16E5 7D          LD        A,L
4824 16E6 E6 F8      AND        0F8H
4825 16E8 3E 01      LD        A,1
4826 16EA 20 0F      JR        NZ,LEFTCL
4827 16EC 18 6C      JR        ONBRD1
4828 16EE             LEFTC:
4829             ;
4830             ; LEFTC - move 1 pixel left
4831             ;
4832 16EE E5          PUSH      HL

```

;move right 1 pixel
 ;within byte, just change CMASK

 ;Load offset to new position
 ;Change CLOC also

 ;Get CLOC and CMASK
 ;Move 1 pixel left
 ;Within byte boundary, just change CMASK
 ;Check if we're on left edge

 ;Assume not
 ;Good assumption
 ;We're on border, set carry and return

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXGRP - (Graphic cursor movements)

3.44 01-Jan-85 PAGE 51-2

4833 16EF CD 15D9          CALL      CHKMOD
4834 16F2 C2 17AC          JP        NZ,MLFTC
4835 16F5 CD 1639          CALL      FETCHC
4836 16F8 07              RLCA
4837 16F9 30 09           JR         NC,HRZMV1
4838 16FB          LEFTC1:
4839 16FB D5              PUSH      DE
4840 16FC 11 FFF8         LD        DE,0FFF8H
4841 16FF          HRZMOV:
4842 16FF 19              ADD       HL,DE
4843 1700 22 F92A         LD        (CLOC),HL
4844 1703 D1              POP       DE
4845 1704          HRZMV1:
4846 1704 32 F92C         LD        (CMASK),A
4847 1707 A7              AND       A
4848 1708 E1              POP       HL
4849 1709 C9              RET
4850 170A          TDOWNC:
4851          ;
4852          ; TDOWNC - move 1 pixel down.
4853          ;
4854          ; Return carry set if already on screen border.
4855          ;
4856 170A E5              PUSH      HL
4857 170B D5              PUSH      DE
4858 170C 2A F92A         LD        HL,(CLOC)
4859 170F CD 15D9          CALL      CHKMOD
4860 1712 C2 17C6         JP        NZ,MTDNC
4861 1715 E5              PUSH      HL
4862 1716 2A F3CB         LD        HL,(GRPCGP)
4863 1719 11 1700         LD        DE,1700H

```

;move left 1 pixel
;within byte boundary, just change CMASK

;Load offset to new position

;Add offset to new position
;Update pattern address

;Update CMASK
;Clear carry

```
4864 171C 19          ADD      HL,DE
4865 171D EB          EX       DE,HL
4866 171E E1          POP      HL
4867 171F E7          RST      20H
4868
4869 1720 38 13       JR       C,DWNC10
4870 1722 7D          LD       A,L
4871 1723 3C          INC      A
4872 1724 E6 07       AND      7
4873 1726 20 0D       JR       NZ,DWNC10
4874 1728 18 2F       JR       ONBRDR
4875
4876 172A             DOWNC:
4877
4878
4879
4880 172A E5           PUSH     HL
4881 172B D5           PUSH     DE
4882 172C 2A F92A      LD       HL,(CLOC)
4883 172F CD 15D9      CALL     CHKMOD
4884 1732 C2 17DC      JP       NZ,MDNC
4885 1735
4886 1735 23           INC      HL
4887 1736 7D          LD       A,L
4888 1737 11 00F8      LD       DE,0F8H
4889 173A 18 31       JR       VRTMOV
4890 173C             TUPC:
4891
4892
4893
4894
```

;
; Test [HL] with [DE]
; Looks like on border?
; No
; Possibly on border
; Really?
; No
; Yes, set carry and return
;

;
; DOWNC - move 1 pixel down
;

;
; move down 1 pixel
; Prepare for boundary check
; Load possible offset to new location
; Check

;
; TUPC - move 1 pixel up.
; Return carry set if already on screen border.
;

(MSX ROM BASIC BIOS) Macro-80	01-Jan-85	PAGE	51-4
- MSXGRP - (Graphic cursor movements)	3.44		
4895 173C E5	PUSH	HL	
4896 173D D5	PUSH	DE	
4897 173E 2A F92A	LD	HL,(CLOC)	
4898 1741 CD 15D9	CALL	CHKMOD	
4899 1744 C2 17E3	JP	NZ,MTUPC	
4900 1747 E5	PUSH	HL	
4901 1748 2A F3CB	LD	HL,(GRPCGP)	
4902 174B 11 0100	LD	DE,0100H	
4903 174E 19	ADD	HL,DE	
4904 174F EB	EX	DE,HL	
4905 1750 E1	POP	HL	
4906 1751 E7	RST	20H	
4907			;Test [HL] with [DE]
4908 1752 30 14	JR	NC,UPC10	;Looks like on border?
4909 1754 7D	LD	A,L	;No
4910 1755 E6 07	AND	7	;Possibly on border
4911 1757 20 0F	JR	NZ,UPC10	;Really?
4912	ONBRDR:		;No
4913 1759 D1	POP	DE	
4914 175A	ONBRD1:		
4915 175A 37	SCF		
4916 175B E1	POP	HL	
4917 175C C9	RET		
4918 175D	UPC:		
4919	; UPC - move 1 pixel up		
4920	; ;		
4921			
4922 175D E5	PUSH	HL	
4923 175E D5	PUSH	DE	
4924 175F 2A F92A	LD	HL,(CLOC)	
4925 1762 CD 15D9	CALL	CHKMOD	
			;get current position

```

4926      1765      C2 17F8
4927      1768
4928      1768      7D
4929      1769      2B
4930      176A      11 FF08
4931      176D
4932      176D      E6 07
4933      176F      20 01
4934      1771      19
4935      1772
4936      1772      22 F92A
4937      1775      A7
4938      1776      D1
4939      1777      E1
4940      1778      C9
4941      1779
4942
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4954
4955
4956

      1779      CD 1639
      177C      A7
      177D      3E 0F
      177F      FA 17C0
      1782      7D
      1783      E6 F8
      1785      FE F8
      1787      20 0B
      1789      18 CF
      178B

      1765      NZ,MUPC
      1768      A,L
      1769      HL
      176A      DE,0FF08H
      176D      VRTMOV:
      176D      AND
      176F      JR NZ,VRTMV1
      1771      ADD HL,DE
      1772      VRTMV1:
      1772      LD (CLOC),HL
      1775      AND A
      1776      POP DE
      1777      POP HL
      1778      RET
      1779      MTRGT:
      ;
      ; Graphics cursor movement in multi-color mode
      ; [ Horizontal movements ]
      ;
      1779      CALL FETCHC
      177C      AND A
      177D      LD A,0FH
      177F      JP M,MH2MV1
      1782      LD A,L
      1783      AND 0F8H
      1785      CP 0F8H
      1787      JR NZ,MRGTCL
      1789      JR ONBRD1
      178B      MRGTC:
      ;

      1768      ;Prepare for boundary check
      1769      ;move up 1 pixel
      176A      ;Load possible offset to new location
      176D      ;Crossed boundary?
      176F      ;No, it's okay
      1771      ;Get new location
      1772      ;Update pattern address
      1775      ;Clear carry

      1783      ;Assume CMASK is even
      1785      ;Within byte, just change CMASK

      1787      ;On right edge?
      1789      ;No, move to next pixel
      178B      ;We're on right edge, set carry and return

```

(MSX ROM BASIC BIOS) Macro-80		3.44	01-Jan-85	PAGE 51-6
- MSXGRP - (Graphic cursor movements)				
4957	178B	CD 1639		
4958	178E	A7	FETCHC	
4959	178F	3E 0F	A	
4960	1791	FA 17C0	A,0FH	;Assume CMASK is even
4961	1794		M,MHZMV1	;Good assumption
4962	1794	D5	DE	
4963	1795	11 0008	DE,8	
4964				
4965	1798	3E F0	A,0F0H	
4966	179A	18 1F	MHCMOV	
4967	179C			
4968				
4969	179C	CD 1639	FETCHC	
4970	179F	A7	A	
4971	17A0	3E F0	A,0F0H	
4972	17A2	F2 17C0	P,MHZMV1	
4973	17A5	7D	A,L	
4974	17A6	E6 F8	0F8H	
4975	17A8	20 0B	NZ,MLFTC1	
4976	17AA	18 AE	ONBRD1	
4977	17AC			
4978				
4979	17AC	CD 1639	FETCHC	
4980	17AF	A7	A	
4981	17B0	3E F0	A,0F0H	
4982	17B2	F2 17C0	P,MHZMV1	
4983	17B5			
4984	17B5	D5	DE	
4985	17B6	11 FFF8	DE,0FFF8H	
4986	17B9	3E 0F	A,0FH	
4987	17BB			

```

( MSX ROM BASIC BIOS ) Macro-80
- MSXGRP - (Graphic cursor movements)

3.44 01-Jan-85 PAGE 51-7

4988 17BB 19          ADD      HL,DE
4989 17BC 22 F92A      LD       (CLOC),HL
4990 17BF D1          POP      DE
4991 17C0          MHZMV1:
4992 17C0 32 F92C      LD       (CMASK),A
4993 17C3 A7          AND      A
4994 17C4 E1          POP      HL
4995 17C5 C9          RET
4996 17C6          MTDNC:
4997          ;
4998          ; [ Vertical movements ]
4999          ;

5000 17C6 E5          PUSH     HL
5001 17C7 2A F3D5      LD       HL,(MLTCGP)
5002 17CA 11 0500      LD       DE,0500H
5003 17CD 19          ADD      HL,DE
5004 17CE E1          POP      HL
5005 17CF E7          RST      20H
5006 17D0 38 0A        JR       C,MDNC
5007 17D2 7D          LD       A,L
5008 17D3 3C          INC      A
5009 17D4 E6 07        AND      7
5010 17D6 20 04        JR       NZ,MDNC
5011 17D8 37          SCF
5012          POP      DE
5013 17D9 D1          POP      HL
5014 17DA E1          RET
5015 17DB C9          MDCNC:
5016 17DC          ;
5017          ;
5018 17DC 23          INC      HL

;Possibly on border?
;No
;Check if least 3 bits are all 1's

;No
;We are at the bottom border,
;set carry and return

;Move down 1 byte

```

(MSX ROM BASIC BIOS) Macro-80	3.44	01-Jan-85	PAGE 51-8
- MSXGRP - (Graphic cursor movements)			
5019 17DD 7D	LD	A,L	
5020 17DE 11 00F8	LD	DE,0F8H	;Load possible offset to next block
5021 17E1 18 1A	JR	MVTMOV	;Check
5022 17E3			
5023	MTUPC:		
5024 17E3 E5			
5025 17E4 2A F3D5	PUSH	HL	
5026 17E7 11 0100	LD	HL,(MLTCGP)	
5027 17EA 19	LD	DE,0100H	;Possibly on border?
5028 17EB E1	ADD	HL,DE	
5029 17EC E7	POP	HL	
5030 17ED 30 09	RST	20H	;Test [HL] with [DE]
5031 17EF 7D	JR	NC,MUPC	;No
5032 17F0 E6 07	LD	A,L	;Check if we're top of a block
5033 17F2 20 04	AND	7	
5034 17F4 37	JR	NZ,MUPC	;No
5035 17F5 D1	SCF		;We're on top border, set carry and return
5036 17F6 E1	POP	DE	
5037 17F7 C9	POP	HL	
5038 17F8	RET		
5039	MUPC:		
5040 17F8 7D		A,L	
5041 17F9 2B	LD	HL	;Move up 1 byte
5042 17FA 11 FF08	DEC		;Load possible offset to next block
5043 17FD	LD	DE,0FF08H	
5044 17FD E6 07	MVTMOV:		
5045 17FF 20 01	AND	7	;Wrapped to next block?
5046 1801 19	JR	NZ,MVTMV1	;No
5047 1802	ADD	HL,DE	;Yes, add up offset to next block
5048 1802 22 F92A		(CLOC),HL	
5049 1805 A7	LD	A	;Clear carry

5050	1806	D1		POP	DE
5051	1807	E1		POP	HL
5052	1808	C9		RET	
5053				SUBTTL	-MSXGRP- (Box fill and Misc.)

```

( MSX ROM BASIC BIOS ) Macro-80          3.44    01-Jan-85    PAGE    52
-MSXGRP- (Box fill and Misc.)

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1809

NSETCX:
;
; NSETCX - Performs SETC, RIGHTC [HL] times
;
; In fact, SETC and RIGHTC are never called to increase speed,
; and for the reason described below.
;
; Since only 2 colors can be displayed in a byte, some special
; handling is required when a full-byte is set when writing left
; or right extras. In this case, we can completely ignore the
; background color for that byte, allowing 2 colors displayed
; in a byte.
;
; All registers may be destroyed.
;

CD 15D9
C2 18BB
E5
CD 1639
E3
87
38 18
F5
01 FFFF
0F

CALL
JP
PUSH
CALL
EX
ADD
JR
PUSH
LD
RRCA

CHKMOD
NZ,MNSETCX
HL
FETCHC
(SP),HL
A,A
C,NSTC20
AF
BC,0FFFFH

;Multi-color mode
;Save count
;Get CLOC and CMASK
;Reget count, save CLOC
;Beginig at leftmost position?
;Yes, no extra dots at the left
;Save mask pattern*2

NSTC10:
HL,BC
NC,NSTCSP
RRCA
JR
NC,NSTC10

;Decrement pixel count
;The whole dots are within a byte

```

5085	1822	F1	POP	A,F		;Restore mask pattern*2
5086	1823	3D	DEC	A		;Form left-extra pattern
5087	1824	E3	EX	(SP),HL		;Reget CLOC, save count
5088	1825	E5	PUSH	HL		;Save CLOC
5089	1826	CD 186C	CALL	PATWRT		;Write to VRAM (pattern and color)
5090	1829	E1	POP	HL		;Restore CLOC
5091	182A	11 0008	Ld	DE,8		;Load an offset to next byte
5092	182D	19	ADD	HL,DE		;Update pattern address
5093	182E	E3	EX	(SP),HL		;Reget count, save CLOC
5094	182F				NSTC20:	
5095	182F	7D	Ld	A,L		;Get low byte of count
5096	1830	E6 07	AND	7		;[A]=count mod 8
5097	1832	4F	Ld	C,A		;save count after byte boundary
5098	1833	7C	Ld	A,H		
5099	1834	0F	RRCA			
5100	1835	7D	Ld	A,L		
5101	1836	1F	RRA			
5102	1837	0F	RRCA			
5103	1838	0F	RRCA			
5104	1839	E6 3F	AND	00111111B		;[HL]=[HL]/8
5105	183B	E1	POP	HL		;Reget CLOC
5106	183C	47	Ld	B,A		;[B]=counter
5107	183D	28 14	JR	Z,NSTC40		;No dots in this part
5108	183F				NSTC30:	
5109	183F	AF	XOR	A		;Make specified color a background color
5110	1840	CD 07CD	CALL	WRTVRM		;Write to VRAM (pattern)
5111	1843	11 2000	Ld	DE,GRPDIF		
5112	1846	19	ADD	HL,DE		;Calculate address of color table
5113	1847	3A F3F2	Ld	A,(ATTRBYT)		;Get specified color
5114	184A	CD 07CD	CALL	WRTVRM		;Write to VRAM (color)
5115	184D	11 2008	Ld	DE,GRPDIF+8		;Load an offset to next byte

```

( MSX ROM BASIC BIOS ) Macro-80
-MSXGRP- (Box fill and Misc.)

3.44 01-Jan-85 PAGE 52-2

5116 1850 19      ADD      HL,DE      ;Bump CLOC
5117 1851 10 EC    DJNZ     NSTC30    ;Loop until done
5118 1853         NSTC40:
5119 1853 0D      DEC       C          ;dot count in char boundary
5120 1854 F8      RET       M          ;No dots in right extra
5121 1855 E5      PUSH     HL          ;Save CLOC
5122 1856 21 185D LD       HL,RGTEXT  ;Load address for 'right-extra' pattern table
5123 1859 09      ADD      HL,BC
5124 185A 7E      LD       A,(HL)
5125 185B 18 0E   JR       NSTC50
5126 185D         RGTEXT:
5127         ;
5128 185D 80 C0 E0 F0 DB      80H,0C0H,0E0H,0F0H
5129 1861 F8 FC FE DB      0F8H,0FCH,0FEH
5130 1864         NSTCSP:
5131         ;
5132 1864 87      ADD      A,A          ;Get mask pattern for the right (111111100)
5133 1865 3D      DEC      A
5134 1866 2F      CPL
5135 1867 47      LD       B,A          ;Save it
5136 1868 F1      POP      AF          ;Get mask pattern for the left (00011111)
5137 1869 3D      DEC      A
5138 186A A0      AND      B          ;Make a pattern to write (000111100)
5139 186B         NSTC50:
5140 186B E1      POP      HL          ;Restore CLOC ex.

```

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5167
5168
5169
5170
5171

186C
186C
186D
1870
1871
1874
1875
1878
1879
187B
187C
187D
187E

47
CD 07D7
4F
11 2000
19
CD 07D7
F5
E6 0F
5F
F1
93
57

LD B,A
CALL RDVRM
LD C,A
LD DE,GRPDIF
ADD HL,DE
CALL RDVRM
PUSH AF
AND 0FH
LD E,A
POP AF
SUB E
LD D,A

PATWRT:
;
; PATWRT - Write a pattern to high-resolution screen
;
; Entry: A - Pattern to be written
; HL - Address of pattern table
; ATRBYT - Color of this pattern
;
LD B,A
CALL RDVRM
LD C,A
LD DE,GRPDIF
ADD HL,DE
CALL RDVRM
PUSH AF
AND 0FH
LD E,A
POP AF
SUB E
LD D,A

;Save pattern to be added
;Read VRAM (pattern)
;Save current pattern
;Form address of color table
;Read from VRAM (color)
;Extract background color
;Save background color
;Restore foreground and background color
;Set foreground color in the upper 4 bit
;[B] has the specified pattern,
;[C] has the current pattern,
;[D] has the current foreground color
; shifted left 4 times,
;[E] has the current background color,
;[HL] has the address of color table.
;Get specified color
;Same with current background?
;Yes

187F
1882
1883
1885

3A F3F2
BB
28 19
87

LD A,(ATRBYT)
CP E
JR Z,SAMEBG
ADD A,A
```

(MSX ROM BASIC BIOS) Macro-80 -MSXGRP- (Box fill and Misc.)	3.44	01-Jan-85	PAGE 53-1	
5172	1886	87	ADD	A,A
5173	1887	87	ADD	A,A
5174	1888	87	ADD	A,A
5175	1889	BA	CP	D
5176	188A	28 16	JR	Z,SAMEFG
5177	188C	F5	PUSH	AF
5178	188D	78	LD	A,B
5179	188E	B1	OR	C
5180	188F	FE FF	CP	OFFH
5181	1891	28 17	JR	Z,PATWRL
5182				
5183	1893	E5	PUSH	HL
5184	1894	D5	PUSH	DE
5185	1895	CD 18A2	CALL	SAMEFG
5186	1898	D1	POP	DE
5187	1899	E1	POP	HL
5188	189A	F1	POP	AF
5189				
5190	189B	B3	OR	E
5191	189C	18 1A	JR	JMPWRT
5192	189E			
5193			SAMEBG:	
5194	189E	78	LD	A,B
5195	189F	2F	CPL	
5196	18A0	A1	AND	C
5197	18A1	11	DB	11H
5198	18A2			
5199	18A2	78	LD	A,B
5200	18A3	B1	OR	C
5201	18A4			
5202	18A4	11 2000	WTPTAB:	
			LD	DE,GRPDIF

;Same with current foreground?
 ;Yes
 ;Save new foreground color

 ;All pixels are going to be set?
 ;Yes, Spock will use a new repair technique
 ;logically...
 ;Save address of color table
 ;Save current background color
 ;Write to VRAM (pattern)
 ;Restore current background in [E]
 ;Restore color table address
 ;Restore new foreground color in upper
 ;4 bits of [Acc]
 ;Form new foreground and background color
 ;Write to color table

 ;Skip next 2 bytes (LXI D)

(MSX ROM BASIC BIOS) Macro-80 -MSXGRP- (Box fill and Misc.)	3.44	01-Jan-85	PAGE 53-2	189
5203 18A7 19	ADD	HL,DE		
5204 18A8 18 0E	JR	JMPWRT		
5205 18AA	PATWRT:		;Write to pattern table	
5206	;			
5207 18AA F1	POP	AF	;Discard new foreground color	
5208 18AB 78	LD	A,B	;Reget specified pattern	
5209 18AC 2F	CPL		;Forget current background color, 'cause	
5210 18AD E5	PUSH	HL	;there's no background, we display	
5211 18AE D5	PUSH	DE	;new pattern as background color.	
5212 18AF CD 18A4	CALL	WTPTAB	;Write to pattern table	
5213 18B2 D1	POP	DE		
5214 18B3 E1	POP	HL		
5215 18B4 3A F3F2	LD	A,(ATTRBYT)	;Get new color (this will be the	
5216			;background color)	
5217 18B7 B2	OR	D	;Add current foreground color	
5218 18B8	JMPWRT:			
5219 18B8 C3 07CD	JP	WRTVRM	;Write to VRAM (color)	

(MSX ROM BASIC BIOS) Macro-80
 -MSXGRP- (Box fill and Misc.)

```

5220
5221      18BB
5222
5223
5224
5225      MNSTCX:
5226      ;
5227      ; NSETCX for multicolor screen
5228      ;
5229      E5      PUSH      HL      ;Save counter
5230      CD 167E  CALL      SETC   ;Set pixel
5231      CD 16C5  CALL      RIGHTC ;Move to right
5232      E1      POP       HL      ;Restore counter
5233      2D      DEC       L
5234      20 F5    JR        NZ,MNSTCX
5235      C9      RET
5236
5237      GTASPC:
5238      ;
5239      ; GTASPC - load aspect ratio for CIRCLE
5240      ;
5241      2A F40B  LD        HL,(ASCPCTL)
5242      EB      EX        DE,HL
5243      2A F40D  LD        HL,(ASCPCTL2)
5244      C9      RET
5245
5246      SUBTTL -MSXGRP - (Routines for paint)

```

```
5241      PNTINI:
5242      18CF
5243      ;
5244      ; PNTINI - Initialize border color
5245      ;
5246      F5      PUSH      AF
5247      CD 15D9  CALL      CHKMOD
5248      28 06   JR        Z,PNTHRS
5249      F1      POP       AF
5250      FE 10   CP        10H
5251      3F      CCF
5252      18 05   JR        PNTIRT
5253      18DB
5254      PNTHRS:
5255      ;
5256      F1      POP       AF
5257      3A F3F2 LD        A,(ATRYBT)
5258      A7      AND       A
5259      32 FCB2 LD        (BRDATR),A
5260      C9      RET
5261      PNTIRT:
5262      SCANR:
5263      ;
5264      ; SCANR - scan pixels to right
5265      ; Maximum number of pixels to test is passed in [DE].
5266      ;
5267      21 0000 LD        HL,0
5268      4D      LD        C,L
5269      CD 15D9 CALL      CHKMOD
5270      20 64   JR        NZ,MSCANR
5271      ;
5272      ; Scan to right in high-resolution mode
5273      ;
5274      ; Save specified color
5275      ; In what mode are we now?
5276      ; High-resolution mode
5277      ; Legal value?
5278      ; Carry means illegal
5279      ; Discard specified color
5280      ; Always ignore specified border
5281      ; Always legal
5282      ; Set border color
5283      ; Return with the condition
5284      ; Initialize PNTCNT
5285      ; Initialize PNTDFL
5286      ; Check current screen mode
5287      ; Multi-color mode
```

```
5272 ; [B] set to 0 is need to suspend painting, 1 otherwise.
5273 ;
5274 ; Work1 = Temporary storage for 'suspend painting'
5275 ; Work2 = Save area for pixel count to draw right
5276 ; Work3 = Save area for 'pixel changed' flag
5277 ;
5278 18ED 78 LD A,B
5279 18EE 32 F866 LD (RUNFLG),A ;Remember to suspend or not
5280 18F1 AF XOR A ;Clear 'pixel changed' flag
5281 18F2 32 F869 LD (WORK3),A
5282 18F5 3A FCB2 LD A,(BRDATR)
5283 18F8 47 LD B,A ;Set border color to [B] for comparison
5284 18F9 SCANR1:
5285 18F9 CD 1647 CALL READC ;Read current color
5286 18FC B8 CP B ;Still on border?
5287 18FD 20 0D JR NZ,SCANR2 ;No, start painting
5288 18FF 1B DEC DE ;All pixels tested?
5289 1900 7A LD A,D
5289 1901 B3 OR E
5290 1902 C8 RET Z ;Yes
5291 1903 CD 16AC CALL TRIGHT ;Advance to right, and check if out of screen
5292 1906 30 F1 JR NC,SCANR1 ;Not yet out of screen, continue
5293 1908 11 0000 LD DE,0 ;All pixels has border attribute on
5294 190B C9 RET ;this row, let BRDCNT be 0, and return
5295 190C SCANR2:
5296 190C ;
5297 190C ; A pixel with non-border attribute is found. Start painting
5298 190C ;
5299 190C ;
5300 190C CD 19AE CALL CHKCHG ;Check if pixel changed
5301 190F D5 PUSH DE ;Save BRDCNT
5302 1910 CD 1639 CALL FETCHC ;Get current CLOC, CMASK
```

```

5303      1913      22 F942      LD      (CSAVEA),HL
5304      1916      32 F944      LD      (CSAVEM),A
5305      1919      11 0000      LD      DE,0
5306      191C      SCANR3:
5307      191C      13          INC      DE
5308      191D      CD 16AC      CALL   TRIGHT
5309      1920      38 0B      JR      C,SCANR4
5310      1922      CD 1647      CALL   READC
5311      1925      B8          CP      B
5312      1926      28 05      JR      Z,SCANR4
5313      1928      CD 19AE      CALL   CHKCHG
5314      192B      18 EF      JR      SCANR3
5315      192D      SCANR4:
5316      ;
5317      192D      D5          PUSH   DE
5318      192E      CD 1639      CALL   FETCHC
5319      1931      E5          PUSH   HL
5320      1932      F5          PUSH   AF
5321      1933      2A F942      LD      HL,(CSAVEA)
5322      1936      3A F944      LD      A,(CSAVEM)
5323      1939      CD 1640      CALL   STOREC
5324      193C      EB          DE,HL
5325      193D      22 F867      LD      (WORK2),HL
5326      1940      3A F866      LD      A,(WORK1)
5327      1943      A7          AND     A
5328      1944      C4 1809      CALL   NZ,NSETCX
5329      1947      F1          POP     AF
5330      1948      E1          POP     HL
5331      1949      CD 1640      CALL   STOREC
5332      194C      E1          POP     HL
5333      194D      D1          POP     DE

;Set first non-border pixel encountered
;Initialize # of painted pixels (PNTCNT)
;Update PNTCNT
;Move 1 pixel right
;Out of screen
;Read color of current pixel
;Reached border?
;Yes
;Check if pixel changed
;Keep on scanning

;Save PNTCNT
;Since NSETCX does not update 'C', these value
; must be saved
;Set where to start painting
;Set CLOC and CMASK
;Set length of line to [HL] (PNTCNT)
;Same as [RUNFLG]
;Draw [HL] pixels to the right if not suspend
;Restore 'last-examined-pixel' information

;Restore PNTCNT
;Restore BRDCNT

```

(MSX ROM BASIC BIOS) Macro-80					
-MSXGRP - (Routines for paint)	3.44	01-Jan-85	PAGE	55-3	194
5334 194E C3 19A9	JP	SCANL4			

```
5335
5336      1951      MSCANR:
5337      ;
5338      ; Scan to right in multi-color mode
5339      ;
5340      1951      CD 19C7
5341      1954      30 0D
5342      1956      1B
5343      1957      7A
5344      1958      B3
5345      1959      C8
5346      195A      CD 16AC
5347      195D      30 F2
5348      195F      11 0000
5349      1962      C9
5350      1963
5351      MSCNR1:
5352      ;
5353      1963      CD 1639
5354      1966      22 F942
5355      1969      32 F944
5356      196C      21 0000
5357      196F
5358      1970      23
5359      1973      CD 16AC
5360      1974      D8
5361      1977      CD 19C7
5362      1979      30 F6
5363      C9
5364
5365      CALL      MTSBRD      ;Is it border color?
5366      JR        NC,MSCNR1  ;No, start painting
5367      DEC       DE          ;All pixels tested?
5368      LD        A,D
5369      OR        E
5370      RET       Z           ;Yes
5371
5372      CALL      TRIGHT      ;Advance to right, and check if out of screen
5373      JR        NC,MSCANR   ;Not yet out of screen, continue
5374      LD        DE,0        ;Out of screen, let BRDCNT be 0, and return
5375      RET
5376
5377      MSCNR2:
5378      ;
5379      1963      CD 1639
5380      1966      22 F942
5381      1969      32 F944
5382      196C      21 0000
5383      196F
5384      1970      23
5385      1973      CD 16AC
5386      1974      D8
5387      1977      CD 19C7
5388      1979      30 F6
5389      C9
5390
5391      CALL      MTSBRD      ;Increment PNTCNT
5392      JR        NC,MSCNR2   ;Advance to right, and check if out of screen
5393      DEC       DE          ;Going out of screen
5394      LD        A,D
5395      OR        E
5396      RET       Z           ;Reached border color?
5397
5398      CALL      TRIGHT      ;Not yet, continue
5399      JR        NC,MSCNR2
5400      DEC       DE          ;Increment PNTCNT
5401      LD        A,D
5402      OR        E
5403      RET       Z           ;Reached border color?
5404
5405      CALL      MTSBRD      ;Increment PNTCNT
5406      JR        NC,MSCNR2   ;Advance to right, and check if out of screen
5407      DEC       DE          ;Going out of screen
5408      LD        A,D
5409      OR        E
5410      RET       Z           ;Reached border color?
5411
5412      CALL      TRIGHT      ;Not yet, continue
5413      JR        NC,MSCNR2
5414      DEC       DE          ;Increment PNTCNT
5415      LD        A,D
5416      OR        E
5417      RET       Z           ;Reached border color?
```

(MSX ROM BASIC BIOS) Macro-80
 -MSXGRP - (Routines for paint)

```

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197A
197A
197D
197E
1981
21 0000
4D
CD 15D9
20 37

1983
1984
1987
198A
198B
198B
198E
1990
1993
1994
1996
1999
199A
199C
199C
199F
199F
19A0

AF
32 F869
3A FCB2
47
CD 16D8
38 0F
CD 1647
B8
28 06
CD 19AE
23
18 EF
CD 16C5
E5
ED 5B F867

SCANL:
;
; SCANL - Scan pixels to left
;
LD HL,0
LD C,L
CALL CHKMOD
JR NZ,MSCANL
;Initialize PNTCNT
;Initialize PNTDFL
;Check current screen mode
;Multi-color mode
;
; Scan to left in high-resolution mode
;
XOR A
LD (WORK3),A
LD A,(BRDATR)
LD B,A
CALL TLEFT
JR C,SCANL3
CALL READC
CP B
JR Z,SCANL2
CALL CHKCHG
INC HL
JR SCANL1
SCANL2:
;
CALL RIGHTC
SCANL3:
PUSH HL
LD DE,(WORK2)
;Save PNTCNT
;Load suspended pixels which remain
; 'C' must specify 'last pixel painted'

```

```
5394 19A4 19          ADD      HL,DE          ;to the right
5395 19A5 CD 1809     CALL     NSETCX         ;Draw [HL] pixel from current 'C'
5396 19A8 E1         POP      HL             ;Restore PNTCNT
5397 19A9          SCANL4:
5398 19A9 3A F869     LD       A,(WORK3)      ;Non 0 if pixels changed attribute
5399 19AC 4F         LD       C,A
5400 19AD C9         RET
5401 19AE          CHKCHG:
5402          ;
5403 19AE E5         PUSH     HL
5404 19AF 21 F3F2     LD       HL,ATRBYT      ;Get specified paint attribute
5405 19B2 BE         CP       (HL)           ;Same?
5406 19B3 E1         POP      HL
5407 19B4 C8         RET      Z             ;Yes, no change of attribute
5408 19B5 3C         INC      A             ;Load non 0 to [Acc]
5409 19B6 32 F869     LD       (WORK3),A     ;Remember this temporarily
5410 19B9 C9         RET
5411 19BA          MSCANL:
5412          ;
5413          ; Scan to left in multi-color mode
5414          ;
5415 19BA CD 16D8     CALL     TLEFT          ;Advance to left, and check if out of screen
5416 19BD D8         RET      C             ;going out of screen
5417 19BE CD 19C7     CALL     MTSBRD        ;Reached border color?
5418 19C1 DA 16C5     JP       C,RIGHTC      ;Yes, adjust CLOC, CMASK and return
5419 19C4 23         INC      HL             ;Increment PNTCNT
5420 19C5 18 F3       JR       MSCANL        ;Continue
5421 19C7          MTSBRD:
5422          ;
5423          ; Test border subroutine for multi-color mode
5424          ;
```

(MSX ROM BASIC BIOS) Macro-80	3.44	01-Jan-85	PAGE 57-2	
-MSXGRP - (Routines for paint)				
5425	19C7	CD 1647	CALL	READC
5426	19CA	47	LD	B,A
5427	19CB	3A FCB2	LD	A,(BRDATR)
5428	19CE	90	SUB	B
5429	19CF	37	SCF	
5430	19D0	C8	RET	Z
5431	19D1	3A F3F2	LD	A,(ATRBVT)
5432	19D4	B8	CP	B
5433	19D5	C8	RET	Z
5434				
5435	19D6	CD 167E	CALL	SETC
5436	19D9	0E 01	LD	C,1
5437	19DB	A7	AND	A
5438	19DC	C9	RET	
5439				

;Get the color of target pixel
 ;Load specified border color
 ;Reached border?
 ;Assume so
 ;Yes, return with carry flag set
 ;Is current pixel same as ATRBYT?
 ;Yes, no changes made.
 ;Return with carry reset
 ;Set this pixel to ATRBYT
 ;Set 'pixel-changed' flag
 ;Tell caller that we plot a dot

SUBTTL -CASET- Cassette drivers stuff

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19DD
C5
F5
01 0000
19E2
19E2
19E3
19E4
19E5
19E7
19E8
19E9
19E9
19EA
19EC
19EE
19EF
19F0
19F1

C5
F5
01 0000
0B
78
B1
20 FB
F1
C1
F5
3E 09
D3 AB
F1
FB
C9

; Cassette read/write stuff
;
; Following driver assumes that T cycle is 279.365 nS
;
; Variables referenced
; PPI.CM To write to cassette
; PSG.DR To read from cassette
; BREAKX Routine to check for [STOP] key pressed
;
; TAPOFF:
;
;
; PUSH BC
; PUSH AF
; LD BC,0
;
; CTWOF1:
; DEC BC
; LD A,B
; OR C
; JR NZ,CTWOF1
; POP AF
; POP BC
;
; TAPIOF:
; PUSH AF
; LD A,00001001B
; OUT (PPI.CM),A
; POP AF
; EI
; RET
;
; TAPOON:
;
```

(MSX ROM BASIC BIOS) Macro-80
-CASET- Cassette drivers stuff

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```

5471      ; Write out header, if [A]=0 then write short header
5472      ; otherwise write long header ( 5sec)
5473      ;
5474      19F1 B7      OR      A      ;set flag for length of header
5475      19F2 F5      PUSH     AF     ;save flag
5476      19F3 3E 08   LD      A,8    ;Motor on
5477      19F5 D3 AB   OUT      (PPI.CM),A
5478      19F7 21 0000 LD      HL,0
5479      19FA          MOTRWT:
5480      19FA 2B      DEC      HL
5481      19FB 7C      LD      A,H
5482      19FC B5      OR      L
5483      19FD 20 FB   JR      NZ,MOTRWT
5484      19FF F1      POP      AF
5485      1A00 3A F40A  LD      A,(HEADER)
5486      1A03 28 02  JR      Z,SYNCWL
5487      1A05 87      ADC      A,A
5488      1A06 87      ADL
5489      1A07          SYNCWL:
5490      1A07 47      LD      B,A
5491      1A08 0E 00   LD      C,0
5492      1A0A F3      DI
5493      1A0B          SYNLP1:
5494      1A0B CD 1A4D  CALL     BITLOT
5495      1A0E CD 1A3F  CALL     RETRET
5496      1A11 0B      DEC      BC
5497      1A12 78      LD      A,B
5498      1A13 B1      OR      C
5499      1A14 20 F5   JR      NZ,SYNLP1
5500      1A16 C3 046F JP      BREAKX
5501      1A19          TAPOUT:

```

```

;set up counter
;Don't disturb during writing to cassette
;Write enough marks
;compensate overhead
;loop till counter exhausts
;check control-stop and return

```

```
5502 1A19          DATAW:
5503              ;
5504              ; Output a byte
5505              ;
5506 1A19 2A F406    LD      HL,(LOW)
5507 1A1C F5         PUSH   AF
5508 1A1D 7D         LD      A,L
5509 1A1E D6 0E      SUB    0EH
5510 1A20 6F        LD      L,A
5511 1A21 CD 1A50    CALL   BITOUT
5512 1A24 F1        POP     AF
5513 1A25 06 08     LD      B,8
5514 1A27          DATAWL:
5515 1A27 0F        RRCA
5516 1A28 DC 1A40    CALL   C,BIT1
5517 1A2B D4 1A39    CALL   NC,BIT0
5518 1A2E 10 F7     DJNZ   DATAWL
5519 1A30 CD 1A40    CALL   BIT1
5520 1A33 CD 1A40    CALL   BIT1
5521 1A36 C3 046F    JP      BREAKX

;get time constants for space
;compensate loss time since last stop bit
;output start bit
;Initialize counter
;next bit to carry
;output mark if the bit is 1
;Output space
;Loop until 8 bits sent
;Output stop bit
;Check if break pressed and return
```

```
5522          1A39          BIT0:
5523          ;
5524          ; Output a bit to cassette
5525          ;
5526          ;
5527          ; Absolute jumps are used to improve accuracy
5528          ;
5529          1A39          2A F406          LD          HL,(LOW)          ;Output 0 (space)          (17 T)
5530          1A3C          CD 1A50          CALL         BITOUT          ;          (18 T)
5531          1A3F          RETRET:
5532          1A3F          C9          RET          ;          (11 T)
5533          1A40          BIT1:
5534          ;
5535          1A40          CD 1A4D          CALL         BIT1OUT          ;          (18 T)
5536          1A43          E3          EX          (SP),HL          ;          (20 T)
5537          1A44          E3          EX          (SP),HL          ;compensate overhead          (20 T)
5538          1A45          00          NOP          ;(Total 60 state)          ( 5 T)
5539          1A46          00          NOP          ;          ( 5 T)
5540          1A47          00          NOP          ;          ( 5 T)
5541          1A48          00          NOP          ;          ( 5 T)
5542          1A49          CD 1A4D          CALL         BIT1OUT          ;To compensate time          (18 T)
5543          1A4C          C9          RET          ;Don't change this          (11 T)
5544          1A4D          BIT1OUT:
5545          ;
5546          ; output a single cycle
5547          ;
5548          ; Total number of states =16 x [L] + 16 x [H] + 71
5549          ; =4.47uS x [L] + 4.47uS x [H] + 19.8uSec
5550          ;
5551          1A4D          2A F408          LD          HL,(HIGH)          ;          (17 T)
5552          1A50          BITOUT:
```

```

( MSX ROM BASIC BIOS ) Macro-80
-CASET- Cassette drivers stuff

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5553 1A50 F5 PUSH AF ; (12 T)
5554 ;
5555 KEEPL:
5556 1A51 2D DEC L ;Keep low level ( 5 T)
5557 1A52 C2 1A51 JP NZ,KEEPL ; (11 T)
5558 1A55 3E 0B LD A,0BH ; ( 8 T)
5559 1A57 D3 AB OUT (PPI.CM),A ;Output high level (11 T)
5560 1A59 KEEPH:
5561 1A59 25 DEC H ;keep high level ( 5 T)
5562 1A5A C2 1A59 JP NZ,KEEPH ; (11 T)
5563 1A5D 3E 0A LD A,0AH ; ( 8 T)
5564 1A5F D3 AB OUT (PPI.CM),A ;Output low level (11 T)
5565 1A61 F1 POP AF ;Restore data (12 T)
5566 ;
5567 1A62 C9 RET ; (11 T)
5568 1A63 TAPION:
5569 ;
5570 ; Detect header block
5571 ;
5572 1A63 3E 08 LD A,8 ;Motor on
5573 1A65 D3 AB OUT (PPI.CM),A
5574 1A67 F3 DI
5575 1A68 3E 0E LD A,0EH ;Select PSG port A
5576 1A6A D3 A0 OUT (PSG.LW),A
5577 1A6C SYN05:
5578 ;
5579 ; First, wait until a series of good pulses are found.
5580 ;
5581 1A6C 21 0457 LD HL,0457H ;Initialize counter
5582 ;Number of pulse to detect header
5583 1A6F SYN10:

```

```
5584 1A6F 51          LD      D,C          ;Remember last value
5585 1A70 CD 1B34     CALL   CNTFUL       ;Count full cycle
5586 1A73 D8         RET      C          ;Aborted
5587 1A74 79        LD      A,C          ;Get count
5588 1A75 FE DE     CP      0DEH        ;0DE = Max count
5589 1A77 30 F3     JR      NC,SYN05     ;Too long, reset number of pulses
5590 1A79 FE 05     CP      5           ;5 = Min count
5591 1A7B          JR      C,SYN05       ;Too short, reset number of pulses
5592
5593                ;
5594                ; Now compare with last pulse width and approve this as a good pulse
5595                ; if this is similar to last one.
5596
5597 1A7D 92          SUB      D          ;current - last
5598 1A7E 30 02     JR      NC,SYN11
5599 1A80 2F        CPL          ;result was negative, negate it
5600 1A81 3C        INC      A
5601 1A82          SYN11:
5602 1A82 FE 04     CP      4           ;within a wow allowance?
5603 1A84 30 E6     JR      NC,SYN05     ;no, reset number of pulse ever seen
5604 1A86 2B        DEC      HL
5605 1A87 7C        LD      A,H
5606 1A88 B5        OR      L
5607 1A89 20 E4     JR      NZ,SYN10     ;Loop till seen enough good pulses
5608 1A8B          ;
5609          SYN20:
5610                ;
5611                ; Next, calculate the mean width of pulse.
5612
5613 1A8B 21 0000     LD      HL,0        ;Initialize sum
5614 1A8E 45          LD      B,L        ;Initialize high byte of [BC] pair
5615 1A8F 55          LD      D,L        ;Loop 256 times
```

```
5615 1A90          SYN30:
5616 1A90          CD 1B34          CALL CNTFUL
5617 1A93          D8              RET C
5618 1A94          09              ADD HL,BC
5619 1A95          15              DEC D
5620 1A96          C2 1A90          JP NZ,SYN30
5621 1A99          01 06AE          LD BC,06AEH ;compensate over head
5622 1A9C          09              ADD HL,BC
5623
5624 ;
5625 ; Set various values for read routine. Those are,
5626 ;
5627 ; LOWLIM - lower limit of the width of start bit. [H]*1.5
5628 ; WINWID - width of window to count the transition.
5629 ;
5630 1A9D          7C              LD A,H ;[H] has mean pulse width
5631 1A9E          1F              RRA
5632 1A9F          E6 7F          AND 7FH
5633 1AA1          57              LD D,A ;[D]=[mean]/2
5634 1AA2          29              ADD HL,HL
5635 1AA3          7C              LD A,H ;[A]=[mean]x2
5636 1AA4          92              SUB D ;[A]=[mean]x1.5
5637 1AA5          57              LD D,A ;save
5638 1AA6          D6 06          SUB 6 ;compensate overhead at DATAR
5639 1AA8          32 FCA4          LD (LOWLIM),A
5640 ;
5641 ; Set width of window 'WINWID'
5642 ; CNTFUL takes 40T for a loop, RDBIT takes 60T for loop
5643 ; set WINWID as 3 times wider than single short pulse ([mean]/2)
5644 ; [WINWID]=[mean] x 1.5 x 40/60
5645 ; = [D] x 2/3
5646 ;
```

	(MSX ROM BASIC BIOS)	Macro-80	3.44	01-Jan-85	PAGE	59-4	
-CASSET- Cassette drivers stuff							
5646	1AAB	7A					
5647	1AAC	87	LD	A,D			;get [mean width]xl.75
5648	1AAD	06 00	ADD	A,A			;x2
5649	1AAF		LD	B,0			;clear quotient
5650	1AAF	D6 03	SULOP:				
5651	1AB1	04	SUB	3			
5652	1AB2	30 FB	INC	B			
5653	1AB4	78	JR	NC,SULOP			;loop till get carry
5654	1AB5	D6 03	LD	A,B			;[A]=[mean]xl.75x2/3
5655	1AB7	32 FCA5	SUB	3			;compensate overhead in RDBIT routine
5656	1ABA	B7	LD	(WINWID),A			
5657	1ABB	C9	OR	A			
			RET				

5658		TAPIN:			
5659	LABC				
5660		; Read a byte from cassette			
5661					
5662					
5663	LABC	LD A,(LOWLIM)			[D] has lower limit for start bit
5664	LABF	LD D,A			
5665	LAC0				
5666	LAC0	CALL BREAKX			
5667	LAC3	RET C			;Aborted
5668	LAC4	IN A,(PSG.DR)			;Get cassette
5669	LAC6	RLCA			;High state?
5670	LAC7	JR NC,DATAR			;No
5671	LAC9				
5672	LAC9	CALL BREAKX			
5673	LACC	RET C			;Aborted
5674	LACD	IN A,(PSG.DR)			;Get cassette
5675	LACF	RLCA			;falling egde?
5676	LAD0	JR C,DATAR0			;No
5677	LAD2	LD E,0			;Initialize edge mask
5678	LAD4	CALL CNTHLF			;Get width in [C]
5679	LAD7				
5680	LAD7	LD B,C			;Save old width
5681	LAD8	CALL CNTHLF			;Get new width in [C]
5682	LADB	RET C			;aborted
5683	LADC	LD A,B			;Add width of 2 pulses
5684	LADD	ADD A,C			
5685	LADE	JP C,DATAR1			;Pulse too long
5686	LAE1	CP D			;Longer than lower limit?
5687	LAE2	JR C,DATAR1			;No
5688					

```
5689 ; Now, a valid start bit has been found.
5690 ; [E] = 0 if NORMAL polarity,
5691 ; =255 if REVERSE polarity.
5692 ;
5693 ; LD L,8 ;Initialize counter
5694 ;
5695 ; DATARL: CALL RDBIT
5696 ; LAE9 FE 04 CP 3+1 ;Legal transitions?
5697 ; LAEB 3F CCF ;
5698 ; LAEC D8 RET C ;Too many transitions
5699 ; LAED FE 02 CP 2 ;
5700 ; LAEF 3F CCF ;Set carry if 2 or 3 transitions
5701 ; LAFO CB 1A RR D ;
5702 ;
5703 ; We've just assembled a bit. A check must be done to make sure
5704 ; that we're at the start of next bit field.
5705 ;
5706 ; LD A,C ;Reget number of transitions
5707 ; RRCA
5708 ; D4 1B23 CALL NC,CNTHL0 ;Wait for next transition if 0 or 2
5709 ; 1AF7 CD 1B1F CALL CNTHLF
5710 ; 1AFA 2D DEC L ;
5711 ; 1AFB C2 1AE6 JP NZ,DATARL ;Loop till done
5712 ; 1AFE CD 046F CALL BREAKX ;return with carry set if broken
5713 ; 1B01 7A LD A,D
5714 ; 1B02 C9 RET
5715 ; RDBIT:
5716 ;
5717 ; Count number of transitions within a period specified by 'WINWID'
5718 ;
5719 ; length of window = 17uSec x [WINWID] + 12.3 uSec
```

```
5720 ;  
5721 ; [D],[H] and [L] are preserved.  
5722 ; [E] is updated to prepare for next edge  
5723 ;  
5724 3A FCA5 LD A,(WINWID) ;Get width of window  
5725 47 LD B,A  
5726 0E 00 LD C,0 ;Clear # of transitions seen  
5727 1B09 RDBITL:  
5728 1B09 IN A,(PSG.DR) ;Get a bit  
5729 1B0B XOR E ;Any changes?  
5730 1B0C F2 1B17 JP P,NOTRAN ;No  
5731 1B0F 7B LD A,E ;Transition seen  
5732 1B10 2F CPL ;Prepare for next transition  
5733 1B11 5F LD E,A  
5734 1B12 0C INC C ;Increment # of transitions  
5735 1B13 10 F4 DJNZ RDBITL  
5736 1B15 79 LD A,C  
5737 1B16 C9 RET ;Get transition count  
5738 1B17 NOTRAN:  
5739 ;  
5740 1B17 00 NOP ;Compensate time  
5741 1B18 00 NOP  
5742 1B19 00 NOP  
5743 1B1A 00 NOP  
5744 1B1B 10 EC DJNZ RDBITL  
5745 ;  
5746 1B1D 79 LD A,C ;Get transition count  
5747 1B1E C9 RET
```

```
5748
5749 1B1F CNTHLF:
5750 ;
5751 ; Count half cycle
5752 ; 1T =279.4nS
5753 ; period=[C] x 11.18 + 35.48uS
5754 ;
5755 1B1F CD 046F ;Break? (87 T)
5756 1B22 D8 RET C ;Yes, aborted ( 6 T)
5757 1B23 CNTHL0: LD C,0 ;Initialize counter ( 8 T)
5758 1B23 0E 00
5759 1B25 CNTHL1: INC C ;# of state for this loop
5760 1B25 0C JR Z,TIMOUT ;40T=11.18uSec ( 5 T)
5761 1B26 28 0A IN A,(PSG.DR) ;Pulse too long ( 8 T)
5762 1B28 DB A2 XOR E ;Read cassette (11 T)
5764 1B2A AB JP P,CNTHL1 ;Desired transition? ( 5 T)
5765 1B2B F2 1B25 LD A,E ;No (11 T)
5766 1B2E 7B CPL ;Complement edge mask ( 5 T)
5767 1B2F 2F LD E,A ;
5768 1B30 5F RET ;
5769 1B31 C9 TIMOUT:
5770 1B32 ;
5771 ;
5772 1B32 0D DEC C ;Load 255
5773 1B33 C9 RET
5774 1B34 CNTFUL:
5775 ;
5776 ; Count full cycle
5777 ;
5778 1B34 CD 046F CALL BREAKX
```

5779	1B37	D8		RET	C	
5780	1B38	DB A2		IN	A,(PSG.DR)	;Aborted
5781	1B3A	07		RLCA		;Get cassette
5782	1B3B	38 F7		JR	C,CNTFUL	;Low state?
5783	1B3D	1E 00		LD	E,0	;No
5784	1B3F	CD 1B23		CALL	CNTHL0	;Initialize edge mask
5785	1B42	C3 1B25		JP	CNTHL1	
5786						

SUBTTL - BIO - OUTDO routine

```

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- BIO - OUTDO routine

5787      OUTDO:
5788      1B45
5789
5790
5791      ; OUTDO ( RST 18H )
5792      ; Prints char in [A], to either terminal or disk
5793      ; or printer depending on the flags:
5794      ; PRTFLG if non-zero print to printer
5795      ; PTRFIL if non-zero print to disk file pointed
5796      ; to by PTRFIL
5797      ;
5798      1B45      F5      PUSH      AF      ;Save character
5799      1B46      CD FEE4  CALL      H.OUTD
5800      1B47      CD 145F  CALL      ISFLIO
5801      1B48      28 08      JR      Z,LPTCOD
5802      1B49      F1      POP      AF      ;Doing I/O to file?
5803      1B4A      DD 21 6C48 LD      IX,FILOU1
5804      1B4B      C3 01FF  JP      CALBAS
5805      ;
5806      1B56      LPTCOD:
5807      1B57      3A F416  LD      A,(PRTFLG)
5808      1B58      B7      OR      A      ;Output to printer?
5809      1B59      28 5F      JR      Z,TTYCHR
5810      1B5A      3A F418  LD      A,(RAWPRT)
5811      1B5B      A7      AND      A      ;Nope, output to console
5812      1B5C      20 49      JR      NZ,LPTCH1
5813      1B5D      F1      POP      AF      ;Print in "RAW" mode?
5814      1B5E      ;
5815      1B5F      OUTDLP:
5816      1B60      F5      PUSH      AF      ;Yes, send char to printer
5817      1B61      ;
5818      1B62      ;
5819      1B63      ;
5820      1B64      ;
5821      1B65      ;
5822      1B66      ;
5823      1B67      ;
5824      1B68      ;
5825      1B69      ;
5826      1B6A      ;
5827      1B6B      ;
5828      1B6C      ;
5829      1B6D      ;
5830      1B6E      ;
5831      1B6F      ;
5832      1B70      ;
5833      1B71      ;
5834      1B72      ;
5835      1B73      ;
5836      1B74      ;
5837      1B75      ;
5838      1B76      ;
5839      1B77      ;
5840      1B78      ;
5841      1B79      ;
5842      1B7A      ;
5843      1B7B      ;
5844      1B7C      ;
5845      1B7D      ;
5846      1B7E      ;
5847      1B7F      ;
5848      1B80      ;
5849      1B81      ;
5850      1B82      ;
5851      1B83      ;
5852      1B84      ;
5853      1B85      ;
5854      1B86      ;
5855      1B87      ;
5856      1B88      ;
5857      1B89      ;
5858      1B8A      ;
5859      1B8B      ;
5860      1B8C      ;
5861      1B8D      ;
5862      1B8E      ;
5863      1B8F      ;
5864      1B90      ;
5865      1B91      ;
5866      1B92      ;
5867      1B93      ;
5868      1B94      ;
5869      1B95      ;
5870      1B96      ;
5871      1B97      ;
5872      1B98      ;
5873      1B99      ;
5874      1B9A      ;
5875      1B9B      ;
5876      1B9C      ;
5877      1B9D      ;
5878      1B9E      ;
5879      1B9F      ;
5880      1BA0      ;
5881      1BA1      ;
5882      1BA2      ;
5883      1BA3      ;
5884      1BA4      ;
5885      1BA5      ;
5886      1BA6      ;
5887      1BA7      ;
5888      1BA8      ;
5889      1BA9      ;
5890      1BAA      ;
5891      1BAB      ;
5892      1BAC      ;
5893      1BAD      ;
5894      1BAE      ;
5895      1BAF      ;
5896      1BB0      ;
5897      1BB1      ;
5898      1BB2      ;
5899      1BB3      ;
5900      1BB4      ;
5901      1BB5      ;
5902      1BB6      ;
5903      1BB7      ;
5904      1BB8      ;
5905      1BB9      ;
5906      1BBA      ;
5907      1BBB      ;
5908      1BBC      ;
5909      1BBD      ;
5910      1BBE      ;
5911      1BBF      ;
5912      1BC0      ;
5913      1BC1      ;
5914      1BC2      ;
5915      1BC3      ;
5916      1BC4      ;
5917      1BC5      ;
5918      1BC6      ;
5919      1BC7      ;
5920      1BC8      ;
5921      1BC9      ;
5922      1BCA      ;
5923      1BCB      ;
5924      1BCC      ;
5925      1BCD      ;
5926      1BCE      ;
5927      1BCF      ;
5928      1BD0      ;
5929      1BD1      ;
5930      1BD2      ;
5931      1BD3      ;
5932      1BD4      ;
5933      1BD5      ;
5934      1BD6      ;
5935      1BD7      ;
5936      1BD8      ;
5937      1BD9      ;
5938      1BDA      ;
5939      1BDB      ;
5940      1BDC      ;
5941      1BDD      ;
5942      1BDE      ;
5943      1BDF      ;
5944      1BE0      ;
5945      1BE1      ;
5946      1BE2      ;
5947      1BE3      ;
5948      1BE4      ;
5949      1BE5      ;
5950      1BE6      ;
5951      1BE7      ;
5952      1BE8      ;
5953      1BE9      ;
5954      1BEA      ;
5955      1BEB      ;
5956      1BEC      ;
5957      1BED      ;
5958      1BEE      ;
5959      1BEF      ;
5960      1BF0      ;
5961      1BF1      ;
5962      1BF2      ;
5963      1BF3      ;
5964      1BF4      ;
5965      1BF5      ;
5966      1BF6      ;
5967      1BF7      ;
5968      1BF8      ;
5969      1BF9      ;
5970      1BFA      ;
5971      1BFB      ;
5972      1BFC      ;
5973      1BFD      ;
5974      1BFE      ;
5975      1BFF      ;
5976      1C00      ;
5977      1C01      ;
5978      1C02      ;
5979      1C03      ;
5980      1C04      ;
5981      1C05      ;
5982      1C06      ;
5983      1C07      ;
5984      1C08      ;
5985      1C09      ;
5986      1C0A      ;
5987      1C0B      ;
5988      1C0C      ;
5989      1C0D      ;
5990      1C0E      ;
5991      1C0F      ;
5992      1C10      ;
5993      1C11      ;
5994      1C12      ;
5995      1C13      ;
5996      1C14      ;
5997      1C15      ;
5998      1C16      ;
5999      1C17      ;
6000      1C18      ;
6001      1C19      ;
6002      1C1A      ;
6003      1C1B      ;
6004      1C1C      ;
6005      1C1D      ;
6006      1C1E      ;
6007      1C1F      ;
6008      1C20      ;
6009      1C21      ;
6010      1C22      ;
6011      1C23      ;
6012      1C24      ;
6013      1C25      ;
6014      1C26      ;
6015      1C27      ;
6016      1C28      ;
6017      1C29      ;
6018      1C2A      ;
6019      1C2B      ;
6020      1C2C      ;
6021      1C2D      ;
6022      1C2E      ;
6023      1C2F      ;
6024      1C30      ;
6025      1C31      ;
6026      1C32      ;
6027      1C33      ;
6028      1C34      ;
6029      1C35      ;
6030      1C36      ;
6031      1C37      ;
6032      1C38      ;
6033      1C39      ;
6034      1C3A      ;
6035      1C3B      ;
6036      1C3C      ;
6037      1C3D      ;
6038      1C3E      ;
6039      1C3F      ;
6040      1C40      ;
6041      1C41      ;
6042      1C42      ;
6043      1C43      ;
6044      1C44      ;
6045      1C45      ;
6046      1C46      ;
6047      1C47      ;
6048      1C48      ;
6049      1C49      ;
6050      1C4A      ;
6051      1C4B      ;
6052      1C4C      ;
6053      1C4D      ;
6054      1C4E      ;
6055      1C4F      ;
6056      1C50      ;
6057      1C51      ;
6058      1C52      ;
6059      1C53      ;
6060      1C54      ;
6061      1C55      ;
6062      1C56      ;
6063      1C57      ;
6064      1C58      ;
6065      1C59      ;
6066      1C5A      ;
6067      1C5B      ;
6068      1C5C      ;
6069      1C5D      ;
6070      1C5E      ;
6071      1C5F      ;
6072      1C60      ;
6073      1C61      ;
6074      1C62      ;
6075      1C63      ;
6076      1C64      ;
6077      1C65      ;
6078      1C66      ;
6079      1C67      ;
6080      1C68      ;
6081      1C69      ;
6082      1C6A      ;
6083      1C6B      ;
6084      1C6C      ;
6085      1C6D      ;
6086      1C6E      ;
6087      1C6F      ;
6088      1C70      ;
6089      1C71      ;
6090      1C72      ;
6091      1C73      ;
6092      1C74      ;
6093      1C75      ;
6094      1C76      ;
6095      1C77      ;
6096      1C78      ;
6097      1C79      ;
6098      1C7A      ;
6099      1C7B      ;
6100      1C7C      ;
6101      1C7D      ;
6102      1C7E      ;
6103      1C7F      ;
6104      1C80      ;
6105      1C81      ;
6106      1C82      ;
6107      1C83      ;
6108      1C84      ;
6109      1C85      ;
6110      1C86      ;
6111      1C87      ;
6112      1C88      ;
6113      1C89      ;
6114      1C8A      ;
6115      1C8B      ;
6116      1C8C      ;
6117      1C8D      ;
6118      1C8E      ;
6119      1C8F      ;
6120      1C90      ;
6121      1C91      ;
6122      1C92      ;
6123      1C93      ;
6124      1C94      ;
6125      1C95      ;
6126      1C96      ;
6127      1C97      ;
6128      1C98      ;
6129      1C99      ;
6130      1C9A      ;
6131      1C9B      ;
6132      1C9C      ;
6133      1C9D      ;
6134      1C9E      ;
6135      1C9F      ;
6136      1CA0      ;
6137      1CA1      ;
6138      1CA2      ;
6139      1CA3      ;
6140      1CA4      ;
6141      1CA5      ;
6142      1CA6      ;
6143      1CA7      ;
6144      1CA8      ;
6145      1CA9      ;
6146      1CAA      ;
6147      1CAB      ;
6148      1CAC      ;
6149      1CAD      ;
6150      1CAE      ;
6151      1CAF      ;
6152      1CB0      ;
6153      1CB1      ;
6154      1CB2      ;
6155      1CB3      ;
6156      1CB4      ;
6157      1CB5      ;
6158      1CB6      ;
6159      1CB7      ;
6160      1CB8      ;
6161      1CB9      ;
6162      1CBA      ;
6163      1CBB      ;
6164      1CBC      ;
6165      1CBD      ;
6166      1CBE      ;
6167      1CBF      ;
6168      1CC0      ;
6169      1CC1      ;
6170      1CC2      ;
6171      1CC3      ;
6172      1CC4      ;
6173      1CC5      ;
6174      1CC6      ;
6175      1CC7      ;
6176      1CC8      ;
6177      1CC9      ;
6178      1CCA      ;
6179      1CCB      ;
6180      1CCC      ;
6181      1CCD      ;
6182      1CCE      ;
6183      1CCF      ;
6184      1CD0      ;
6185      1CD1      ;
6186      1CD2      ;
6187      1CD3      ;
6188      1CD4      ;
6189      1CD5      ;
6190      1CD6      ;
6191      1CD7      ;
6192      1CD8      ;
6193      1CD9      ;
6194      1CDA      ;
6195      1CDB      ;
6196      1CDC      ;
6197      1CDE      ;
6198      1CDF      ;
6199      1CE0      ;
6200      1CE1      ;
6201      1CE2      ;
6202      1CE3      ;
6203      1CE4      ;
6204      1CE5      ;
6205      1CE6      ;
6206      1CE7      ;
6207      1CE8      ;
6208      1CE9      ;
6209      1CEA      ;
6210      1CEB      ;
6211      1CEC      ;
6212      1CED      ;
6213      1CEE      ;
6214      1CEF      ;
6215      1CF0      ;
6216      1CF1      ;
6217      1CF2      ;
6218      1CF3      ;
6219      1CF4      ;
6220      1CF5      ;
6221      1CF6      ;
6222      1CF7      ;
6223      1CF8      ;
6224      1CF9      ;
6225      1CFA      ;
6226      1CFB      ;
6227      1CFC      ;
6228      1CFD      ;
6229      1CFE      ;
6230      1CFF      ;
6231      1D00      ;
6232      1D01      ;
6233      1D02      ;
6234      1D03      ;
6235      1D04      ;
6236      1D05      ;
6237      1D06      ;
6238      1D07      ;
6239      1D08      ;
6240      1D09      ;
6241      1D0A      ;
6242      1D0B      ;
6243      1D0C      ;
6244      1D0D      ;
6245      1D0E      ;
6246      1D0F      ;
6247      1D10      ;
6248      1D11      ;
6249      1D12      ;
6250      1D13      ;
6251      1D14      ;
6252      1D15      ;
6253      1D16      ;
6254      1D17      ;
6255      1D18      ;
6256      1D19      ;
6257      1D1A      ;
6258      1D1B      ;
6259      1D1C      ;
6260      1D1D      ;
6261      1D1E      ;
6262      1D1F      ;
6263      1D20      ;
6264      1D21      ;
6265      1D22      ;
6266      1D23      ;
6267      1D24      ;
6268      1D25      ;
6269      1D26      ;
6270      1D27      ;
6271      1D28      ;
6272      1D29      ;
6273      1D2A      ;
6274      1D2B      ;
6275      1D2C      ;
6276      1D2D      ;
6277      1D2E      ;
6278      1D2F      ;
6279      1D30      ;
6280      1D31      ;
6281      1D32      ;
6282      1D33      ;
6283      1D34      ;
6284      1D35      ;
6285      1D36      ;
6286      1D37      ;
6287      1D38      ;
6288      1D39      ;
6289      1D3A      ;
6290      1D3B      ;
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6745      1F03      ;
6746      1F04      ;
6747      1F05      ;
6748      1F06      ;
6749      1F07     
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5818 1B64 FE 09 CP
5819 1B66 20 0E JR
5820
5821 ;
5822 ; MORSP1:
5823 1B68 3E 20 LD
5824 1B6A CD 1B63 CALL
5825 1B6D 3A F415 LD
5826 1B70 E6 07 AND
5827 1B72 20 F4 JR
5828 1B74 F1 POP
5829 1B75 C9 RET
5830
5831 ;
5832 ; NOTABL:
5833 1B76 D6 0D SUB
5834 1B78 28 0A JR
5835 1B7A 38 0B JR
5836 1B7C FE 13 CP
5837 1B7E 38 07 JR
5838 1B80 3A F415 LD
5839 1B83 3C INC
5840
5841 ;
5842 ; ZERLPL:
5843 1B84 32 F415 LD
5844
5845 ;
5846 ; LPTCH0:
5847 1B87 3A F417 LD
5848 1B89 A7 AND
5849 1B8B 28 1E JR
5850 1B8D F1 POP
5851 1B8E CD 089D CALL
5852
5853 ;TAB?
5854 ;No
5855
5856 ;Print a space
5857 ;Get current LPOS
5858 ;At TAB stop?
5859 ;No, back for more space
5860 ;Discard character
5861
5862 ;Check if CR. If so load a zero
5863 ;It is, clear LPTPOS and send CR
5864 ;Code is 0..0CH, just send.
5865 ;without modify LPTPOS
5866 ;See if control character
5867 ;Code is 0EH..1FH, ditto
5868 ;Get LPOS
5869
5870 ;Update LPOS
5871
5872 ;Output to MSX standard printer
5873 ;No mapping for KATAKANA to HIRAGANA
5874 ;restore char to print
5875 ;See if graphic header

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( MSX ROM BASIC BIOS ) Macro-80
- BIO - OUTDO routine

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5849      1B91      D0      RET      NC      ;Yep
5850      1B92      20 23      JR      NZ,MAPSPC      ;Graphic symbol, map to space
5851      1B94      A7      AND      A      ;
5852      1B95      F2 1BAC      JP      P,LPTCHR
5853      1B98      FE 86      CP      86H      ;Graphic symbol?
5854      1B9A      38 1B      JR      C,MAPSPC      ;Yes, map this to space too!
5855      1B9C      FE A0      CP      0A0H      ;A HIRAGANA(part 1)?
5856      1B9E      30 04      JR      NC,NTHIRA
5857      1BA0      C6 20      ADD     A,' '
5858      1BA2      18 08      JR      LPTCHR      ;Map to KATAKANA
5859      1BA4      NTHIRA:
5860      1BA4      FE E0      CP      0E0H      ;HIRAGANA(part 2)?
5861      1BA6      38 04      JR      C,LPTCHR      ;No
5862      1BA8      D6 20      SUB     ' '      ;Map to KATAKANA
5863      1BAA      38      DB      38H      ;'JRC' instruction (Skip next byte)
5864      1BAB      LPTCH1:
5865      1BAB      F1      POP     AF      ;Restore char
5866      1BAC      ;
5867      1BAC      LPTCHR:
5868      1BAC      CD 085D      CALL    LPTOUT
5869      1BAF      D0      RET     NC      ;Send character out
5870      1BB0      DD 21 73B2      LD      IX,DIOERR      ;Sent successful
5871      1BB4      C3 01FF      JP      CALBAS      ;Direct I/O error
5872      1BB7      MAPSPC:
5873      1BB7      3E 20      LD      A,' '
5874      1BB9      18 F1      JR      LPTCHR
5875      1BBB      TTYCHR:
5876      ;
5877      ; Output to console
5878      ;
5879      1BBB      F1      POP     AF      ;Get the character

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(MSX ROM BASIC BIOS) Macro-80
- BIO - OUTDO routine

5880 1BBC C3 08BC
5881

1BBC C3 08BC

5913	1C25	50 00 7C			
5914	1C28	20 7C 44 7C	DB	20H,7CH,44H,7CH,44H,7CH,00H	
5915	1C2C	44 7C 00			
5916	1C2F	0C 70 10 FE	DB	0CH,70H,10H,0FEH,10H,10H,10H	
5917	1C33	10 10 10			
5918	1C36	00 7E 10 1E	DB	00H,7EH,10H,1EH,12H,22H,44H	
5919	1C3A	12 22 44			
5920	1C3D	08 00 00 7C	DB	08H,00H,00H,7CH,28H,28H,28H	
5921	1C41	28 28 28			
5922	1C44	4E 00 00 10	DB	4EH,00H,00H,10H,10H,10H,0FFH	
5923	1C48	10 10 FF			
5924	1C4B	00 00 00 00	DB	00H,00H,00H,00H,00H,00H,00H	
5925	1C4F	00 00 00			
5926	1C52	FF 10 10 10	DB	0FFH,10H,10H,10H,10H,10H,10H	
5927	1C56	10 10 10			
5928	1C59	10 F0 10 10	DB	10H,0F0H,10H,10H,10H,10H,10H	
5929	1C5D	10 10 10			
5930	1C60	10 10 1F 10	DB	10H,10H,1FH,10H,10H,10H,10H	
5931	1C64	10 10 10			
5932	1C67	10 10 10 FF	DB	10H,10H,10H,0FFH,10H,10H,10H	
5933	1C6B	10 10 10			
5934	1C6E	10 10 10 10	DB	10H,10H,10H,10H,10H,10H,10H	
5935	1C72	10 10 10			
5936	1C75	10 10 00 00	DB	10H,10H,00H,00H,00H,0FFH,00H	
5937	1C79	00 FF 00			
5938	1C7C	00 00 00 00	DB	00H,00H,00H,00H,00H,00H,1FH	
5939	1C80	00 00 1F			
5940	1C83	10 10 10 10	DB	10H,10H,10H,10H,10H,00H,00H,00H	
5941	1C87	00 00 00			
5942	1C8A	F0 10 10 10	DB	0F0H,10H,10H,10H,10H,10H,10H,10H	
5943	1C8E	10 10 10			

5944	1C91	10 1F 00 00	DB	10H,1FH,00H,00H,00H,00H,10H
5945	1C95	00 00 10		
5946	1C98	10 10 F0 00	DB	10H,10H,0F0H,00H,00H,00H,00H
5947	1C9C	00 00 00		
5948	1C9F	81 42 24 18	DB	81H,42H,24H,18H,18H,24H,42H
5949	1CA3	18 24 42		
5950	1CA6	81 10 7C 10	DB	81H,10H,7CH,10H,10H,28H,44H
5951	1CAA	10 28 44		
5952	1CAD	82 00 10 10	DB	82H,00H,10H,10H,0FEH,92H,0FEH
5953	1CB1	FE 92 FE		
5954	1CB4	10 10 00 10	DB	10H,10H,00H,10H,10H,54H,54H
5955	1CB8	10 54 54		
5956	1CBB	92 10 30 00	DB	92H,10H,30H,00H,00H,00H,00H
5957	1CBF	00 00 00		
5958	1CC2	00 00 00 00	DB	00H,00H,00H,00H,00H,20H,20H
5959	1CC6	00 20 20		
5960	1CC9	20 20 00 00	DB	20H,20H,00H,00H,20H,00H,50H
5961	1CCD	20 00 50		
5962	1CD0	50 50 00 00	DB	50H,50H,00H,00H,00H,00H,00H
5963	1CD4	00 00 00		
5964	1CD7	50 50 F8 50	DB	50H,50H,0F8H,50H,0F8H,50H,50H
5965	1CDB	F8 50 50		
5966	1CDE	00 20 78 A0	DB	00H,20H,78H,0A0H,70H,28H,0F0H
5967	1CE2	70 28 F0		
5968	1CE5	20 00 C0 C8	DB	20H,00H,0C0H,0C8H,10H,20H,40H
5969	1CE9	10 20 40		
5970	1CEC	98 18 00 40	DB	98H,18H,00H,40H,0A0H,40H,0A8H
5971	1CF0	A0 40 A8		
5972	1CF3	90 98 60 00	DB	90H,98H,60H,00H,10H,20H,40H
5973	1CF7	10 20 40		
5974	1CFA	00 00 00 00	DB	00H,00H,00H,00H,00H,10H,20H

5975	1CFE	00 10 20			
5976	1D01	40 40 40 20	DB	40H,40H,40H,20H,10H,00H,40H	
5977	1D05	10 00 40			
5978	1D08	20 10 10 10	DB	20H,10H,10H,10H,20H,40H,00H	
5979	1D0C	20 40 00			
5980	1D0F	20 A8 70 20	DB	20H,0A8H,70H,20H,70H,0A8H,20H	
5981	1D13	70 A8 20			
5982	1D16	00 00 20 20	DB	00H,00H,20H,20H,0F8H,20H,20H	
5983	1D1A	F8 20 20			
5984	1D1D	00 00 00 00	DB	00H,00H,00H,00H,00H,00H,00H	
5985	1D21	00 00 00			
5986	1D24	20 20 40 00	DB	20H,20H,40H,00H,00H,00H,78H	
5987	1D28	00 00 78			
5988	1D2B	00 00 00 00	DB	00H,00H,00H,00H,00H,00H,00H	
5989	1D2F	00 00 00			
5990	1D32	00 00 60 60	DB	00H,00H,60H,60H,00H,00H,00H	
5991	1D36	00 00 00			
5992	1D39	08 10 20 40	DB	08H,10H,20H,40H,80H,00H,70H	
5993	1D3D	80 00 70			
5994	1D40	88 98 A8 C8	DB	88H,98H,0A8H,0C8H,88H,70H,00H	
5995	1D44	88 70 00			
5996	1D47	20 60 A0 20	DB	20H,60H,0A0H,20H,20H,20H,0F8H	
5997	1D4B	20 20 F8			
5998	1D4E	00 70 88 08	DB	00H,70H,88H,08H,10H,60H,80H	
5999	1D52	10 60 80			
6000	1D55	F8 00 70 88	DB	0F8H,00H,70H,88H,08H,30H,08H	
6001	1D59	08 30 08			
6002	1D5C	88 70 00 10	DB	88H,70H,00H,10H,30H,50H,90H	
6003	1D60	30 50 90			
6004	1D63	F8 10 10 00	DB	0F8H,10H,10H,00H,0F8H,80H,0E0H	
6005	1D67	F8 80 E0			

(MSX ROM BASIC BIOS) Macro-80 -MSXCHR- MSX character set		3.44	01-Jan-85	PAGE	63-4
6006	1D6A 10 08 10 E0	DB	10H,08H,10H,0E0H,00H,30H,40H		
6007	1D6E 00 30 40				
6008	1D71 80 F0 88 88	DB	80H,0F0H,88H,88H,70H,00H,0F8H		
6009	1D75 70 00 F8				
6010	1D78 88 10 20 20	DB	88H,10H,20H,20H,20H,20H,00H		
6011	1D7C 20 20 00				
6012	1D7F 70 88 88 70	DB	70H,88H,88H,70H,88H,88H,70H		
6013	1D83 88 88 70				
6014	1D86 00 70 88 88	DB	00H,70H,88H,88H,78H,08H,10H		
6015	1D8A 78 08 10				
6016	1D8D 60 00 00 00	DB	60H,00H,00H,00H,20H,00H,00H		
6017	1D91 20 00 00				
6018	1D94 20 00 00 00	DB	20H,00H,00H,00H,00H,20H,00H		
6019	1D98 00 20 00				
6020	1D9B 00 20 20 40	DB	00H,20H,20H,40H,18H,30H,60H		
6021	1D9F 18 30 60				
6022	1DA2 C0 60 30 18	DB	0C0H,60H,30H,18H,00H,00H,00H		
6023	1DA6 00 00 00				
6024	1DA9 F8 00 F8 00	DB	0F8H,00H,0F8H,00H,00H,00H,0C0H		
6025	1DAD 00 00 C0				
6026	1DB0 60 30 18 30	DB	60H,30H,18H,30H,60H,0C0H,00H		
6027	1DB4 60 C0 00				
6028	1DB7 70 88 08 10	DB	70H,88H,08H,10H,20H,00H,20H		
6029	1DBB 20 00 20				
6030	1DBE 00 70 88 08	DB	00H,70H,88H,08H,68H,0A8H,0A8H		
6031	1DC2 68 A8 A8				
6032	1DC5 70 00 20 50	DB	70H,00H,20H,50H,88H,88H,0F8H		
6033	1DC9 88 88 F8				
6034	1DCC 88 88 00 F0	DB	88H,88H,00H,0F0H,48H,48H,70H		
6035	1DD0 48 48 70				
6036	1DD3 48 48 F0 00	DB	48H,48H,0F0H,00H,30H,48H,80H		

6037	1DD7	30 48 80			
6038	1DDA	80 80 48 30	DB	80H,80H,48H,30H,00H,0E0H,50H	
6039	1DDE	00 E0 50			
6040	1DE1	48 48 48 50	DB	48H,48H,48H,50H,0E0H,00H,0F8H	
6041	1DE5	E0 00 F8			
6042	1DE8	80 80 F0 80	DB	80H,80H,0F0H,80H,80H,0F8H,00H	
6043	1DEC	80 F8 00			
6044	1DEF	F8 80 80 F0	DB	0F8H,80H,80H,0F0H,80H,80H,80H	
6045	1DF3	80 80 80			
6046	1DF6	00 70 88 80	DB	00H,70H,88H,80H,0B8H,88H,88H	
6047	1DFA	B8 88 88			
6048	1DFD	70 00 88 88	DB	70H,00H,88H,88H,88H,0F8H,88H	
6049	1E01	88 F8 88			
6050	1E04	88 88 00 70	DB	88H,88H,00H,70H,20H,20H,20H	
6051	1E08	20 20 20			
6052	1E0B	20 20 70 00	DB	20H,20H,70H,00H,38H,10H,10H	
6053	1E0F	38 10 10			
6054	1E12	10 90 90 60	DB	10H,90H,90H,60H,00H,88H,90H	
6055	1E16	00 88 90			
6056	1E19	A0 C0 A0 90	DB	0A0H,0C0H,0A0H,90H,88H,00H,80H	
6057	1E1D	88 00 80			
6058	1E20	80 80 80 80	DB	80H,80H,80H,80H,80H,0F8H,00H	
6059	1E24	80 F8 00			
6060	1E27	88 D8 A8 A8	DB	88H,0D8H,0A8H,0A8H,88H,88H,88H	
6061	1E2B	88 88 88			
6062	1E2E	00 88 C8 C8	DB	00H,88H,0C8H,0C8H,0A8H,98H,98H	
6063	1E32	A8 98 98			
6064	1E35	88 00 70 88	DB	88H,00H,70H,88H,88H,88H,88H	
6065	1E39	88 88 88			
6066	1E3C	88 70 00 F0	DB	88H,70H,00H,0F0H,88H,88H,0F0H	
6067	1E40	88 88 F0			

6068	1E43	80 80 80 00	DB	80H,80H,80H,00H,70H,88H,88H
6069	1E47	70 88 88		
6070	1E4A	88 A8 90 68	DB	88H,0A8H,90H,68H,00H,0F0H,88H
6071	1E4E	00 F0 88		
6072	1E51	88 F0 A0 90	DB	88H,0F0H,0A0H,90H,88H,00H,70H
6073	1E55	88 00 70		
6074	1E58	88 80 70 08	DB	88H,80H,70H,08H,88H,70H,00H
6075	1E5C	88 70 00		
6076	1E5F	F8 20 20 20	DB	0F8H,20H,20H,20H,20H,20H,20H
6077	1E63	20 20 20		
6078	1E66	00 88 88 88	DB	00H,88H,88H,88H,88H,88H,88H
6079	1E6A	88 88 88		
6080	1E6D	70 00 88 88	DB	70H,00H,88H,88H,88H,88H,50H
6081	1E71	88 88 50		
6082	1E74	50 20 00 88	DB	50H,20H,00H,88H,88H,88H,0A8H
6083	1E78	88 88 A8		
6084	1E7B	A8 D8 88 00	DB	0A8H,0D8H,88H,00H,88H,88H,50H
6085	1E7F	88 88 50		
6086	1E82	20 50 88 88	DB	20H,50H,88H,88H,00H,88H,88H
6087	1E86	00 88 88		
6088	1E89	88 70 20 20	DB	88H,70H,20H,20H,20H,00H,0F8H
6089	1E8D	20 00 F8		
6090	1E90	08 10 20 40	DB	08H,10H,20H,40H,80H,0F8H,00H
6091	1E94	80 F8 00		
6092	1E97	70 40 40 40	DB	70H,40H,40H,40H,40H,40H,70H
6093	1E9B	40 40 70		
6094	1E9E	00 88 50 20	DB	00H,88H,50H,20H,70H,20H,70H
6095	1EA2	70 20 70		
6096	1EA5	20 00 70 10	DB	20H,00H,70H,10H,10H,10H,10H
6097	1EA9	10 10 10		
6098	1EAC	10 70 00 20	DB	10H,70H,00H,20H,50H,88H,00H

6099	1EB0	50 88 00			
6100	1EB3	00 00 00 00			
6101	1EB7	00 00 00	DB	00H,00H,00H,00H,00H,00H,00H	
6102	1EBA	00 00 00 F8	DB	00H,00H,00H,0F8H,00H,40H,20H	
6103	1EBE	00 40 20	DB	10H,00H,00H,00H,00H,00H,00H	
6104	1EC1	10 00 00 00	DB	00H,70H,08H,78H,88H,78H,00H	
6105	1EC5	00 00 00	DB	80H,80H,0B0H,0C8H,88H,0C8H,0B0H	
6106	1EC8	00 70 08 78	DB	00H,00H,00H,70H,88H,80H,88H	
6107	1ECC	88 78 00	DB	70H,00H,08H,08H,68H,98H,88H	
6108	1ECF	80 80 B0 C8	DB	98H,68H,00H,00H,00H,70H,88H	
6109	1ED3	88 C8 B0	DB	0F8H,80H,70H,00H,10H,28H,20H	
6110	1ED6	00 00 00 70	DB	0F8H,20H,20H,20H,00H,00H,00H	
6111	1EDA	88 80 88	DB	68H,98H,98H,68H,08H,70H,80H	
6112	1EDD	70 00 08 08	DB	80H,0F0H,88H,88H,88H,88H,00H	
6113	1EE1	68 98 88	DB	20H,00H,60H,20H,20H,20H,70H	
6114	1EE4	98 68 00 00	DB	00H,10H,00H,30H,10H,10H,10H	
6115	1EE8	00 70 88	DB	90H,60H,40H,40H,48H,50H,60H	
6116	1EEB	F8 80 70 00			
6117	1EEF	10 28 20			
6118	1EF2	F8 20 20 20			
6119	1EF6	00 00 00			
6120	1EF9	68 98 98 68			
6121	1EFD	08 70 80			
6122	1F00	80 F0 88 88			
6123	1F04	88 88 00			
6124	1F07	20 00 60 20			
6125	1F0B	20 20 70			
6126	1F0E	00 10 00 30			
6127	1F12	10 10 10			
6128	1F15	90 60 40 40			
6129	1F19	48 50 60			

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6130 1F1C 50 48 00 60	DB	50H,48H,00H,60H,20H,20H,20H		
6131 1F20 20 20 20				
6132 1F23 20 20 70 00	DB	20H,20H,70H,00H,00H,00H,0D0H		
6133 1F27 00 00 D0				
6134 1F2A A8 A8 A8 A8	DB	0A8H,0A8H,0A8H,0A8H,00H,00H,00H		
6135 1F2E 00 00 00				
6136 1F31 B0 C8 88 88	DB	0B0H,0C8H,88H,88H,88H,00H,00H		
6137 1F35 88 00 00				
6138 1F38 00 70 88 88	DB	00H,70H,88H,88H,88H,70H,00H		
6139 1F3C 88 70 00				
6140 1F3F 00 00 B0 C8	DB	00H,00H,0B0H,0C8H,0C8H,0B0H,80H		
6141 1F43 C8 B0 80				
6142 1F46 80 00 00 68	DB	80H,00H,00H,68H,98H,98H,68H		
6143 1F4A 98 98 68				
6144 1F4D 08 08 00 00	DB	08H,08H,00H,00H,0B0H,0C8H,80H		
6145 1F51 B0 C8 80				
6146 1F54 80 80 00 00	DB	80H,80H,00H,00H,00H,78H,80H		
6147 1F58 00 78 80				
6148 1F5B F0 08 F0 00	DB	0F0H,08H,0F0H,00H,40H,40H,0F0H		
6149 1F5F 40 40 F0				
6150 1F62 40 40 48 30	DB	40H,40H,48H,30H,00H,00H,00H		
6151 1F66 00 00 00				
6152 1F69 90 90 90 90	DB	90H,90H,90H,90H,68H,00H,00H		
6153 1F6D 68 00 00				
6154 1F70 00 88 88 88	DB	00H,88H,88H,88H,50H,20H,00H		
6155 1F74 50 20 00				
6156 1F77 00 00 88 A8	DB	00H,00H,88H,0A8H,0A8H,0A8H,50H		
6157 1F7B A8 A8 50				
6158 1F7E 00 00 00 88	DB	00H,00H,00H,88H,50H,20H,50H		
6159 1F82 50 20 50				
6160 1F85 88 00 00 00	DB	88H,00H,00H,00H,88H,88H,98H		

6161	1F89	88 88 98		
6162	1F8C	68 08 70 00	DB	68H,08H,70H,00H,00H,0F8H,10H
6163	1F90	00 F8 10		
6164	1F93	20 40 F8 00	DB	20H,40H,0F8H,00H,18H,20H,20H
6165	1F97	18 20 20		
6166	1F9A	40 20 20 18	DB	40H,20H,20H,18H,00H,20H,20H
6167	1F9E	00 20 20		
6168	1FA1	20 00 20 20	DB	20H,00H,20H,20H,20H,00H,0C0H
6169	1FA5	20 00 C0		
6170	1FA8	20 20 10 20	DB	20H,20H,10H,20H,20H,0C0H,00H
6171	1FAC	20 C0 00		
6172	1FAF	40 A8 10 00	DB	40H,0A8H,10H,00H,00H,00H,00H
6173	1FB3	00 00 00		
6174	1FB6	00 00 00 00	DB	00H,00H,00H,00H,00H,00H,00H
6175	1FBA	00 00 00		
6176	1FBD	00 00 10 38	DB	00H,00H,10H,38H,7CH,0FEH,0FEH
6177	1FC1	7C FE FE		
6178	1FC4	38 7C 00 6C	DB	38H,7CH,00H,6CH,0FEH,0FEH,0FEH
6179	1FC8	FE FE FE		
6180	1FCB	7C 38 10 00	DB	7CH,38H,10H,00H,38H,38H,0FEH
6181	1FCF	38 38 FE		
6182	1FD2	FE D6 10 7C	DB	0FEH,0D6H,10H,7CH,00H,10H,38H
6183	1FD6	00 10 38		
6184	1FD9	7C FE 7C 38	DB	7CH,0FEH,7CH,38H,10H,00H,00H
6185	1FDD	10 00 00		
6186	1FE0	78 84 84 84	DB	78H,84H,84H,84H,84H,78H,00H
6187	1FE4	84 78 00		
6188	1FE7	00 78 FC FC	DB	00H,78H,0FCH,0FCH,0FCH,0FCH,78H
6189	1FEB	FC FC 78		
6190	1FEE	00 40 FE 48	DB	00H,40H,0FEH,48H,70H,48H,82H
6191	1FF2	70 48 82		

6192	1FF5	7C 00 00 00	DB	7CH,00H,00H,00H,10H,7EH,3CH
6193	1FF9	10 7E 3C		
6194	1FFC	5A 34 00 00	DB	5AH,34H,00H,00H,00H,40H,42H
6195	2000	00 40 42		
6196	2003	42 52 20 00	DB	42H,52H,20H,00H,00H,00H,1CH
6197	2007	00 00 1C		
6198	200A	1C 22 02 0C	DB	1CH,22H,02H,0CH,00H,00H,00H
6199	200E	00 00 00		
6200	2011	18 7E 18 30	DB	18H,7EH,18H,30H,6EH,00H,00H
6201	2015	6E 00 00		
6202	2018	00 12 7E 3C	DB	00H,12H,7EH,3CH,52H,34H,00H
6203	201C	52 34 00		
6204	201F	00 00 28 7C	DB	00H,00H,28H,7CH,2AH,22H,24H
6205	2023	2A 22 24		
6206	2026	00 00 00 08	DB	00H,00H,00H,08H,5CH,6AH,0CH
6207	202A	5C 6A 0C		
6208	202D	30 00 00 00	DB	30H,00H,00H,00H,08H,0EH,38H
6209	2031	08 0E 38		
6210	2034	4C 3A 00 00	DB	4CH,3AH,00H,00H,00H,00H,3CH
6211	2038	00 00 3C		
6212	203B	02 02 1C 00	DB	02H,02H,1CH,00H,00H,00H,00H
6213	203F	00 00 00		
6214	2042	00 00 00 00	DB	00H,00H,00H,00H,00H,20H,0FEH
6215	2046	00 20 FE		
6216	2049	20 7C AA B2	DB	20H,7CH,0AAH,0B2H,64H,00H,00H
6217	204D	64 00 00		
6218	2050	80 82 82 82	DB	80H,82H,82H,82H,90H,60H,00H
6219	2054	90 60 00		
6220	2057	1C 00 7C 02	DB	1CH,00H,7CH,02H,02H,04H,18H
6221	205B	02 04 18		
6222	205E	00 38 00 FE	DB	00H,38H,00H,0FEH,08H,30H,50H

6223	2062	08 30 50			
6224	2065	9E 00 20 FA	DB		9EH,00H,20H,0FAH,22H,7CH,0A2H
6225	2069	22 7C A2			
6226	206C	A2 4C 00 40	DB		0A2H,4CH,00H,40H,44H,0F2H,4AH
6227	2070	44 F2 4A			
6228	2073	48 88 30 00	DB		48H,88H,30H,00H,10H,0FCH,08H
6229	2077	10 FC 08			
6230	207A	3E 04 80 7C	DB		3EH,04H,80H,7CH,00H,18H,18H
6231	207E	00 18 18			
6232	2081	30 60 60 30	DB		30H,60H,60H,30H,18H,00H,04H
6233	2085	18 00 04			
6234	2088	84 BE 84 84	DB		84H,0BEH,84H,84H,84H,48H,00H
6235	208C	84 48 00			
6236	208F	00 FC 02 00	DB		00H,0FCH,02H,00H,40H,80H,7EH
6237	2093	40 80 7E			
6238	2096	00 10 16 F8	DB		00H,10H,16H,0F8H,08H,7CH,80H
6239	209A	08 7C 80			
6240	209D	78 00 80 80	DB		78H,00H,80H,80H,80H,80H,84H
6241	20A1	80 80 84			
6242	20A4	88 70 00 08	DB		88H,70H,00H,08H,0FEH,08H,38H
6243	20A8	FE 08 38			
6244	20AB	48 38 08 00	DB		48H,38H,08H,00H,04H,44H,0FEH
6245	20AF	04 44 FE			
6246	20B2	44 44 40 3E	DB		44H,44H,40H,3EH,00H,64H,28H
6247	20B6	00 64 28			
6248	20B9	30 FE 20 40	DB		30H,0FEH,20H,40H,3CH,00H,00H
6249	20BD	3C 00 00			
6250	20C0	00 00 00 00	DB		00H,00H,00H,00H,00H,00H,00H
6251	20C4	00 00 00			
6252	20C7	00 00 00 00	DB		00H,00H,00H,00H,60H,90H,60H
6253	20CB	60 90 60			

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6254	20CE 00 38 20 20	DB	00H,38H,20H,20H,20H,00H,00H		
6255	20D2 20 00 00				
6256	20D5 00 00 00 00	DB	00H,00H,00H,00H,00H,20H,20H		
6257	20D9 00 20 20				
6258	20DC 20 E0 00 00	DB	20H,0E0H,00H,00H,00H,00H,00H		
6259	20E0 00 00 00				
6260	20E3 80 40 20 00	DB	80H,40H,20H,00H,00H,00H,00H		
6261	20E7 00 00 00				
6262	20EA 30 30 00 00	DB	30H,30H,00H,00H,00H,0F8H,08H		
6263	20EE 00 F8 08				
6264	20F1 F8 08 10 20	DB	0F8H,08H,10H,20H,40H,00H,00H		
6265	20F5 40 00 00				
6266	20F8 00 F0 10 60	DB	00H,0F0H,10H,60H,40H,80H,00H		
6267	20FC 40 80 00				
6268	20FF 00 10 20 60	DB	00H,10H,20H,60H,0A0H,20H,20H		
6269	2103 A0 20 20				
6270	2106 00 00 20 F0	DB	00H,00H,20H,0F0H,90H,10H,20H		
6271	210A 90 10 20				
6272	210D 40 00 00 00	DB	40H,00H,00H,00H,0F0H,20H,20H		
6273	2111 F0 20 20				
6274	2114 20 F0 00 00	DB	20H,0F0H,00H,00H,20H,0F0H,60H		
6275	2118 20 F0 60				
6276	211B A0 A0 20 00	DB	0A0H,0A0H,20H,00H,00H,40H,0F8H		
6277	211F 00 40 F8				
6278	2122 48 50 40 40	DB	48H,50H,40H,40H,00H,00H,00H		
6279	2126 00 00 00				
6280	2129 70 10 10 10	DB	70H,10H,10H,10H,0F8H,00H,00H		
6281	212D F8 00 00				
6282	2130 00 F0 10 F0	DB	00H,0F0H,10H,0F0H,10H,0F0H,00H		
6283	2134 10 F0 00				
6284	2137 00 00 A8 A8	DB	00H,00H,0A8H,0A8H,08H,10H,20H		

6285	213B	08 10 20			
6286	213E	00 00 00 00	DB	00H,00H,00H,00H,0F8H,00H,00H	
6287	2142	F8 00 00			
6288	2145	00 00 F8 08	DB	00H,00H,0F8H,08H,28H,30H,20H	
6289	2149	28 30 20			
6290	214C	20 40 00 08	DB	20H,40H,00H,08H,10H,20H,60H	
6291	2150	10 20 60			
6292	2153	A0 20 20 00	DB	0A0H,20H,20H,00H,20H,0F8H,88H	
6293	2157	20 F8 88			
6294	215A	88 08 10 20	DB	88H,08H,10H,20H,00H,00H,0F8H	
6295	215E	00 00 F8			
6296	2161	20 20 20 20	DB	20H,20H,20H,20H,0F8H,00H,10H	
6297	2165	F8 00 10			
6298	2168	F8 10 30 50	DB	0F8H,10H,30H,50H,90H,10H,00H	
6299	216C	90 10 00			
6300	216F	20 F8 28 28	DB	20H,0F8H,28H,28H,28H,48H,88H	
6301	2173	28 48 88			
6302	2176	00 20 F8 20	DB	00H,20H,0F8H,20H,0F8H,20H,20H	
6303	217A	F8 20 20			
6304	217D	20 00 78 48	DB	20H,00H,78H,48H,88H,08H,08H	
6305	2181	88 08 08			
6306	2184	10 20 00 40	DB	10H,20H,00H,40H,78H,50H,90H	
6307	2188	78 50 90			
6308	218B	10 10 20 00	DB	10H,10H,20H,00H,00H,0F8H,08H	
6309	218F	00 F8 08			
6310	2192	08 08 08 F8	DB	08H,08H,08H,0F8H,00H,50H,0F8H	
6311	2196	00 50 F8			
6312	2199	50 50 10 10	DB	50H,50H,10H,10H,20H,00H,00H	
6313	219D	20 00 00			
6314	21A0	C0 08 C8 08	DB	0C0H,08H,0C8H,08H,10H,0E0H,00H	
6315	21A4	10 E0 00			

6316	21A7	00 F8 08 10	DB	00H,0F8H,08H,10H,20H,50H,88H
6317	21AB	20 50 88		
6318	21AE	00 40 F8 48	DB	00H,40H,0F8H,48H,50H,40H,40H
6319	21B2	50 40 40		
6320	21B5	38 00 88 88	DB	38H,00H,88H,88H,48H,08H,10H
6321	21B9	48 08 10		
6322	21BC	20 40 00 78	DB	20H,40H,00H,78H,48H,78H,88H
6323	21C0	48 78 88		
6324	21C3	08 10 20 00	DB	08H,10H,20H,00H,10H,0E0H,20H
6325	21C7	10 E0 20		
6326	21CA	F8 20 20 40	DB	0F8H,20H,20H,40H,00H,00H,0A8H,0A8H
6327	21CE	00 A8 A8		
6328	21D1	A8 08 08 10	DB	0A8H,08H,08H,10H,20H,00H,70H
6329	21D5	20 00 70		
6330	21D8	00 F8 20 20	DB	00H,0F8H,20H,20H,20H,40H,00H
6331	21DC	20 40 00		
6332	21DF	40 40 60 50	DB	40H,40H,60H,50H,48H,40H,40H
6333	21E3	48 40 40		
6334	21E6	00 20 F8 20	DB	00H,20H,0F8H,20H,20H,20H,20H
6335	21EA	20 20 20		
6336	21ED	40 00 00 70	DB	40H,00H,00H,70H,00H,00H,00H
6337	21F1	00 00 00		
6338	21F4	00 F8 00 00	DB	00H,0F8H,00H,00H,0F8H,08H,0D0H
6339	21F8	F8 08 D0		
6340	21FB	20 50 88 00	DB	20H,50H,88H,00H,20H,0F8H,08H
6341	21FF	20 F8 08		
6342	2202	30 E8 20 20	DB	30H,0E8H,20H,20H,00H,08H,08H
6343	2206	00 08 08		
6344	2209	08 10 20 40	DB	08H,10H,20H,40H,80H,00H,20H
6345	220D	80 00 20		
6346	2210	10 48 48 48	DB	10H,48H,48H,48H,48H,88H,00H

6347	2214	48 88 00			
6348	2217	80 80 F8 80	DB		80H,80H,0F8H,80H,80H,80H,78H
6349	221B	80 80 78			
6350	221E	00 F8 08 08	DB		00H,0F8H,08H,08H,10H,20H
6351	2222	08 10 20			
6352	2225	40 00 00 40	DB		40H,00H,00H,40H,0A0H,10H,08H
6353	2229	A0 10 08			
6354	222C	08 00 00 20	DB		08H,00H,00H,20H,0F8H,20H,20H
6355	2230	F8 20 20			
6356	2233	A8 A8 20 00	DB		0A8H,0A8H,20H,00H,00H,00H,0F8H,08H
6357	2237	00 F8 08			
6358	223A	08 50 20 10	DB		08H,50H,20H,10H,00H,0F0H,00H
6359	223E	00 F0 00			
6360	2241	60 00 00 F0	DB		60H,00H,00H,0F0H,08H,00H,10H
6361	2245	08 00 10			
6362	2248	20 40 80 90	DB		20H,40H,80H,90H,88H,0F8H,00H
6363	224C	88 F8 00			
6364	224F	08 08 08 50	DB		08H,08H,08H,50H,20H,50H,80H
6365	2253	20 50 80			
6366	2256	00 78 20 F8	DB		00H,78H,20H,0F8H,20H,20H,20H
6367	225A	20 20 20			
6368	225D	18 00 40 F8	DB		18H,00H,40H,0F8H,48H,48H,50H
6369	2261	48 48 50			
6370	2264	40 40 00 00	DB		40H,40H,00H,00H,70H,10H,10H
6371	2268	70 10 10			
6372	226B	10 10 F8 00	DB		10H,10H,0F8H,00H,00H,0F8H,08H
6373	226F	00 F8 08			
6374	2272	F8 08 08 F8	DB		0F8H,08H,08H,0F8H,00H,70H,00H
6375	2276	00 70 00			
6376	2279	F8 08 08 10	DB		0F8H,08H,08H,10H,20H,00H,48H
6377	227D	20 00 48			

6378	2280	48 48 48 48	DB	48H,48H,48H,48H,10H,20H,00H
6379	2284	10 20 00		
6380	2287	10 50 50 50	DB	10H,50H,50H,50H,50H,58H,90H
6381	228B	50 58 90		
6382	228E	00 40 40 40	DB	00H,40H,40H,40H,48H,48H,50H
6383	2292	48 48 50		
6384	2295	60 00 00 F8	DB	60H,00H,00H,0F8H,88H,88H,88H
6385	2299	88 88 88		
6386	229C	88 F8 00 F8	DB	88H,0F8H,00H,0F8H,88H,88H,08H
6387	22A0	88 88 08		
6388	22A3	08 10 20 00	DB	08H,10H,20H,00H,00H,0C0H,00H
6389	22A7	00 C0 00		
6390	22AA	08 08 10 E0	DB	08H,08H,10H,0E0H,00H,90H,48H
6391	22AE	00 90 48		
6392	22B1	00 00 00 00	DB	00H,00H,00H,00H,00H,00H,60H
6393	22B5	00 00 60		
6394	22B8	90 60 00 00	DB	90H,60H,00H,00H,00H,00H,00H
6395	22BC	00 00 00		
6396	22BF	40 FE 40 5E	DB	40H,0FEH,40H,5EH,80H,0A0H,9EH
6397	22C3	80 A0 9E		
6398	22C6	00 20 FE 40	DB	00H,20H,0FEH,40H,0F8H,04H,04H
6399	22CA	F8 04 04		
6400	22CD	78 00 00 00	DB	78H,00H,00H,00H,0FCH,02H,02H
6401	22D1	FC 02 02		
6402	22D4	04 38 00 00	DB	04H,38H,00H,00H,0FEH,0CH,30H
6403	22D8	FE 0C 30		
6404	22DB	40 40 38 00	DB	40H,40H,38H,00H,10H,12H,1CH
6405	22DF	10 12 1C		
6406	22E2	30 40 40 3E	DB	30H,40H,40H,3EH,00H,24H,0F2H
6407	22E6	00 24 F2		
6408	22E9	48 48 9C AA	DB	48H,48H,9CH,0AAH,10H,00H,80H

6409	22ED	10 00 80			
6410	22F0	9E 80 80 A0	DB	9EH,80H,80H,0A0H,0BEH,0C0H,00H	
6411	22F4	BE C0 00			
6412	22F7	44 4C 7A AA	DB	44H,4CH,7AH,0AAH,0A6H,0AAH,6CH	
6413	22FB	A6 AA 6C			
6414	22FE	00 40 EC 52	DB	00H,40H,0ECH,52H,62H,0CEH,4AH	
6415	2302	62 CE 4A			
6416	2305	4C 00 00 38	DB	4CH,00H,00H,38H,54H,92H,0A2H	
6417	2309	54 92 A2			
6418	230C	A2 4C 00 04	DB	0A2H,4CH,00H,04H,0BEH,84H,84H	
6419	2310	BE 84 84			
6420	2313	9E A4 5C 00	DB	9EH,0A4H,5CH,00H,08H,4CH,0C6H	
6421	2317	08 4C C6			
6422	231A	46 44 44 38	DB	46H,44H,44H,38H,00H,20H,18H	
6423	231E	00 20 18			
6424	2321	20 16 8A CA	DB	20H,16H,8AH,0CAH,18H,00H,00H	
6425	2325	18 00 00			
6426	2328	20 70 D8 8C	DB	20H,70H,0D8H,8CH,06H,02H,00H	
6427	232C	06 02 00			
6428	232F	3E 84 BE 84	DB	3EH,84H,0BEH,84H,9CH,0A6H,18H	
6429	2333	9C A6 18			
6430	2336	00 08 7E 08	DB	00H,08H,7EH,08H,7EH,38H,4CH	
6431	233A	7E 38 4C			
6432	233D	3A 00 E0 24	DB	3AH,00H,0E0H,24H,24H,7EH,0A4H	
6433	2341	24 7E A4			
6434	2344	A4 68 00 20	DB	0A4H,68H,00H,20H,0FCH,24H,62H	
6435	2348	FC 24 62			
6436	234B	A0 62 3C 00	DB	0A0H,62H,3CH,00H,04H,44H,7CH	
6437	234F	04 44 7C			
6438	2352	C6 AA 92 64	DB	0C6H,0AAH,92H,64H,00H,20H,20H	
6439	2356	00 20 20			

(MSX ROM BASIC BIOS) Macro-80
-MSXCHR- MSX character set

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6440	2359	78 20 78 22	DB	78H,20H,78H,22H,1CH,00H,00H
6441	235D	1C 00 00		
6442	2360	48 FC 4A 42	DB	48H,0FCH,4AH,42H,4CH,40H,00H
6443	2364	4C 40 00		
6444	2367	08 BC CA 8A	DB	08H,0BCH,0CAH,8AH,0BCH,08H,30H
6445	236B	BC 08 30		
6446	236E	00 08 08 0E	DB	00H,08H,08H,0EH,08H,78H,8CH
6447	2372	08 78 8C		
6448	2375	72 00 38 84	DB	72H,00H,38H,84H,80H,0FCH,0C2H
6449	2379	80 FC C2		
6450	237C	02 38 00 00	DB	02H,38H,00H,00H,42H,42H,42H
6451	2380	42 42 42		
6452	2383	62 04 18 00	DB	62H,04H,18H,00H,7CH,08H,30H
6453	2387	7C 08 30		
6454	238A	DC 62 92 7C	DB	0DCH,62H,92H,7CH,00H,20H,2CH
6455	238E	00 20 2C		
6456	2391	F4 24 64 E4	DB	0F4H,24H,64H,0E4H,26H,00H,7CH
6457	2395	26 00 7C		
6458	2398	18 20 5C 82	DB	18H,20H,5CH,82H,02H,7CH,00H
6459	239C	02 7C 00		
6460	239F	40 60 DC 62	DB	40H,60H,0DCH,62H,42H,0C2H,5CH
6461	23A3	42 C2 5C		
6462	23A6	00 10 30 20	DB	00H,10H,30H,20H,70H,48H,0CEH
6463	23AA	70 48 CE		
6464	23AD	84 00 00 00	DB	84H,00H,00H,00H,00H,00H,00H
6465	23B1	00 00 00		
6466	23B4	00 00 00 00	DB	00H,00H,00H,00H,00H,00H,00H
6467	23B8	00 00 00		
6468	23BB	00 00 00 00	DB	00H,00H,00H,00H

SUBTTL - MSXINL, Screen editor - Line input and function character

Address	Operation	Comments
6470		
23BF	PINLIN:	
6471	;	
6472	; Main entry point	
6473	;	
6474		
23BF	CD FDBB	
23C2	3A F6AA	
23C5	A7	
23C6	20 0D	
6478		
6479		
23C8	2E 00	
23CA	18 14	
23CC		
6483		
6484		
6485		
23CC	CD FDE0	
23CF	3E 3F	
23D1	DF	
23D2	3E 20	
23D4	DF	
23D5		
23D5	CD FDE5	
23D8	2A F3DC	
23DB	2D	
23DC	C4 0C29	
23DF	2C	
23E0		
23E0	22 FBCA	
23E3	AF	
23E4	32 FC9B	
6500		

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
 - MSXINL, Screen editor - Line input and function character

```

6501 23E7
6502 23E7 CD 10CB CHGET
6503 23EA 21 2437 LD HL,SCITBL-2
6504 23ED 0E 0B LD C,0BH ;SCI Max
6505 23EF CD 0919 CALL INDJMP ;Do functions
6506 23F2 F5 PUSH AF ;Output a character
6507 23F3 C4 23FF CALL NZ,INLOUT
6508 23F6 F1 POP AF
6509 23F7 30 EE JR NC,INLIN2 ;Not a terminator
6510
6511 ; ; return to BASIC (break or CR)
6512 ;
6513 23F9 21 F55D LD HL,BUFMIN
6514 23FC C8 RET Z
6515 23FD 3F CCF
6516 23FE RETURN:
6517 23FE C9 RET

```

;Cnt-C, return with carry set
 ;No, return carry clear

```

6518
6519      23FF
6520
6521      23FF      F5      PUSH      AF
6522      2400      FE 09      CP      9
6523      2402      20 0F      JR      NZ,OUTNTB
6524      2404      F1      POP      AF
6525      2405
6526      2405      3E 20      LD
6527      2407      CD 23FF      CALL
6528      240A      3A F3DD      LD      A,(CSRX)
6529      240D      3D      DEC      A
6530      240E      E6 07      AND      7
6531      2410      20 F3      JR      NZ,OUTTAB
6532      2412      C9      RET
6533      2413
6534
6535      2413      F1      POP
6536      2414      21 FCA8      LD      HL,INSFLG
6537      2417      FE 01      CP      1
6538      2419      28 0B      JR      Z,INLOT0
6539      241B      FE 20      CP      ' '
6540      241D      38 09      JR      C,INLOT1
6541      241F      F5      PUSH      AF
6542      2420      7E      LD      A,(HL)
6543      2421      A7      AND      A
6544      2422      C4 24F2      CALL      NZ,INSERT
6545      2425      F1      POP      AF
6546      2426
6547      2426      DF      INLOT0:      RST      18H
6548      2427      C9      RET

```

INLOT:

;

;Save character to output
 ;TAB?

;Nope

;Discard stack

;Map to space

;Make it zero based.

;Reached TAB stop?

;Not yet, continue...

OUTNTB:

;

;Restore character
 ;points insert mode flag
 ;Graphic header byte?
 ;Yes, send as is
 ;control char?
 ;branch if so. - Reset insert mode
 ;save char to output
 ;get insert mode flag
 ;test
 ;if insert mode, make room to insert
 ;restore char to output

;output char

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
 - MSXINL, Screen editor - Line input and function character

```

6549      2428      INLOT1:
6550      ;
6551      2428      36 00      LD      (HL),0
6552      242A      DF      RST      18H
6553      242B      3E      DB      3EH
6554      242C      SETINS:
6555      242C      3E      DB      3EH
6556      242D      SETOVW:
6557      242D      AF      XOR      A
6558      242E      F5      PUSH     AF
6559      242F      CD 0A2E  CALL     CKERCS
6560      2432      F1      POP      AF
6561      2433      32 FCAA  LD      (CSTYLE),A
6562      2436      C3 09E1  JP      CKDPCS

```

;reset insert mode
 ;send this control char

;Set insert mode and exit

;Set overwrite mode

```
6563
6564      2439      SCITBL:
6565      ;
6566      ; Table of function characters
6567      ;
6568      2439      08      DB      08H      ;Delete previous char
6569      243A      2561      DW      DELETE
6570      243C      12      DB      12H      ;Toggle insert flag
6571      243D      24E5      DW      TGLINS
6572      243F      1B      DB      1BH      ;Escape
6573      2440      23FE      DW      RETURN
6574      2442      02      DB      02H      ;Back word
6575      2443      260E      DW      LBCKWD
6576      2445      06      DB      06H      ;Next word
6577      2446      25F8      DW      LNX.TWD
6578      2448      0E      DB      0EH
6579      2449      25D7      DW      LAPPND
6580      244B      05      DB      05H      ;Erase to end of line
6581      244C      25B9      DW      TRUNC
6582      244E      03      DB      03H      ;Abort
6583      244F      24C5      DW      LBREAK
6584      2451      0D      DB      0DH      ;Carriage return
6585      2452      245A      DW      LCRRET
6586      2454      15      DB      15H      ;Delete whole line
6587      2455      25AE      DW      LERASE
6588      2457      7F      DB      7FH      ;Delete character at cursor
6589      2458      2550      DW      LDELNX
6590      SUBTTL - MSXINL, Screen editor - Process special characters
```



```

6622 2479 28 14 JR Z,LCRNUL ;Yes, ignore this
6623 247B FE 20 CP ' ' ;Special graphic character?
6624 247D 30 0B JR NC,LCRNRM ;No, proceed normally
6625 247F 05 DEC B ;Decrement BUF size counter before storing
6626 2480 28 1D JR Z,LBLKSP ;At end of BUF, so ignore this
6627 2482 4F LD C,A ;Store header byte for graphic symbol
6628 2483 3E 01 LD A,1
6629 2485 12 LD (DE),A
6630 2486 13 INC DE
6631 2487 79 LD A,C
6632 2488 C6 40 ADD A,'@'
6633 248A LCRNRM:
6634 248A 12 LD (DE),A
6635 248B 13 INC DE
6636 248C 05 DEC B ;Store byte in buffer
6637 248D 28 10 JR Z,LBLKSP ;Bump buffer pointer
6638 248F LCRNUL: ;Decrement BUF size counter
6639 248F INC H ;At end of BUF
6640 2490 3A F3B0 LD A,(LINLEN) ;Next column
6641 2493 BC CP H ;Max column reached?
6642 2494 30 DB JR NC,LCR2 ;
6643 2496 D5 PUSH DE ;Not yet
6644 2497 CD 0C1D CALL GETTRM ;Save buffer pointer
6645 249A D1 POP DE ;Is this line terminated?
6646 249B 26 01 LD H,1 ;Restore buffer pointer
6647 249D 28 D1 JR Z,LCR1 ;Assume not, start from top of next line
6648 249F LBLKSP: ;No
6649 ;
6650 ; Suppress trailing blanks, [DE]=last+1
6651 ;
6652 249F 1B DEC DE ;Back up buffer pointer

```

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
 - MSXINL, Screen editor - Process special characters

```

6653 24A0 1A LD A,(DE) ;Get stored character
6654 24A1 FE 20 CP ;Is it space?
6655 24A3 28 FA JR Z,LBLKSP ;Yes, ignore this
6656 24A5 E5 PUSH HL
6657 24A6 D5 PUSH DE
6658 24A7 CD 09E1 CALL CKDPCS
6659 24AA D1 POP DE
6660 24AB E1 POP HL
6661 ;
6662 ; Terminate
6663 ;
6664 24AC 13 INC DE ;Point past last valid character
6665 24AD AF XOR A ;Load terminator
6666 24AE 12 LD (DE),A ;Put it in BUF
6667 24AF FAKECR:
6668 24AF 3E 0D LD A,0DH ;Load character to echo to console
6669 24B1 A7 AND A ;Reset Z-flag, (say not break)
6670 24B2 LNXTLN:
6671 24B2 F5 PUSH AF ;Save this flag
6672 24B3 CD 0C29 CALL TERMIN
6673 24B6 CD 088E CALL POSIT
6674 24B9 3E 0A LD A,0AH ;Save current cursor position
6675 24BB DF RST 18H ;Move cursor to start of next line
6676 24BC AF XOR A ;Clear possible INSFLG
6677 24BD 32 FCA8 LD (INSFLG),A
6678 24C0 F1 POP AF ;Restore flags
6679 24C1 37 POP SCF ;Set carry indicating end of input
6680 24C2 E1 POP HL ;Discard return address (XRA A;RET)
6681 24C3 C9 RET ;If break, Z flag is set
6682 24C4 LBREK0:
6683 ;

```

```

6684      ; Control-C input
6685      ;
6686      2C      24C4      2C
6687      24C5      LBREAK:
6688      24C5      CD 0C1D      CALL      GETTRM
6689      24C8      28 FA      JR      Z,LBREK0
6690      24CA      CD 242D      CALL      SETOVW
6691      24CD      AF      XOR      A
6692      24CE      32 F55E      LD      (BUF),A
6693      24D1      26 01      LD      H,1
6694      24D3      E5      PUSH      HL
6695      24D4      CD 04BD      CALL      GICINI
6696      24D7      CD 0454      CALL      CKSTTP
6697      24DA      E1      POP      HL
6698      24DB      38 D2      JR      C,FAKECR
6699      24DD      3A FBBl      LD      A,(BASROM)
6700      24E0      A7      AND      A
6701      24E1      20 CC      JR      NZ,FAKECR
6702      24E3      18 CD      JR      LNXTLN

```

;Bump line counter
 ;Line terminated?
 ;No, check next line
 ;Set to overwrite mode
 ;Load 0 in Acc, and set Z flag
 ;Say no character in BUF
 ;Set to first column
 ;Save cursor position
 ;Initialize sound chip and queue
 ;Check if STOP trap is active or not
 ;Yes, fake CR
 ;Executing BASIC program in ROM?
 ;Yes, fake CR

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
 - MSXINL, Screen editor - Process special characters

```

6703
6704      24E5      TGLINS:
6705      ;
6706      ; Toggle insert mode flag
6707      ;
6708      24E5      21 FCA8      LD HL,INSFLG
6709      24E8      7E          LD A,(HL)
6710      24E9      EE FF      XOR 0FFH
6711      24EB      77          LD (HL),A
6712      24EC      CA 242D      JP Z,SETOVW
6713      24EF      C3 242C      JP SETINS
6714      24F2      INSERT:
6715      ;
6716      ; Insert a blank
6717      ;
6718      24F2      CD 0A2E      CALL CKERCS
6719      24F5      2A F3DC      LD HL,(CSRY)
6720      24F8      0E 20      LD C,' '
6721      24FA      INSL:
6722      24FA      E5          PUSH HL
6723      24FB      INS2:
6724      24FB      C5          PUSH BC
6725      24FC      CD 0BD8      CALL GETVRM
6726      24FF      D1          POP DE
6727      2500      C5          PUSH BC
6728      2501      4B          LD C,E
6729      2502      CD 0BE6      CALL PUTVRM
6730      2505      C1          POP BC
6731      2506      3A F3B0      LD A,(LINLEN)
6732      2509      24          INC H
6733      250A      BC          CP H

```

;Get current insert flag
 ;Toggle insert status and affect Z flag
 ;Set to overwrite mode
 ;Set to insert mode
 ;Erase cursor before operation
 ;Load raw code for space
 ;Save current cursor position
 ;Save previous character
 ;Get current character in C
 ;Restore previous character in [E]
 ;Save current character
 ;C=previous character
 ;Put it on screen
 ;Restore current character in C
 ;Check if end of line
 ;Bump column counter
 ;End of line?

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
 - MSXINL, Screen editor - Process special characters

```

6734 250B 7A      LD      A,D      ;Get current attribute in Acc
6735 250C 30 ED    JR      NC,INS2  ;If not, continue till end of line
6736                      ;
6737                      ; Now we just finished a line, code of character wrapped to next
6738                      ; line is held in [C].
6739                      ;
6740 250E E1      POP     HL      ;Restore current cursor position
6741 250F CD 0C1D  CALL    GETTRM  ;Is this line terminated?
6742 2512 28 37    JR      Z,INS6   ;Line not terminated on this visual
6743                      ;
6744                      ; The current line is terminated. A check must be made to
6745                      ; determine if a wrapped character is a space, or we're inserting
6746                      ; at the end-of-line. If so, we have to open a next line to
6747                      ; insert.
6748                      ;
6749 2514 79      LD      A,C      ;Move last character to A for comparison
6750 2515 FE 20    CP      ' '
6751 2517 F5      PUSH    AF      ;Save the condition
6752 2518 20 0A    JR      NZ,INS3  ;No, open next line
6753 251A 3A F3B0  LD      A,(LINLEN) ;Are we trying to insert at the EOL?
6754 251D BC      CP      H      ;
6755 251E 28 04    JR      Z,INS3  ;Yes, open next line
6756 2520 F1      POP     AF      ;Discard stack
6757 2521 C3 09E1  JP      CKDPCS  ;Display cursor again
6758 2524          INS3:
6759                      ;
6760 2524 CD 0C2A   CALL    UNTERM  ;Terminate this line
6761 2527 2C      INC      L      ;Go to next row
6762 2528 C5      PUSH    BC      ;Save character code
6763 2529 E5      PUSH    HL      ;Save position of character in operation
6764 252A CD 0C32  CALL    GETLEN  ;Bottom of screen?

```

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
 - MSXINL, Screen editor - Process special characters

```

6765 252D BD CP L ;
6766 252E 38 05 JR C,INS4 ;Yes
6767
6768 ;
6769 ; Scroll down starting at line L
6770 2530 CD 0AB7 CALL INSLN0 ;Insert a blank line there
6771 2533 18 0F JR INS5
6772 2535
6773 INS4:
6774 ;
6775 ; Scroll up
6776
6777 2535 21 F3DC LD HL,CSRY
6778 2538 35 DEC (HL)
6779 2539 20 01 JR NZ,INS45
6780 253B 34 INC (HL)
6781 253C
6782 253C 2E 01 LD L,1
6783 253E CD 0A88 CALL DELLN0
6784 2541 E1 POP HL
6785 2542 2D DEC L
6786 2543 E5 PUSH HL
6787 2544
6788 2544 E1 POP HL
6789 2545 C1 POP BC
6790 2546 F1 POP AF
6791 2547 CA 09E1 JP Z,CKDPCS ;Restore flags
6792 254A 2D DEC L ;If we were trying to insert at the
6793 254B ; ;end-of-line, nothing else to do
6794 ; ;Cancel next 'INR L'
6795 ; Not end of logical line, pass character to next line

```

6796		:			
6797	254B	2C	INC	L	;Bump row counter
6798	254C	26 01	LD	H,1	;Start from first column
6799	254E	18 AA	JR	INSL	;Pass character to next line

```

6800
6801      2550      LDELNX:
6802      ;
6803      ; Delete current character
6804      ;
6805      2550      3A F3B0      LD      A,(LINLEN)
6806      2553      BC          CP      H
6807      2554      20 05      JR      NZ,LDELX1
6808      2556      CD 0C1D      CALL   GETTRM
6809      2559      20 3A      JR      NZ,DELET5
6810      255B      LDELX1:
6811      255B      3E 1C      LD      A,lCH
6812      255D      DF          RST     18H
6813      255E      2A F3DC      LD      HL,(CSRY)
6814      2561      DELETE:
6815      ;
6816      ; Delete previous character
6817      ;
6818      2561      E5          PUSH   HL
6819      2562      CD 0A2E      CALL   CKERCS
6820      2565      E1          POP    HL
6821      2566      25          DEC    H
6822      2567      C2 257A      JP     NZ,DELET2
6823      256A      24          INC    H
6824      256B      E5          PUSH   HL
6825      256C      2D          DEC    L
6826      256D      28 0A      JR      Z,DELET1
6827      256F      3A F3B0      LD      A,(LINLEN)
6828      2572      67          LD      H,A
6829      2573      CD 0C1D      CALL   GETTRM
6830      2576      20 01      JR      NZ,DELET1

;At rightmost position?
;Nope
;Is this a terminated line?
;Yes, place a space there.

;Move cursor right
;Fall into 'delete prev. character'

;Are we at top of line?
;No
;Yes
;Save current cursor position
;Look a line above
;At top of screen

;Is previous line terminated?
;Yes

```

6831	2578	E3	EX	(SP),HL	;No, substitute by current HL
6832	2579		DELET1:		
6833	2579	E1	POP	HL	;Get saved cursor position
6834	257A		DELET2:		
6835	257A	22 F3DC	LD	(CSRY),HL	;Set new cursor position
6836	257D		DELET3:		
6837	257D	3A F3B0	LD	A,(LINLEN)	
6838	2580	BC	CP	H	
6839	2581	28 12	JR	Z,DELET5	
6840	2583	24	INC	H	;Just over strike with blank
6841	2584		DELET4:		
6842	2584	CD 0BD8	CALL	GETVRM	;Get current character and attribute
6843	2587	25	DEC	H	
6844	2588	CD 0BE6	CALL	PUTVRM	;Output it to left of current position
6845	258B	24	INC	H	
6846	258C	24	INC	H	
6847	258D	3A F3B0	LD	A,(LINLEN)	
6848	2590	3C	INC	A	
6849	2591	BC	CP	H	
6850	2592	20 F0	JR	NZ,DELET4	
6851	2594	25	DEC	H	;Do next till end of visual
6852	2595		DELET5:		
6853	2595	0E 20	LD	C,' '	
6854	2597	CD 0BE6	CALL	PUTVRM	;Load raw code for space
6855	259A	CD 0C1D	CALL	GETVRM	
6856	259D	C2 09E1	JP	NZ,CKDPCS	
6857	25A0	E5	PUSH	HL	;End of line, all done
6858	25A1	2C	INC	L	
6859	25A2	26 01	LD	H,1	
6860	25A4	CD 0BD8	CALL	GETVRM	;Get first character next visual
6861	25A7	E3	EX	(SP),HL	

(MSX ROM BASIC BIOS) Macro-80 3.44
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6862	25A8	CD 0BE6	CALL	PUTVRM	
6863	25AB	E1	POP	HL	
6864	25AC	18 CF	JR	DELET3	;Put at last position last line

```
6865
6866      25AE      LERASE:
6867      ;
6868      ; Erase logical line
6869      ;
6870      25AE      CD 0A2E      CALL CKERCS
6871      25B1      CD 266C      CALL GTFIRST
6872      25B4      22 F3DC      LD (CSRY),HL
6873      25B7      18 05      JR TRUNCL
6874      25B9
6875
6876      TRUNC:
6877      ;
6878      ; Truncate logical line
6879      ;
6880      25B9      E5      PUSH HL
6881      25BA      CD 0A2E      CALL CKERCS
6882      25BD      E1      POP HL
6883
6884      TRUNCL:
6885      25BE      CD 0C1D      CALL GETTRM
6886      25C1      F5      PUSH AF
6887      25C2      CD 0AEE      CALL EOL
6888      25C5      F1      POP AF
6889      25C6      20 05      JR NZ,DPCSOW
6890      25C8      26 01      LD H,1
6891      25CA      2C      INC L
6892      25CB      18 F1      JR TRUNCL
6893
6894      DPCSOW:
6895      ;
6896      25CD      CD 09E1      CALL CKDPCS
6897      25D0      AF      XOR A
6898      25D1      32 FCA8      LD (INSFLG),A
6899      25D4      C3 242D      JP SETOVW
```

;Set L=first visual this logical line

;Is this line terminated?
;Save the condition
;Erase to end-of-line
;Restore condition
;Yes
;Go to next line
;Bump row counter
;And continue

```
6896      25D7      LAPPND:
6897      ;
6898      ; Append to current line
6899      ;
6900      25D7      CD 0A2E      CALL      CKERCS
6901      25DA      2A F3DC      LD        HL,(CSRY)
6902      25DD      2D          DEC        L
6903      25DE      L          L
6904      25DE      2C          INC
6905      25DF      CD 0C1D      CALL      GETTRM
6906      25E2      28 FA      JR         Z,LAP1
6907      25E4      3A F3B0      LD        A,(LINLEN)
6908      25E7      67          LD        H,A
6909      25E8      24          INC        H
6910      25E9      L          L
6911      25E9      25          DEC
6912      25EA      28 07      JR         Z,LAP3
6913      25EC      CD 0BD8      CALL      GETVRM
6914      25EF      FE 20      CP         ,
6915      25F1      28 F6      JR         Z,LAP2
6916      25F3      L          L
6917      25F3      CD 0A5B      CALL      ADVCUR
6918      25F6      18 D5      JR         DPCSOW
6919      25F8      LNXSTD:
6920      ;
6921      ; Move to next word
6922      ;
6923      25F8      CD 0A2E      CALL      CKERCS
6924      25FB      CD 2634      CALL      PRVCHK
6925      25FE      LNW1:
6926      25FE      CD 2624      CALL      NXTCHK
```

;Erase cursor
;Get current cursor position

;Line terminated?
;No, look at next line

;Reached start of line?
;Yes
;Get a character at the cursor
;Space?
;Yes, skip this

;Advance cursor to point to end of line
;Re-display cursor

;Still in word?

(MSX ROM BASIC BIOS) Macro-80 3.44 01-Jan-85
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```

6927 2601 28 CA JR Z,DPCSOW ;Reached screen bottom, abort
6928 2603 38 F9 JR C,LNW1 ;Yes
6929 2605 LNW2:
6930 2605 CD 2624 CALL NXTCHK ;Reached word?
6931 2608 28 C3 JR Z,DPCSOW ;Reached screen bottom, abort
6932 260A 30 F9 JR NC,LNW2 ;Not yet
6933 260C 18 BF JR DPCSOW
6934 260E LBCKWD:
6935 ;
6936 ; Move to previous word
6937 ;
6938 260E CD 0A2E CALL CKERCS
6939 2611 LBW1:
6940 2611 CD 2634 CALL PRVCHK ;Still in separator?
6941 2614 28 B7 JR Z,DPCSOW ;Reached screen top, abort
6942 2616 30 F9 JR NC,LBW1 ;Yes
6943 2618 LBW2:
6944 2618 CD 2634 CALL PRVCHK ;Reached separator?
6945 261B 28 B0 JR Z,DPCSOW ;Reached screen top, abort
6946 261D 38 F9 JR C,LBW2 ;Not yet
6947 261F CD 0A5B CALL ADVCUR
6948 2622 18 A9 JR DPCSOW
6949 2624 NXTCHK:
6950 ;
6951 ; Move right and check
6952 ;
6953 2624 2A F3DC LD HL,(CSRY) ;Get current cursor position
6954 2627 CD 0A5B CALL ADVCUR ;Advance cursor
6955 262A CD 0C32 CALL GETLEN ;Get an actual height of screen
6956 262D 5F LD E,A ;[D],[E] hold the dead end position
6957 262E 3A F3B0 LD A,(LINLEN)

```

```

6958      2631      57          LD      D,A
6959      2632      18 09          JR      PRVCK1
6960      2634          PRVCHK:
6961      ;
6962      ; Move left and check
6963      ;
6964      2634      2A F3DC          LD      HL,(CSRY)
6965      2637      CD 0A4C          CALL   BS
6966      263A      11 0101         LD      DE,0101H
6967      263D          PRVCK1:
6968      ;
6969      ; Check current character
6970      ; Carry set if the character is regarded as separator
6971      ;
6972      263D      2A F3DC          LD      HL,(CSRY)
6973      2640      E7              RST     20H
6974      2641      C8              RET     Z
6975      2642      11 2668          LD      DE,RESZRO
6976      2645      D5              PUSH   DE
6977      2646      CD 0BD8          CALL   GETVRM
6978      2649      FE 30          CP      '0'
6979      264B      3F              CCF
6980      264C      D0              RET     NC
6981      264D      FE 3A          CP      ':'
6982      264F      D8              RET     C
6983      2650      FE 41          CP      'A'
6984      2652      3F              CCF
6985      2653      D0              RET     NC
6986      2654      FE 5B          CP      'Z'+1
6987      2656      D8              RET     C
6988      2657      FE 61          CP      'a'

```

;Get current cursor position
;Regress cursor
;[D],[E] hold the dead end position
;Get updated cursor position
;Reached dead end?
;Yes, return with Z flag
;Jump to RESZRO when done
;Get ASCII code of character at [H],[L]
;Set carry if "0".."9"
;Set carry if "A".."Z"
;Set carry if "a".."z"

```

6989      2659      3F      CCF
6990      265A      D0      RET
6991      265B      FE 7B      NC
6992      265D      D8      CP      'z'+1
6993      265E      FE 86      C
6994      2660      3F      CP      86H
6995      2661      D0      CCF
6996      2662      FE A0      RET
6997      2664      D8      CP      0A0H
6998      2665      FE A6      C
6999      2667      3F      CP      0A6H
7000      2668      RESZRO:
7001      2668      3E 00      LD      A,0
7002      266A      3C      INC      A
7003      266B      C9      RET
7004
7005      ;
7006      ; Set H,L to first visual line in logical line
7007      ;
7008      266C      GTFRST:
7009      266C      2D      DEC      L
7010      266D      28 05      JR      Z,GTFTST1
7011      266F      CD 0C1D      CALL   GETTRM
7012      2672      28 F8      JR      Z,GTFRST
7013      2674      2C      GTFTST1:
7014      2674      2C      INC      L
7015      2675      3A FBCA      LD      A,(FSTPOS)
7016      2678      BD      CP      L
7017      2679      26 01      LD      H,1
7018      267B      C0      RET      NZ
7019      267C      2A FBCA      LD      HL,(FSTPOS)
7020      267F      C9      RET

;Look a line just above
;If we're at top of screen, all done
;Get terminator
;More to get above in this logical
;L=line number of first visual
;Get first line
;Same?
;Assume not
;Good assumption
;Get first line and column

```

- MSXINL, Screen editor - Process special characters

7020

END

MSX BIOS CROSS REFERENCE

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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- BIOS CROSS REFERENCE LISTING -

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265

GPRT60	1#	4447	4452#			
GPRT70	1#	4462	4465#			
GPRT80	1#	4468	4470#			
GRPACX	1#	4410	4443	4453	4459	
GRPACY	1#	4408	4461	4471		
GRPATR	1#	1254				
GRPCGP	1#	1576	4612	4862	4901	
GRPCOL	1#	1573				
GRPCR	1#	4401	4446	4451	4456#	
GRPDIF	1#	4688	5111	5115	5153	5202
GRPHED	1#	1787				
GRPNAM	1#	1256	1283			
GRPPAT	1#	1252				
GRPPRT	1#	138	4389#			
GRPTAB	1#	3365	3377#			
GSPAD1	1#	1422	1425#			
GSPSIZ	1#	137	1420	1440#		
GTASPC	1#	228	5232#			
GTFRST	1#	6598	6871	7007#	7011	
GTFST1	1#	7009	7012#			
GTPAD	1#	186	3867#			
GTPAD0	1#	3893	3899#			
GTPAT1	1#	1526#	1540			
GTPDL	1#	187	3807#			
GTPDP1	1#	3888	3891#			
GTROW8	1#	2689	3698	3726#	3804	
GTSTCK	1#	184	3683#			
GTTRIG	1#	185	3783#			
H.CHGE	1#	3410				
H.CHPU	1#	1819				
H.DSPC	1#	2070				
H.DSPF	1#	2370				

(MSX BASIC ROM BIOS) Macro-80
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H.ERAC	1#	2125			
H.ERAF	1#	2350			
H.FORM	1#	4203			
H.INIP	1#	1470			
H.INLI	1#	6492			
H.ISFL	1#	4139			
H.KEYC	1#	2993			
H.KEYI	1#	2621			
H.KYEA	1#	3160			
H.LPTO	1#	1730			
H.LPTS	1#	1759			
H.NMI	1#	4061			
H.OUTD	1#	5798			
H.PHYD	1#	4199			
H.PINL	1#	6475			
H.QINL	1#	6486			
H.TIMI	1#	2625			
H.TOTE	1#	1703			
HEADER	1#	5485			
HIGH	1#	5551			
HRSSCL	1#	4521	4526#		
HRZMOV	1#	4811	4841#		
HRZMVL	1#	4791	4807	4822	4837 4845#
ILN	1#	1951	2251#		
INDJMP	1#	1889#	1897	1994	6505
INESC	1#	1846	1987#		
INESC1	1#	1989	1995#		
INESC2	1#	2005	2009#		
INGI	1#	1051	3476#	3723	3915 3987 4012
INIFNK	1#	99	4065#		
INIGRI	1#	1260#	1263	1264	
INIGRP	1#	129	1245#	1722	

(MSX BASIC ROM BIOS) Macro-80
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INIML1	1#	1300#	1313	
INIML2	1#	1302#	1310	
INIML3	1#	1305#	1308	
INIMLT	1#	130	1286#	1723
INIPAT	1#	1143	1166	1466#
INIPT1	1#	1477#	1490	
INIT	1#	919		
INIT32	1#	128	1146#	1720
INITIO	1#	98	1038#	
INITQ	1#	1088	4328#	
INITXT	1#	127	1127#	1719
INLIN	1#	164	6478	6491#
INLIN1	1#	6481	6497#	
INLIN2	1#	6501#	6509	
INLOT0	1#	6538	6546#	
INLOT1	1#	6540	6549#	
INLOUT	1#	6507	6519#	6527
INS1	1#	6721#	6799	
INS2	1#	6723#	6735	
INS3	1#	6752	6755	6758#
INS4	6766	6772#		
INS45	1#	6778	6780#	
INS5	1#	6771	6786#	
INS6	1#	6742	6793#	
INSERT	1#	6544	6714#	
INSFLG	1#	6536	6677	6708 6894
INSLN0	1#	2258#	6770	
INSLN1	1#	2279#	2288	
INTCNT	1#	2638	2647	
INTFLG	1#	927	944	3217 3419 6500
INTRET	1#	2624	2672	2720 2723 2731#
INTVAL	1#	2645		

(MSX BASIC ROM BIOS) Macro-80
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ISCNTC	1#	168	922#	3437
ISFLIO	1#	247	4135#	5799
JFLVRM	1#	1579#	1593	
JIFFY	1#	2651	2653	
JMPBC	1#	1887	1902	1905#
JMPWRT	1#	5191	5204	5218#
JPPPAL	1#	4398	4403	4412
JPUTCH	1#	3025	3032#	3060
KAUEO	1#	3260	3265#	
KANAMD	1#	1053	3256	
KANANO	1#	3267	3290#	
KANASF	1#	3269	3307#	
KANAST	1#	3002	3173	
KANJNO	1#	3261	3324#	
KANJSF	1#	3263	3341#	
KEEPH	1#	5560#	5562	
KEEPL	1#	5555#	5557	
KEYANY	1#	2795	2828#	
KEYBUF	1#	3401		
KEYCHK	1#	2719	2746#	
KEYCK1	1#	2753#	2760	
KEYCK2	1#	2773#	2779	
KEYCK3	1#	2778	2781#	
KEYCK4	1#	2730	2780	2785#
KEYCK5	1#	2789#	2798	
KEYCOD	1#	2848	2983#	
KEYINT	1#	97	2603#	
KEYNOM	1#	2896	3053#	
KEYSFT	1#	2895	3050#	
KEYTRG	1#	3786	3802#	
KILBUF	1#	251	962	1002#
KSTKTB	1#	3703	3765#	

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KYLCNT	2900	2914#			
KY1NOM	2902	2903#			
KY1SFC	2899	2924#			
KY1SFT	2901	2908#			
KYALP	1#	2865	3034#		
KYANY1	1#	2844#	2852		
KYCLTB	1#	2898#	3063		
KYCLA0	1#	2999	3005#		
KYCLAS	1#	2995	3007#	3017	
KYCLS	2881	3150#			
KYCOD1	1#	2863	3061#		
KYEASY	1#	2867	2875	2879	2883
KYFNC1	1#	3086	3090#		3156#
KYFNC2	1#	3098#	3120		
KYFNC3	1#	3107#	3113		
KYFUNC	1#	2873	3080#		
KYGRAP	1#	3001	3360#		
KYJTAB	1#	2859#	2992		
KYKAN1	1#	3262	3264	3268	3270#
KYKANA	1#	3004	3252#		
KYKLOK	1#	2871	3169#		
KYLOCK	1#	2869	3189#		
KYNUM	1#	2861	3018#		
KYSTCK	1#	3686	3696#		
KYSTOP	1#	2877	3206#		
KYSTP1	1#	3214	3216#		
LAP1	1#	6903#	6906		
LAP2	1#	6910#	6915		
LAP3	1#	6912	6916#		
LAPPND	1#	6579	6896#		
LBCKWD	1#	6575	6934#		
LBLKSP	1#	6626	6637	6648#	6655

LBREAK	1#	6583	6687#
LBREK0	1#	6682#	6689
LBW1	1#	6939#	6942
LBW2	1#	6943#	6946
LCR1	1#	6613#	6647
LCR2	1#	6615#	6642
LCRNRM	1#	6624	6633#
LCRNUL	1#	6622	6638#
LCRRET	1#	6585	6592#
LDELNX	1#	6589	6801#
LDELEXI	1#	6807	6810#
LDIMV1	1#	1457#	1464
LDIRMV	1#	116	1452#
LDIRVM	1#	117	1493#
LDIVM1	1#	1497#	1504
LEFT	1#	1963	2148
LEFTC	1#	213	4828#
LEFTC1	1#	4826	4838#
LERASE	1#	6587	6866#
LF	1#	1860#	1920
LFTQ	1#	205	4347#
LINL32	1#	1154	
LINL40	1#	1135	
LINLEN	1#	1136	1155
LINLTB	6731	6753	6805
	1#	1561	
LNWL	1#	6925#	6928
LNW2	1#	6929#	6932
LNXTLN	1#	6670#	6702
LNXTWD	1#	6577	6919#
LOC	1#	1955	1979#
LOW	1#	5506	5529

(MSX BASIC ROM BIOS) Macro-80
- BIOS CROSS REFERENCE LISTING -

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271

LOWLIM	1#	5638	5663		
LPT.DW	1#	623#	1740		
LPT.SB	1#	624#	1055	1742	1744
LPT.ST	1#	625#			
LPTABO	1#	1734	1748#		
LPTCH0	1#	5833	5836	5843#	
LPTCH1	1#	5811	5846	5864#	
LPTCHR	1#	5852	5858	5861	5867#
LPTCOD	1#	5800	5805#		5874
LPTOUT	1#	159	1726#	5868	
LPTPOS	1#	1751	5824	5837	5841
LPTSTT	1#	160	1735	1757#	
MAPSPC	1#	5850	5854	5872#	
MAPXYC	1#	219	4413	4540#	
MDNC	1#	4884	5006	5010	5016#
MHCMOV	1#	4966	4987#		
MHZMV1	1#	4949	4960	4972	4982
MLFTC	1#	4834	4977#		4991#
MLFTC1	1#	4975	4983#		
MLTATR	1#	1295			
MLTCGP	1#	1591	4650	5001	5025
MLTNAM	1#	1297	1333		
MLTPAT	1#	1293			
MPXY1	1#	4631	4633#		
MPXYC	1#	4590	4624#		
MNSTCX	1#	5071	5221#	5230	
MORACT	1#	3577#	3596	3619	3631
MORSPL	1#	5821#	5826		
MOTRON	1#	4045	4048#		
MOTRWT	1#	5479#	5483		
MREADC	1#	4684	4706#		
MRGTC	1#	4804	4955#		

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MRGTC1	1#	4953	4961#		
MSCANL	1#	5371	5411#	5420	
MSCANR	1#	5269	5336#	5347	
MSCNR1	1#	5341	5350#		
MSCNR2	1#	5356#	5361		
MSETC	1#	4738	4745#		
MSETC1	1#	4759	4764#		
MTDNC	1#	4860	4996#		
MTLFT	1#	4819	4967#		
MTRGT	1#	4788	4941#		
MTSBRD	1#	5340	5360	5417	5421#
MTUPC	1#	4899	5022#		
MUPC	1#	4926	5030	5033	5038#
MUSCLL	1#	1073#	1076		
MUSICF	1#	1070	2657	3642	3657 3670
MUSINT	1#	2660#	2669		
MUSITB	1#	1098	1114#		
MVTMOV	1#	5021	5043#		
MVTMV1	1#	5045	5047#		
NAMBAS	1#	1138	1157	1553	2557
NEWKEY	1#	2752	2771	2788	
NMI	1#	124	4057#		
NMSFTB	1#	2885#	3027		
NOKEY	1#	3187#			
NONEG1	1#	3935	3938#		
NONEG2	1#	3943	3946#		
NOSTOP	1#	2763	2766	2769#	
NOTABL	1#	5819	5830#		
NOTAUT	1#	6601	6603#		
NOTRAN	1#	5730	5738#		
NSETCX	1#	227	5055#	5328	5395
NSTC10	1#	5080#	5084		

NSTC20	1#	5076	5094#			
NSTC30	1#	5108#	5117			
NSTC40	1#	5107	5118#			
NSTC50	1#	5125	5139#			
NSTCSP	1#	5082	5130#			
NTBKS2	1#	5817#				
NTBOTM	1#	2379	2381#			
NTHIRA	1#	5856	5859#			
NTINTT	1#	2642	2646#			
NTMSXP	1#	5844				
NXTCHK	1#	6926	6930	6949#		
OLDKEY	1#	1029	1031	2725	2726	2787
OLDSCR	1#	1134	1153	1702		
ONBRD1	1#	4797	4827	4914#	4954	4976
ONBRDR	1#	4874	4912#			
ONGSBF	1#	3145	3147			
OUTDLP	1#	248	5814#	5823		
OUTDO	1#	55	5788#			
OUTGI	1#	3986	3995	3999	4010	4017#
OUTNTB	1#	6523	6533#			
OUTTAB	1#	6525#	6531			
PADX	1#	3895	3950			
PADX1	1#	3926	3928	3932	3953#	
PADY	1#	3897	3952			
PATBAS	1#	1161	1253	1294	1380	1427
PATWR1	1#	5181	5205#			
PATWRK	1#	1523	4414			
PATWRT	1#	4740	5089	5142#		
PBDHRT	1#	1832#	2853	3430	3679	
PDL1	1#	3833#	3835			
PDL2	1#	3856#	3861			
PDL3	1#	3859	3863#			

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PUTVRM	1#	1854	2105	2131	2300	2512#	6729	6844	6854	6862
QINLIN	1#	166	6482#							
QSTART	1#	4333	4368	4377#						
QUEBAK	1#	4324								
QUEUEN	1#	3560	3674							
QUEUES	1#	4384								
RAMLOW	1#	296	871	872						
RAWPRT	1#	5809								
RDBIT	1#	5695	5715#							
RDBITL	1#	5727#	5735	5744						
RDESLT	1#	291	299#							
RDPSG	1#	148	3481#	3712	3846	3909				
RDSLT	1#	49	289#	304	1482	1531				
RDVDP	1#	241	4112#							
RDVRM	1#	111	1606#	4685	4690	4708	4750	5151	5155	
READC	1#	225	4674#	5285	5310	5382	5425			
READC0	1#	4696#	4712							
READC1	1#	4694	4701#	4711						
READYR	1#	2767								
REDCOD	1#	3927	3930	3959#						
REDLOP	1#	3983#	3996							
REDPAD	1#	3924	3925	3964	3969	3975#				
REPCNT	1#	1033	2721	2784						
REQSTP	1#	965	988	991						
REQTRP	1#	967	2634	2644	2701	2704	2707	2710	2713	3127#
RESZRO	1#	6975	7000#							
RETRET	1#	5495	5531#							
RETURN	1#	6516#	6573							
RG0SAV	1#	1205	1214	1232	1273	1322				
RG1SAV	1#	1173	1180	1219	1237	1278	1327	1376	1400	1444
RGHTC1	1#	4796	4808#							
RGTEXT	1#	5122	5126#							

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RIGHT	1#	1855	1961	2135#	2170			
RIGHTC	1#	212	4798#	5227	5390	5418		
RSET10	1#	2038	2043#					
RSLREG	1#	239	4116#					
RSTFL1	1#	3645#	3647					
RSTMOD	1#	1969	1976#					
RUBOUT	1#	1853	2293#					
RUNFLG	1#	3902	4023	5279				
SAMEBG	1#	5170	5192#					
SAMEFG	1#	5176	5185	5198#				
SAVSTK	1#	979						
SCALXY	1#	218	4411	4475#				
SCANL	1#	231	5364#					
SCANL1	1#	5379#	5387					
SCANL2	1#	5384	5388#					
SCANL3	1#	5381	5391#					
SCANL4	1#	5334	5397#					
SCANR	1#	230	5261#					
SCANR1	1#	5284#	5293					
SCANR2	1#	5287	5296#					
SCANR3	1#	5306#	5314					
SCANR4	1#	5309	5312	5315#				
SCITBL	1#	6503	6564#					
SCLXOK	1#	4513	4518#					
SCLYOK	1#	4497	4502#					
SCNCNT	1#	2670						
SCRMOD	1#	1133	1152	1251	1292	1551	1648	2455
SELEXP	1#	301	342	420	486	544#	2540	4537
SELPRM	1#	290	331	411	477	500#		
SETATR	1#	224	4714#					
SETC	1#	226	4425	4727#	5226	5435		
SETCHK	1#	2352	2372	2446#				

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STOREC	1#	222	4435	4665#	5323	5331
STRTMS	1#	149	3651#			
STSTYL	1#	2027	2040#			
SULOP	1#	5649#	5652			
SYN05	1#	5577#	5589	5591	5602	
SYN10	1#	5583#	5606			
SYN11	1#	5597	5600#			
SYN20	1#	5608#				
SYN30	1#	5615#	5620			
SYNCHR	1#	46				
SYNCW1	1#	5486	5489#			
SYNLP1	1#	5493#	5499			
T32ATR	1#	1162				
T32CGP	1#	1158				
T32COL	1#	1662				
T32NAM	1#	1156	1242			
T32PAT	1#	1160				
TAB	1#	1918	2190#	2199		
TAPIN	1#	194	5659#			
TAPIOF	1#	195	5462#			
TAPION	1#	193	5568#			
TAPOFF	1#	198	5450#			
TAPOON	1#	196	5469#			
TAPOUT	1#	197	5501#			
TDOWNC	1#	217	4436	4850#		
TERMIN	1#	2314	2579#	6495	6672	
TGLINS	1#	6571	6704#			
TIMOUT	1#	5762	5770#			
TLEFT	1#	4812#	5380	5415		
TOTEXT	1#	176	973	1696#		
TRGFLG	1#	2694				
TRIG1	1#	3794	3796#			

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